# Relativistic Scalar–Vector Plenum:
# Mathematical Foundations of Lamphrodynamic Field Theory

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# Preface

This volume is the first of a three–volume monograph developing the Relativistic Scalar–Vector Plenum (RSVP) as a rigorous mathematical framework. Its primary audience is mathematicians and theoretical physicists with a background in differential geometry, category theory, and quantum field theory. A secondary aim is pedagogical: the material is written so that physicists willing to learn derived geometry can follow, provided they are comfortable with homological algebra and gauge theory.

The central goal of Volume I is to construct the lamphrodynamic field theory underlying RSVP, to formulate it in the language of derived algebraic geometry and shifted symplectic structures, and to establish fundamental analytic results (well–posedness, energy estimates, regularity). Later volumes will build on this foundation to explore cosmology, emergent gravity, geometric computation, and philosophical implications.

We have tried to adhere to the following design principles:

- *Rigor first in foundations.* The core constructions (derived stacks, shifted symplectic forms, AKSZ action) are presented with as much mathematical rigor as possible.
- *Motivation before formalism.* Each chapter begins with an informal explanation of why the subsequent technical machinery is needed.
- *Explicit status of claims.* Theorems are proved or referenced; conjectures and open problems are clearly labeled and accompanied by evidence or heuristic arguments.

Volume I is structured in four parts. Part I reviews the necessary background in derived geometry and shifted symplectic structures, oriented toward physicists. Part II introduces the lamphrodynamic plenum and constructs the relevant derived moduli stacks and AKSZ action. Part III develops the analytic foundations of the lamphrodynamic PDEs. Part IV investigates geometric and topological properties, including entropic singularities and symmetries.

Subsequent volumes will not repeat this material, but will refer back to it frequently.

**Acknowledgements.** The author thanks numerous interlocutors — human and artificial — for many conversations that shaped this framework. Any errors are, of course, the authors responsibility.

# Contents

# Notation and Conventions

Unless otherwise stated, we adopt the following conventions:

- $(M, g)$ denotes a smooth oriented Lorentzian or Riemannian manifold, with metric $g$ of signature $(-, +, \ldots, +)$ in the Lorentzian case.
- Greek indices $\mu, \nu, \ldots$ range over spacetime coordinates; Latin indices $i, j, \ldots$ are reserved for spatial coordinates when a splitting is chosen. We freely use the abstract index notation of Penrose.
- The Levi–Civita connection of $g$ is denoted by $\nabla$ and the associated Laplace–Beltrami operator by $\Delta = \nabla_a \nabla^a$.
- The symbol $\mathrm{Map}(X, Y)$ denotes a (derived) mapping stack when $X$ and $Y$ are derived stacks, and an ordinary mapping space when they are topological or smooth manifolds; the intended sense is clear from context.
- All tensor contractions use the Einstein summation convention.
- The core lamphrodynamic fields of RSVP are denoted $(\Phi, \boldsymbol{v}, S)$: a scalar potential, a vector flow, and an entropy density. We often group them into a single configuration field $\Psi := (\Phi, \boldsymbol{v}, S)$.

We recommend that readers unfamiliar with derived stacks or shifted symplectic geometry skim Chapters 1 and 2 first, and then consult Chapter 3 and **??** as needed when these notions appear later.

# Part I

# Derived Algebraic Geometry for Physicists

# Chapter 1

# Introduction and Motivation

The Relativistic Scalar–Vector Plenum (RSVP) is a proposal for a unified field–theoretic framework in which gravity, cosmology, and certain aspects of information and agency arise from the dynamics of a triplet of fields

$$(\Phi, \boldsymbol{v}, S)$$

living on a spacetime manifold $(M, g)$ or, more generally, on a derived geometric object that refines $(M, g)$. The scalar field $\Phi$ encodes a form of lamphrodynamic potential, $\boldsymbol{v}$ represents a flux or flow of negentropic influence, and $S$ is an entropy density that couples to geometry and matter.

At a heuristic level, RSVP treats spacetime not as a static stage on which fields live, but as a *plenum* whose local structure is determined by the coupled evolution of $(\Phi, \boldsymbol{v}, S)$. Geometrical quantities such as curvature, torsion, and metric relations are then derived from the lamphrodynamic configuration rather than posited *a priori*.

This introductory chapter explains why such a framework naturally leads us to derived algebraic geometry and shifted symplectic structures, and sets the stage for the rigorous constructions that follow.

## 1.1 From Classical Fields to Derived Geometry

In classical field theory, a field configuration is a section of some bundle over spacetime $M$, and the space of fields is modeled as an infinite dimensional manifold or Fréchet space. Functional integrals, symmetries, and constraints are handled with varying degrees of rigor depending on context.

However, many modern developments — such as moduli spaces of solutions to gauge equations, intersection theory on spaces of instantons, and the BV formalism for gauge theories — reveal that naive smooth manifolds are insufficient. One encounters:

- Singularities when solutions collide or degenerate.
- Higher automorphisms from gauge symmetries.
- Obstructions and derived intersections.

Derived algebraic geometry was developed precisely to give these moduli spaces a well–behaved home. Instead of regarding a moduli space as a naive quotient or set of solutions, one encodes it as a derived stack equipped with additional structure (cotangent complex, obstruction theory, shifted symplectic form).

**Intuition.** Even if the base spacetime $M$ is an ordinary smooth manifold, the *space of fields* is almost never a manifold in the classical sense. Gauge symmetry, constraints, and singularities force us to work in a homotopical setting. Derived stacks and $\infty$–categories are the natural language for this.

In RSVP, lamphrodynamic fields are subject to:

- Gauge transformations mixing $\Phi$ and $\boldsymbol{v}$,
- Constraints relating $S$ to curvature and matter content,
- Entropic singularities where $S$ becomes non–smooth or diverges.

These features suggest that the moduli of lamphrodynamic configurations should be modeled as a derived stack, which we denote $\mathsf{Plenum}$. A significant portion of Volume I is devoted to constructing $\mathsf{Plenum}$, equipping it with a $(-1)$–shifted symplectic structure, and understanding its tangent and obstruction theory.

## 1.2 The Lamphrodynamic Triplet

We give a preliminary, heuristic description of the fields before formalizing them in **????**.

- $\Phi$ is a real–valued scalar field on $M$ (or a section of some line bundle) representing a lamphrodynamic potential. Regions of high $\Phi$ can be thought of as reservoirs of "available negentropy".
- $\boldsymbol{v}$ is a vector field on $M$, or more generally a section of $TM \otimes E$ for some auxiliary bundle $E$, encoding the directed transport of lamphrodynamic influence. It plays a role analogous to a current or velocity field.
- $S$ is a scalar entropy density that couples to both geometry and matter fields. In contrast to thermodynamic entropy defined on macrostates, $S$ is intended as a local field that participates dynamically in the evolution equations.

The basic lamphrodynamic field equations are schematically of the form

$$\partial_t \Phi = \mathcal{D}_\Phi \Delta \Phi + \mathcal{F}_\Phi(\Phi, \boldsymbol{v}, S, g), \tag{1.1}$$

$$\partial_t S = \mathcal{D}_S \Delta S + \mathcal{F}_S(\Phi, \boldsymbol{v}, S, g), \tag{1.2}$$

$$\partial_t \boldsymbol{v} = \mathcal{D}_v \Delta \boldsymbol{v} + \mathcal{F}_v(\Phi, \boldsymbol{v}, S, g), \tag{1.3}$$

where $\mathcal{D}_\Phi, \mathcal{D}_S, \mathcal{D}_v$ are diffusion constants and the nonlinear terms $\mathcal{F}_{\Phi,S,v}$ encode lamphrodynamic interactions and coupling to curvature. Precise versions of these equations will be derived from an action principle in **????**.

**Warning.** Equations (1.1)–(1.3) are merely schematic at this stage. They suggest that RSVP is, at its analytic core, a system of parabolic or mixed–type PDEs; however, the actual lamphrodynamic equations will include nonlocal and higher–order terms arising from the BV–AKSZ construction.

## 1.3 Why a $(-1)$–Shifted Symplectic Structure?

In classical Hamiltonian mechanics, the phase space of a system is a symplectic manifold $(X, \omega)$ and the dynamics are encoded by a Hamiltonian function $H$ via the equation

$$\iota_{X_H}\omega = \mathrm{d}H.$$

For gauge theories and field theories, the BV formalism generalizes this picture to a graded setting: one replaces $(X, \omega)$ by a graded manifold equipped with an odd symplectic form, and the Hamiltonian $H$ by a master action $S$ satisfying $\{S, S\} = 0$.

In derived geometry, these graded symplectic manifolds are captured by *shifted symplectic structures* on derived stacks. A $k$–shifted symplectic structure is, roughly, a closed, nondegenerate 2–form of cohomological degree $k$ on the derived stack. The case $k = -1$ is particularly important for the BV formalism: it corresponds to odd symplectic forms and is tailored to encode gauge theories.

**Definition 1.1.** Let $\mathcal{X}$ be a derived Artin stack over a field of characteristic zero. A $(-1)$*–shifted symplectic structure* on $\mathcal{X}$ is a closed 2–form

$$\omega \in \mathcal{A}^{2,cl}(\mathcal{X})[-1]$$

whose underlying pairing on the cotangent complex is nondegenerate in the derived sense.

The key insight for RSVP is that the moduli stack $\mathsf{Plenum}$ of lamphrodynamic configurations should carry such a $(-1)$–shifted symplectic structure, with the master action functional $S_{RSVP}$ playing the role of a Hamiltonian satisfying the classical master equation

$$\{S_{RSVP}, S_{RSVP}\} = 0.$$

Once this structure is constructed, the AKSZ formalism provides a natural way to build $S_{RSVP}$ from geometric data associated to $\mathsf{Plenum}$ and the spacetime source manifold.

**Intuition.** The choice of $k = -1$ is not arbitrary: lamphrodynamic fields are subject to gauge redundancies and constraints, so their moduli naturally form a BV phase space rather than a naive symplectic manifold. This forces the symplectic structure to be of odd degree, hence $k = -1$.

A central technical goal of Volume I is to:

- Construct $\mathsf{Plenum}$ as a derived Artin stack.
- Equip $\mathsf{Plenum}$ with a $(-1)$–shifted symplectic structure $\omega$.

- Show that $\omega$ can be expressed in terms of $(\Phi, \boldsymbol{v}, S)$ and curvature data, and that it is closed and nondegenerate.

These results will be established in **????**.

## 1.4 Analytic Questions: WellPosedness and Regularity

The derived geometric formulation of RSVP provides a powerful conceptual framework, but physics also demands that the associated PDEs be analytically well–behaved. Concretely, we must address:

1. *Well–posedness.* Given initial data $(\Phi_0, \boldsymbol{v}_0, S_0)$, does there exist a unique solution $(\Phi, \boldsymbol{v}, S)$, at least for short time, in appropriate Sobolev spaces?
2. *Energy estimates.* Are there a priori bounds controlling the growth of norms of the fields, and do these bounds depend continuously on the initial data?
3. *Regularity and singularities.* Under what conditions can singularities form in finite time, and how are such singularities related to entropic phenomena in RSVP?
4. *Long–time behavior.* Do solutions converge to lamphrodynamic equilibria or attractors, and can one classify these asymptotic states?

These questions are addressed in Part III of this volume. We will prove global well–posedness for linearized lamphrodynamic equations and establish energy estimates and local well–posedness for certain nonlinear regimes. Global existence for the fully nonlinear system remains conjectural; we will state precise open problems and present partial evidence.

## 1.5 Organization of the Three Volumes

For orientation, we briefly summarize the roles of the three volumes.

### Volume I: Mathematical Foundations

This volume develops:

- The background in derived algebraic geometry and shifted symplectic structures needed to formulate RSVP rigorously.
- The construction of the lamphrodynamic moduli stack $\mathsf{Plenum}$ and its $(-1)$–shifted symplectic structure.
- The AKSZ–BV action for lamphrodynamic fields and its classical master equation.
- Fundamental analytic results on well–posedness, energy estimates, and regularity of the lamphrodynamic PDEs.

### Volume II: Physical Applications

The second volume applies this foundation to physics:

- Derivation of modified Einstein equations with an explicit lamphrodynamic stress–energy contribution $\Theta_{ab}$.
- Development of a cosmological model in which redshift, structure formation, and cosmic microwave background anisotropies are governed by lamphrodynamic fields rather than pure metric expansion.
- Detailed comparison with observational data (supernovae, BAO, Planck CMB measurements) and identification of potential tests that distinguish RSVP from ΛCDM.

### Volume III: Extensions and Interpretations

The third volume explores:

- The SpherePop calculus as a geometric model of computation, including categorical and homotopical formalization and a proof of Turing completeness.
- L–systems, semantic fields, and distributed computation (CRDTs) as instances of lamphrodynamic or related field theories.
- Entropy–weighted spectral geometry and "cymatic" resonance modes.
- Philosophical and creative dimensions, including the interpretation of RSVP within structural realism and process philosophy.

## 1.6 What is Proven, What is Conjectural

Given the ambitious scope of RSVP, it is important to distinguish carefully between results that are established in this monograph and those that remain conjectural.

- *Proven in Volume I:* We provide complete proofs for the construction of $\mathsf{Plenum}$ as a derived Artin stack, for the existence of a $(-1)$–shifted symplectic form on $\mathsf{Plenum}$ under specified conditions, and for well–posedness results in the linear and certain nonlinear lamphrodynamic regimes.
- *Proven in Volume II:* We derive modified Einstein equations from the lamphrodynamic action, obtain concrete formulas for cosmological observables (including a redshift function $f(S, \boldsymbol{v})$), and perform quantitative comparisons with existing data. The degree of observational agreement is investigated but not treated as a fully conclusive fit.
- *Proven in Volume III:* We rigorously formalize SpherePop as a homotopy–theoretic model of computation, prove that merge operations are homotopy colimits, and establish Turing completeness via explicit encodings.

- *Conjectural:* Global existence of solutions for the full nonlinear lamphrodynamic system; completeness of the cosmological model as a full replacement for ΛCDM; certain speculative connections between lamphrodynamic spectral modes and cognition or consciousness.

These conjectural aspects are clearly labeled as such, and in each case we provide reasons for their plausibility and suggest directions for future research.

## 1.7 How to Read this Volume

Different readers may wish to approach the material in different orders.

- **Mathematical physicists** familiar with homological algebra and derived geometry may skim Part I and focus on Parts II and III, where the lamphrodynamic stack and PDE analysis are developed.
- **Physicists new to derived geometry** are encouraged to read Chapters 2 and 3 carefully, referring to appendices as needed, before tackling the AKSZ construction in **????**.
- **Analysts** interested in PDE aspects may wish to jump directly to Part III, using the early chapters primarily for motivation and notation.

Throughout, we have tried to separate background material, core constructions, and speculative interpretations. The reader is invited to treat this volume as a foundation that can be built upon independently of whether they ultimately accept the full physical picture proposed by RSVP.

In the next chapter, we begin our technical journey by reviewing the basic language of model categories, $\infty$–categories, and derived stacks, with an eye toward their eventual application to lamphrodynamic field theory.

# Chapter 2

# $\infty$–Categories and Homotopy Theory

This chapter reviews the homotopical and $\infty$–categorical structures needed to formulate derived geometry and, ultimately, the lamphrodynamic moduli stack $\mathsf{Plenum}$ developed in later chapters. We assume that the reader is familiar with basic category theory and classical homotopy theory. Our goal is not to give a comprehensive treatment, but to present a minimal collection of definitions, constructions, and results that will be invoked in the sequel.

Throughout, we work over a field of characteristic zero (typically $\mathbb{R}$ or $\mathbb{C}$) unless otherwise specified. References include [**Lurie2009HTT**], [**Lurie2017HA**], and [**ToenVezzosi2008**].

## 2.1 Model Categories and Homotopical Structures

The motivation for $\infty$–categories begins with the desire to regard homotopy equivalent objects as "the same" while retaining control over higher homotopies. Classical category theory does not remember higher homotopies; instead, it collapses homotopies to identities. In homotopy theory, however, we need to keep track of:

- homotopies between maps,
- homotopies between homotopies,
- and so on, ad infinitum.

A traditional approach is provided by *model categories*, which endow a category with distinguished classes of weak equivalences, fibrations, and cofibrations satisfying Quillen's axioms [**Quillen1967**]. The basic example is the model structure on topological spaces, where weak equivalences are weak homotopy equivalences, fibrations are Serre fibrations, and cofibrations are retracts of relative CW–complex inclusions.

While model categories provide a powerful toolkit, they are ultimately an *encoding* of $\infty$–categorical structures, and the language of $\infty$–categories gives a more conceptual, coordinate–free approach.

## 2.2 $\infty$–Categories

An $\infty$–category is a category enriched in spaces rather than sets, or, equivalently, a homotopical object that remembers all higher morphisms. A concrete model is given by *quasi–categories* (simplicial sets satisfying inner horn conditions) introduced by Boardman and Vogt and developed extensively by Joyal and Lurie.

**Definition 2.1.** A *quasi–category* is a simplicial set $C$ such that every inner horn $\Lambda^n_i \to C$ admits a filler $\Delta^n \to C$ for $0 < i < n$.

This condition generalizes the Kan condition for simplicial sets corresponding to $\infty$–groupoids (spaces). Every topological space has a singular complex which is a Kan complex; every Kan complex is a model for an $\infty$–groupoid. The quasi–category condition weakens Kan filling so that objects need not have all inverses, thus modeling $\infty$–categories rather than $\infty$–groupoids.

**Intuition.** Ordinary categories forget higher homotopies between morphisms. Quasi–categories remember not only morphisms but also homotopies and homotopies between homotopies, all the way up. This makes them ideal for encoding moduli spaces, where higher automorphisms naturally appear.

## 2.3 $\infty$–Topoi and Sheaves of Spaces

Derived geometry requires a setting where "sheaves of homotopical data" can be treated systematically. This motivates the notion of an $\infty$–topos: a homotopical enhancement of Grothendieck topoi.

**Definition 2.2.** An *$\infty$–topos* is an $\infty$–category that is equivalent to an $\infty$–category of sheaves of spaces on a Grothendieck site. Equivalently, it is a presentable $\infty$–category satisfying descent for hypercovers [**Lurie2009HTT**].

Classical sheaf theory associates sets or vector spaces to open subsets of a manifold. An $\infty$–topos associates *spaces* (i.e. $\infty$–groupoids) to open subsets, thereby encoding higher homotopical information. Derived stacks are then defined as (higher) sheaves of $\infty$–groupoids satisfying appropriate representability and local geometricity conditions.

*Remark* 2.3. The $\infty$–topos of sheaves of spaces on the site $(\mathrm{Sch}, \mathrm{Zariski})$ is denoted $\mathrm{Shv}_\infty(\mathrm{Sch})$. Derived algebraic geometry works mostly inside suitable subcategories of this $\infty$–topos.

## 2.4 Derived Stacks

Derived stacks were developed by ToënVezzosi, Lurie, and others to handle moduli problems with obstructions, singularities, and nontrivial automorphism groups. Roughly speaking, derived stacks are functors from derived rings to $\infty$–groupoids satisfying descent and representability conditions.

**Definition 2.4.** A *derived stack* is a functor

$$F : \mathrm{dAlg}^{op} \longrightarrow \infty\text{–Gpd}$$

satisfying $\infty$–geometricity and descent conditions.

Derived stacks generalize classical schemes and algebraic stacks, adding:

- derived intersections,
- cotangent complexes,
- obstruction theories,
- and higher automorphisms.

**Intuition.** In physics, moduli spaces of solutions to gauge equations (e.g. Yang–Mills, GR) often have singularities and higher gauge symmetries. Derived stacks treat these automatically, removing the need for ad hoc patching arguments.

In RSVP, the moduli stack $\mathsf{Plenum}$ of lamphrodynamic configurations will be constructed as a derived Artin stack equipped with a $(-1)$–shifted symplectic structure. This stack encodes field configurations, gauge orbits, and entropic singularities in a unified homotopical object.

## 2.5 Tangent Complexes and Obstruction Theory

Classical manifolds possess tangent spaces at each point, controlling infinitesimal deformations. Derived stacks generalize this to a *tangent complex*, which is typically a chain complex, possibly with cohomology in multiple degrees.

**Definition 2.5.** Let $x \in F$ be a point of a derived stack $F$. The *tangent complex* $T_xF$ is the derived mapping space

$$T_xF := \mathrm{Map}_{\mathrm{dStk}}(\mathrm{Spec}\, k[\epsilon]/(\epsilon^2), F)_x,$$

considered as a complex in degrees $(-\infty, \dots, \infty)$.

The cohomology groups of $T_xF$ encode:

- $H^0(T_xF)$ classical tangent directions,
- $H^{-1}(T_xF)$ infinitesimal automorphisms,
- $H^1(T_xF)$ obstructions to extending deformations,
- higher degrees controlling higher obstructions and symmetries.

In particular, if $F$ is smooth (in derived sense), then $H^1(T_xF)$ may vanish, ensuring unobstructed deformations; if $H^{-1}(T_xF)$ is nonzero, $x$ has nontrivial automorphisms (gauge symmetries).

**Intuition.** The tangent complex of $\mathsf{Plenum}$ will encode:

- infinitesimal variations of $(\Phi, \boldsymbol{v}, S)$,
- gauge symmetries mixing $\Phi$ and $\boldsymbol{v}$,
- obstructions related to entropic singularities.

## 2.6 Cotangent Complex and Shifted Symplectic Structures

The cotangent complex $L_F$ of a derived stack $F$ is a fundamental invariant generalizing the classical cotangent bundle. It plays a central role in the definition of shifted symplectic structures.

**Definition 2.6.** The *cotangent complex* $L_F$ is a perfect complex on $F$ (when $F$ is derived Artin) whose dual $T_F := L_F^\vee$ generalizes the tangent bundle.

Shifted symplectic structures are defined as certain closed bilinear forms on $L_F$ (or $T_F$) of cohomological degree $k$. The case $k = -1$ is the setting of the BV formalism and will be central to RSVP.

We discuss shifted symplectic structures in detail in Chapter 3.

## 2.7 $\infty$–Groupoids and Fields

In ordinary geometry, a point in a moduli space of fields corresponds to a single classical solution. In derived geometry, such points carry higher homotopical structure: their automorphism groups form $\infty$–groupoids, which encode gauge equivalences and higher symmetries.

Consequently, the moduli stack $\mathsf{Plenum}$ should be thought of as an object in an $\infty$–category of derived stacks, with points representing field configurations, morphisms representing gauge transformations, and higher morphisms encoding redundancy among transformations themselves.

This viewpoint is essential to the BV formalism: ghosts, antifields, and the master equation all arise naturally from the derived structure of $\mathsf{Plenum}$.

## 2.8 Summary and Outlook

This chapter introduced the basic machinery underlying derived geometry: model categories, $\infty$–categories, $\infty$–topoi, and derived stacks, with an emphasis on tangent and cotangent complexes and higher automorphisms. These tools will be used in Chapter 3 and **??** to define and analyze shifted symplectic structures and the AKSZ formalism.

In **??**, we will construct the lamphrodynamic moduli stack $\mathsf{Plenum}$ explicitly as a derived Artin stack, then equip it with a $(-1)$–shifted symplectic form in **??**. The present chapter thus serves as the homotopical foundation for all subsequent developments.

# Chapter 3

# Shifted Symplectic Structures

This chapter develops the basic notions of shifted symplectic geometry used to formulate the moduli stack $\mathsf{Plenum}$ of lamphrodynamic configurations as a $(-1)$–shifted symplectic derived Artin stack. We assume familiarity with the cotangent complex of a derived stack and its dual tangent complex as introduced in Chapter 2. Throughout, we follow [**PTVV2013**], [**Calaque2015**] and related sources.

## 3.1 Preliminaries: Classical Symplectic Geometry

We begin with a brief review of classical symplectic geometry to fix notation and highlight the generalization to derived settings.

**Definition 3.1.** A *symplectic manifold* is a pair $(X, \omega)$ where $X$ is a smooth manifold and $\omega \in \Omega^2(X)$ is a closed (i.e. $\mathrm{d}\omega = 0$) and nondegenerate 2–form.

A central example is $T^*Q$ for a smooth manifold $Q$, equipped with the canonical 2–form $\omega = \mathrm{d}p_i \wedge \mathrm{d}q^i$. Classical Hamiltonian mechanics is framed on $(X, \omega)$ with Hamiltonian $H \in C^\infty(X)$.

In gauge theory and in field theory, however, the moduli space of classical solutions is typically singular, infinite dimensional, and equipped with higher automorphisms, so classical manifolds are no longer adequate. A homotopical refinement is needed, leading to the notion of shifted symplectic structures.

## 3.2 Graded Geometry and Cohomological Degree

In derived geometry, differential forms are replaced by cohomological complexes and their degrees must be tracked carefully. A $k$–shifted symplectic form is, informally, a closed 2–form of cohomological degree $k$ on the derived stack. Heuristically:

$$\omega \in H^k(\mathcal{A}^{2,cl}(F)),$$

where $F$ is a derived Artin stack and $\mathcal{A}^{2,cl}(F)$ is the sheaf of closed 2–forms on $F$.

**Definition 3.2.** Let $F$ be a derived Artin stack. A *$k$–shifted symplectic structure* on $F$ is a closed 2–form $\omega$ of cohomological degree $k$, represented by a morphism of complexes

$$\omega : \wedge^2 L_F \longrightarrow \mathcal{O}_F[k],$$

where $L_F$ is the cotangent complex of $F$, such that the induced pairing $L_F^\vee \otimes L_F^\vee \to \mathcal{O}_F[k]$ is nondegenerate in the derived sense.

Here $L_F^\vee = T_F$ generalizes the tangent bundle. The condition that $L_F$ be perfect (locally of finite Tor dimension) ensures that $T_F$ behaves well as a dual.

**Intuition.** The shift $k$ controls the "oddness" or "evenness" of the symplectic form. Classical symplectic structures correspond to $k = 0$. The case $k = -1$ corresponds to odd symplectic geometry and plays the role of BV phase spaces.

## 3.3 $(-1)$–Shifted Symplectic Structures and BV

The case $k = -1$ is the setting of the BatalinVilkovisky (BV) formalism for gauge theories. In this context, the derived stack $F$ encodes classical fields and gauge symmetries; the shifted symplectic form $\omega$ is odd; and the BV bracket

$$\{-, -\} : \mathcal{O}_F \otimes \mathcal{O}_F \to \mathcal{O}_F[-(-1)] = \mathcal{O}_F[1]$$

is defined from $\omega$.

**Definition 3.3.** Given a $(-1)$–shifted symplectic structure $\omega$ on $F$, the *BatalinVilkovisky bracket* $\{-, -\}$ on functions (or, more precisely, on appropriate cochains) is the unique odd Poisson bracket induced by $\omega$.

The BV mechanism encodes gauge symmetries as degenerate directions of the action, formally resolved by ghosts and higher homotopies. The *classical master equation*

$$\{S, S\} = 0$$

then expresses invariance under these symmetries. Later, in **????**, we will show how the lamphrodynamic action $S_{RSVP}$ arises from an AKSZ construction on the moduli stack $\mathsf{Plenum}$ equipped with a $(-1)$–shifted symplectic structure.

## 3.4 Closedness and Nondegeneracy in the Derived Sense

Closedness of $\omega$ is expressed in terms of the differential and the cohomological structure on $L_F$. Nondegeneracy means that the induced morphism

$$T_F \longrightarrow L_F[-k]$$

is an isomorphism in the derived category. For $k = -1$, this becomes

$$T_F \longrightarrow L_F[1],$$

which is automatically compatible with the BV grade.

*Remark* 3.4. For $k < 0$, the shifted symplectic form is "odd" and induces higher order compatibilities among tangent and cotangent complexes, matching precisely the structure required for the BV bracket.

Closedness means that $\omega$ lifts to a cocycle in a suitable truncated de Rham complex on $F$. We do not reproduce the full homotopical definition here (see [**PTVV2013**]), but emphasize that closedness is a global condition involving sheaf descent and higher coherences.

## 3.5 Examples and Fundamental Results

Key results from PantevToënVaquiéVezzosi [**PTVV2013**] include:

**Theorem 3.5** (PTVV). *Let $X$ be a smooth scheme and let $T^*kX$ be the $k$–shifted cotangent stack. Then $T^*kX$ carries a canonical $k$–shifted symplectic structure.*

This theorem provides the fundamental example of shifted symplectic structures and plays a role analogous to the cotangent bundle in classical mechanics. In derived geometry, one generalizes the phase space to $T^*kX$ for any integer $k$, with $k = -1$ corresponding to BV theory.

Another key result concerns mapping stacks:

**Theorem 3.6** (PTVV). *Let $M$ be an oriented smooth manifold of dimension $d$, and let $F$ be a derived Artin stack with an $n$–shifted symplectic structure. Then the mapping stack* $\mathrm{Map}(M, F)$ *carries an $(n - d)$–shifted symplectic structure.*

This result is central to the AKSZ construction, where $M$ is typically the source manifold (e.g. the spacetime worldvolume) and $F$ is the target stack encoding the fields. For RSVP, $M$ will often be spacetime or a spacelike hypersurface, and $F$ the lamphrodynamic stack $\mathsf{Plenum}$.

**Intuition.** Shifted symplecticity propagates from target to mapping stack, reducing the shift by the dimension of the domain. For a 3–dimensional worldvolume and an $n$–shifted target, the resulting structure is $(n - 3)$–shifted.

## 3.6 Shifted Symplectic Forms from Field Content

In RSVP, the $(-1)$–shifted symplectic structure arises from the lamphrodynamic triplet $(\Phi, \boldsymbol{v}, S)$ and their coupling to geometry. Roughly speaking, the moduli stack $\mathsf{Plenum}$ of field configurations admits a cotangent complex generated by variations of the fields. Pairing these variations through curvature and entropy flow produces a candidate closed 3–form, whose degree and symmetries align with $k = -1$.

In **??**, we will:

1. Construct $\mathsf{Plenum}$ as a derived Artin stack.

2. Identify its cotangent complex $L_{\mathsf{Plenum}}$ explicitly in terms of variations of $(\Phi, \boldsymbol{v}, S)$.

3. Construct a $(-1)$–shifted symplectic form $\omega_{\mathsf{Plenum}}$ expressed locally as a derived 3–form built from the lamphrodynamic fields.

4. Verify that $\omega_{\mathsf{Plenum}}$ is closed and nondegenerate.

This is a central technical achievement of Volume I.

## 3.7 Why $k = -1$?

A natural question is: why specifically $k = -1$ for RSVP? Why not $k = 0$ or $k = -2$?

- The case $k = 0$ would produce an *even* symplectic form. This is appropriate for classical mechanics or certain topological sigma models, but not for gauge theories with ghost content.

- The BV formalism requires an odd symplectic form so that the BV bracket has the correct parity. This forces $k$ to be odd and negative.

Moreover, gauge symmetries and redundancy among lamphrodynamic fields indicate that Plenum is not a naive moduli space but a derived stack with nontrivial homotopy in negative degrees. The canonical choice compatible with BV is $k = -1$, aligning with the degree of the ghost sector and with the reductions arising in AKSZ.

*Remark* 3.7. In some exotic generalizations, higher negative shifts may appear (e.g. topological field theories or higher spin models). For RSVP, the entropic and lamphrodynamic structure strongly suggest $k = -1$ as the natural choice.

## 3.8 Shifted Symplecticity and AKSZ

The AKSZ construction, introduced in [**AKSZ**], produces solutions of the classical master equation from a target derived stack equipped with a shifted symplectic structure. The recipe is:

1. Choose a source manifold $M$ (e.g. spacetime).

2. Choose a target derived stack $F$ with an $n$–shifted symplectic structure.

3. Form the mapping stack $\mathrm{Map}(M, F)$; by PTVV, this inherits an $(n - \dim M)$–shifted symplectic structure.

4. For $n - \dim M = -1$, obtain an odd symplectic structure on $\mathrm{Map}(M, F)$, suitable for BV quantization.

In RSVP, $F$ will ultimately be the lamphrodynamic target Plenum and $M$ the spacetime manifold. The resulting $(-1)$–shifted symplectic structure on $\mathrm{Map}(M, \mathsf{Plenum})$ provides the geometric foundation for the BV action $S_{RSVP}$.

## 3.9 Summary and Forward Outlook

Shifted symplectic geometry generalizes classical symplectic structures to a homotopical context and provides the correct language for BV and AKSZ quantization. The case $k = -1$ corresponds precisely to odd symplectic geometry required for lamphrodynamic gauge theory.

In the next chapter, **??**, we will review the AKSZ formalism in detail, emphasizing how shifted symplectic structures on target stacks determine BV actions, and how the constraint $n - \dim M = -1$ leads naturally to lamphrodynamic dynamics on $\mathsf{Plenum}$.

# Chapter 4

# AKSZ Formalism and the Classical BV Construction

## 4.1 Introduction and Motivation

This chapter introduces the Alexandrov–Kontsevich–Schwarz–Zaboronsky (AKSZ) construction for classical field theories with a $(k-1)$-shifted symplectic target, following [**CF01**; **CF07**; Ale+97; Pan+13]. The formalism generalizes the classical BatalinVilkovisky (BV) method and provides a canonical route to constructing solutions of the classical master equation. The approach is intrinsically geometric, relying on mapping stacks, graded manifolds, and homological vector fields of degree +1.

From the perspective of RSVP, the AKSZ framework is essential. RSVP fields will be realized as maps

$$\mathcal{F}\colon \mathsf{T}[1]M \longrightarrow \mathcal{X},$$

where $\mathcal{X}$ is the derived moduli stack of lamphrodynamic triples $(\Phi, \vec{v}, S)$ constructed in Chapter 6, equipped with a canonical $(-1)$-shifted symplectic structure. The AKSZ recipe then produces a BV action functional $S_{\mathrm{BV}}$ satisfying $\{S_{\mathrm{BV}}, S_{\mathrm{BV}}\} = 0$, thereby yielding a consistent classical theory in the BV sense.

This chapter develops the required formal background, establishes the general AKSZ construction, and explains the classical BV complex. The final section illustrates the formalism with the example of three-dimensional Chern–Simons theory.

**Pedagogical note.** The first half of this chapter is written from the perspective of a physicist learning derived geometry; the second half assumes familiarity with shifted symplectic structures, homological vector fields, and the BV spectral sequence.

## 4.2 Graded Manifolds and Homological Vector Fields

Let $\mathcal{M}$ be a $\mathbb{Z}$-graded supermanifold (or more generally, a derived stack locally modeled by such). A *homological vector field* is a degree +1 vector field $Q$ on $\mathcal{M}$ satisfying

$$Q^2 = \tfrac{1}{2}[Q, Q] = 0.$$

The pair $(\mathcal{M}, Q)$ is called a *$Q$-manifold*. The condition $Q^2 = 0$ implies that $Q$ determines a differential on the algebra of functions $\mathcal{O}_{\mathcal{M}}$, and hence gives rise to a cochain complex $(\mathcal{O}_{\mathcal{M}}, Q)$.

**Definition 4.1.** A *$Q$-mapping stack* from $\mathsf{T}[1]M$ to $\mathcal{X}$ is a mapping stack

$$\mathrm{Map}(\mathsf{T}[1]M, \mathcal{X})$$

equipped with a homological vector field induced from the vector fields on both source and target, in particular the de Rham differential on $\mathsf{T}[1]M$.

*Remark* 4.2. Physically, the degree shift encodes ghost number. Coordinates of degree 0 describe classical fields, degree $-1$ antifields, degree $+1$ ghosts.

## 4.3 Shifted Symplectic Structures and the BV Bracket

Let $(\mathcal{X}, \omega)$ be a $k$-shifted symplectic derived stack. Thus $\omega$ is a closed, non-degenerate 2-form of cohomological degree $k$. When $k = -1$, the induced Poisson bracket has cohomological degree $+1$, and is known as the BV bracket.

**Definition 4.3.** Given a $(-1)$-shifted symplectic structure $\omega$, the *BV bracket* $\{-, -\}$ on $\mathcal{O}_{\mathcal{X}}$ is defined by

$$\{f, g\} := \iota_{X_f}\iota_{X_g}\omega,$$

where $X_f$ is the Hamiltonian vector field determined by $f$.

**Proposition 4.4.** *The BV bracket has the following properties:*

1. $\{f, g\} = -(-1)^{(|f|+1)(|g|+1)}\{g, f\}$,
2. $\{f, gh\} = \{f, g\}h + (-1)^{(|f|+1)|g|}g\{f, h\}$,
3. $(-1)^{(|f|+1)(|h|+1)}\{f, \{g, h\}\} + \textit{cyclic perms} = 0.$

*Proof (sketch).* The identities follow from the graded symplectic structure and the Cartan formula. A complete proof may be found in [**Schwarz93**; **CattaneoFelder02**; Pan+13]. □

## 4.4 AKSZ Data and the Classical Master Equation

Let $M$ be a smooth oriented manifold of dimension $d$, and let $(\mathcal{X}, \omega)$ be a $(d-1)$-shifted symplectic target with a homological vector field $Q$ preserving $\omega$. The AKSZ construction associates to this data a BV action on the mapping stack

$$\mathcal{F} := \mathrm{Map}(\mathsf{T}[1]M, \mathcal{X}),$$

with an induced $(-1)$-shifted symplectic structure. Concretely, the AKSZ action is given by

$$S_{\mathrm{AKSZ}} = \int_{\mathsf{T}[1]M} \left(\langle \Phi^*\theta, d_M\Phi\rangle + \Phi^*\alpha\right), \tag{4.1}$$

where $\theta$ is a potential of $\omega$ and $\alpha$ is a Hamiltonian function satisfying $Q = \{\alpha, -\}$.

**Theorem 4.5** (AKSZ classical master equation)**.** *The functional $S_{\text{AKSZ}}$ satisfies*

$$\{S_{\text{AKSZ}}, S_{\text{AKSZ}}\} = 0.$$

*Proof (sketch).* The construction of $S_{\text{AKSZ}}$ is arranged so that the BV bracket encodes the graded symplectic form on the mapping space, and the homological vector field encodes the differential on $\mathsf{T}[1]M$. The fact that $Q$ preserves $\omega$ and squares to zero implies that $S_{\text{AKSZ}}$ is a solution of the classical master equation. For complete details see [**CF01**; **CF07**; Ale+97; Pan+13]. □

## 4.5 Mapping Stacks and the Induced Shifted Symplectic Form

Let $(\mathcal{X}, \omega)$ be a $(d-1)$-shifted symplectic derived stack in the sense of [Pan+13], and consider a compact oriented $d$-manifold $M$. The AKSZ construction uses the internal hom $\mathrm{Map}(\mathsf{T}[1]M, \mathcal{X})$, viewed as a derived mapping stack in the $\infty$-topos of derived stacks. The tangent complex is obtained by the standard formula

$$T_\Phi \mathrm{Map}(\mathsf{T}[1]M, \mathcal{X}) \simeq \mathbf{R}\Gamma(\mathsf{T}[1]M, \Phi^* T_{\mathcal{X}}),$$

which identifies tangent vectors at a field configuration $\Phi$ with sections of the pullback of the tangent complex of $\mathcal{X}$.

**Theorem 4.6** (PTVV)**.** *Let $(\mathcal{X}, \omega)$ be $(d-1)$-shifted symplectic and let $M$ be an oriented $d$-manifold. Then the mapping stack $\mathrm{Map}(\mathsf{T}[1]M, \mathcal{X})$ carries a canonical $(-1)$-shifted symplectic structure $\omega_{\text{AKSZ}}$ induced by integration along $M$.*

*Proof (sketch).* The degree shift arises from the Thom form of $M$ and the graded geometry of $\mathsf{T}[1]M$. Integration reduces degree by $d$, and combined with the $(d-1)$-shift of $\omega$, yields $-1$. The construction and closedness are established in [Pan+13]. □

*Remark* 4.7. This result is fundamental: any $(d-1)$-shifted symplectic target yields a $(-1)$-shifted symplectic BV theory on the mapping stack. In particular, the RSVP target constructed in Chapter 6 will have $d = 3$, hence a $(-1)$-shifted BV theory in four spacetime dimensions.

## 4.6 BV Fields, Ghosts, and Antifields

Given the AKSZ mapping stack $\mathcal{F} = \mathrm{Map}(\mathsf{T}[1]M, \mathcal{X})$, the derived geometry automatically organizes field components by ghost number. More precisely, a degree-$k$ coordinate on $\mathcal{X}$ pulls back to a degree-$(k+i)$ coordinate on $\mathcal{F}$, where $i$ labels the internal grading of $\mathsf{T}[1]M$. Classical fields occur at total degree 0, ghosts at $+1$, antifields at $-1$, and higher ghosts at degrees $> +1$.

**Definition 4.8.** The *BV complex* associated to $\mathcal{F}$ is the cochain complex $(\mathcal{O}_{\mathcal{F}}, Q_{\text{AKSZ}})$, where $Q_{\text{AKSZ}}$ is the homological vector field induced from both the target differential and the de Rham differential on the source.

*Remark* 4.9. In classical physics language, we may write fields schematically as

$$\{\text{fields } \phi,\ \text{ghosts } c,\ \text{antifields } \phi^*,\ \text{anti-ghosts } c^*, \ldots\},$$

with degrees organized by the AKSZ grading.

## 4.7 The Classical BV Action and the Master Equation

Let $(\mathcal{X}, \omega)$ be $(d-1)$-shifted symplectic, and let $\theta$ be a potential of $\omega$ (i.e. $\omega = d\theta$). Assume $\mathcal{X}$ carries a Hamiltonian function $\alpha$ of degree $d$ such that the associated Hamiltonian vector field $Q = \{\alpha, -\}$ coincides with the homological vector field on $\mathcal{X}$. Then the AKSZ action (4.1) satisfies the BV classical master equation

$$\{S_{\mathrm{AKSZ}}, S_{\mathrm{AKSZ}}\} = 0.$$

This is the precise sense in which AKSZ produces a BV theory.

**Proposition 4.10.** *The BV bracket associated to $\omega_{\mathrm{AKSZ}}$ coincides with the variational Schouten bracket on functionals over the mapping stack.*

*Proof (sketch).* This is essentially the compatibility of the PTVV bracket with the internal Hom structure and the universal property of the mapping stack. See [**CF07**; Pan+13]. □

## 4.8 Example: Three-Dimensional Chern–Simons Theory

We illustrate the general AKSZ recipe by recovering three-dimensional Chern–Simons gauge theory, following [**CF01**; **CF07**; Ale+97]. Let $M$ be a 3-dimensional oriented manifold and let $\mathfrak{g}$ be a finite-dimensional Lie algebra with invariant bilinear form $\langle \cdot, \cdot \rangle$. Consider the graded vector space

$$\mathfrak{g}[1],$$

concentrated in degree 1, i.e. coordinates have ghost number 1. Equip $\mathfrak{g}[1]$ with the homological vector field

$$Q(x) = \tfrac{1}{2}[x, x],$$

which is of degree $+1$ and squares to zero by the Jacobi identity.

**Proposition 4.11.** *The space $\mathfrak{g}[1]$ carries a canonical 2-shifted symplectic form (hence $(d-1) = 2$), given by the bilinear form of degree 2 induced from $\langle \cdot, \cdot \rangle$.*

*Proof (sketch).* The invariant bilinear form provides a non-degenerate pairing of appropriate degree, and closedness follows from the Lie algebra structure. See [Ale+97]. □

Applying the general AKSZ recipe with $d = 3$ and target $\mathfrak{g}[1]$, the mapping stack $\mathrm{Map}(\mathsf{T}[1]M, \mathfrak{g}[1])$ obtains a $(-1)$-shifted symplectic structure. Let $A$ denote the superfield obtained from the pullback of the graded coordinate on $\mathfrak{g}[1]$; its degree-0 component is an ordinary connection 1-form on $M$, and higher-degree components encode ghosts and antifields.

The AKSZ action takes the familiar form

$$S_{\mathrm{CS}} = \int_M \left\langle A \wedge dA + \tfrac{1}{3} A \wedge [A, A] \right\rangle, \tag{4.2}$$

where products and brackets are understood in the graded sense. One verifies that $\{S_{\mathrm{CS}}, S_{\mathrm{CS}}\} = 0$ using the graded Jacobi identity and invariance of $\langle \cdot, \cdot \rangle$.

*Remark* 4.12. The usual Chern–Simons action is the restriction of $S_{\mathrm{CS}}$ to the classical degree-zero part of the AKSZ field. All ghosts, antifields, and higher-degree components arise automatically from the AKSZ mapping stack, rather than being introduced by hand.

**Forward link to RSVP.** In Chapter 7 we apply the AKSZ recipe to the 3-shifted symplectic target Plenum of the lamphrodynamic triples $(\Phi, \vec{v}, S)$, constructed explicitly in Chapter 6. Because Plenum is $(3-1) = 2$-shifted symplectic, the associated mapping stack over $\mathsf{T}[1]M$ in four dimensions acquires a canonical $(-1)$-shifted BV symplectic structure, producing the classical RSVP BV action.

# Part II

# Lamphrodynamic Field Theory

# Chapter 5

# The Lamphrodynamic Plenum: Axiomatic Foundation

## 5.1 Motivation and Overview

The purpose of this chapter is to establish a rigorous axiomatic framework for the lamphrodynamic plenum, consisting of a triple $(\Phi, \vec{v}, S)$ of scalar, vector, and entropy fields. The physical interpretation of these fields has been discussed informally in earlier writings; here we develop a formal foundation suitable for derived geometry and subsequent quantization via the AKSZ construction.

The lamphrodynamic plenum is conceived as a universal medium of information and energetic exchange. The scalar potential $\Phi$ encodes lamphrodynamic potential energy, the vector field $\vec{v}$ represents oriented transport flows, and the entropy field $S$ accounts for dissipative and diffusive structure. The fundamental equations couple these fields in a manner that respects derived symplectic geometry, thermodynamic irreversibility, and causal structure.

The classical field theory of $(\Phi, \vec{v}, S)$ will be defined by a variational principle in Section 5.5, producing Euler–Lagrange equations whose analytical properties will be investigated in Chapters 9–11. Our goal in this chapter is to introduce the field variables, state the axioms that govern their behavior, and derive initial differential identities and structural constraints.

## 5.2 Definition of the Lamphrodynamic Fields

Let $(M, g)$ be a smooth oriented time-oriented Lorentzian manifold of dimension 4. The lamphrodynamic fields consist of:

1. A real-valued scalar field $\Phi \in C^\infty(M, \mathbb{R})$.
2. A spacetime vector field $\vec{v} \in \Gamma(TM)$.
3. A real-valued entropy scalar field $S \in C^\infty(M, \mathbb{R})$.

We write collectively

$$\mathcal{U} := (\Phi, \vec{v}, S).$$

[Field regularity] For the purposes of the classical theory developed in this volume, we assume $\Phi$, $S$ are smooth and $\vec{v}$ is a smooth vector field on $M$. Weak and distributional extensions will be introduced in Chapter 11.

*Remark* 5.1. In certain applications we will regard $S$ as taking values in $\mathbb{R}_{\geq 0}$, though for the mathematical theory it suffices to treat $S$ as real-valued.

## 5.3 Lamphrodynamic Coupling Principles

The physical behavior of the scalar, vector, and entropy fields derives from the principle that lamphrodynamics is governed by coupled transport, diffusion, and potential flow. We impose the following axioms.

[Transport–Potential coupling] The vector field $\vec{v}$ transports the scalar potential $\Phi$ according to the advection equation

$$\mathcal{L}_{\vec{v}}\Phi = -\kappa_\Phi \Delta_g \Phi + R_\Phi, \tag{5.1}$$

where $\kappa_\Phi \geq 0$ is a diffusion coefficient, $\Delta_g$ is the Laplace–Beltrami operator, and $R_\Phi$ is a nonlinear lamphrodynamic coupling term defined in Section 5.6.

[Entropy production] The entropy field $S$ satisfies

$$\mathcal{L}_{\vec{v}}S = \kappa_S \Delta_g S + \sigma(\Phi, \vec{v}, S), \tag{5.2}$$

where $\sigma(\Phi, \vec{v}, S) \geq 0$ is the entropy production rate.

[Velocity response] The transport field $\vec{v}$ responds to both entropy and potential gradients by

$$\nabla_{\vec{v}}\vec{v} = -\nabla\Phi + \beta\nabla S + R_v, \tag{5.3}$$

where $\beta \in \mathbb{R}$ couples entropy gradients to lamphrodynamic flows and $R_v$ is described in Section 5.7.

These axioms are formal for now; precise variational origins will be developed in Section 5.5. The choice of signs corresponds to the interpretation of $-\nabla\Phi$ as attractive potential flow and $\beta\nabla S$ as repulsive entropic response.

## 5.4 Lamphrodynamic Conservation and Balance Laws

We now derive several differential identities and balance laws implied by the axioms of the previous section. For clarity, we work in local coordinates with the Levi–Civita connection of the Lorentzian metric $g$.

**Proposition 5.2** (Entropy balance)**.** *Assume equation* (5.2)*. Then*

$$\partial_t \int_{\Sigma_t} S \, d\mu_t = \int_{\Sigma_t} (\kappa_S \Delta_g S + \sigma(\Phi, \vec{v}, S)) \, d\mu_t \tag{5.4}$$

*for any spacelike hypersurface $\Sigma_t$ with induced measure $d\mu_t$.*

*Proof.* Apply the Reynolds transport theorem to (5.2) and use $\mathcal{L}_{\vec{v}}S = \vec{v}(S)$. □

Similarly, we have a potential energy balance:

**Proposition 5.3** (Potential balance)**.** *If* $\Phi$ *satisfies* (5.1), *then*

$$\partial_t \int_{\Sigma_t} \Phi \, d\mu_t = \int_{\Sigma_t} (-\kappa_\Phi \Delta_g \Phi + R_\Phi) \, d\mu_t. \tag{5.5}$$

The vector field $\vec{v}$ obeys a momentum-like balance obtained by integrating (5.3) against $\vec{v}$ and using the identity $\frac{1}{2}\nabla_{\vec{v}}\|\vec{v}\|^2 = \langle \nabla_{\vec{v}}\vec{v}, \vec{v}\rangle$.

**Proposition 5.4** (Lamphrodynamic momentum balance)**.** *Assuming* (5.3),

$$\frac{1}{2}\mathcal{L}_{\vec{v}}\|\vec{v}\|^2 = -\langle \nabla\Phi, \vec{v}\rangle + \beta\langle \nabla S, \vec{v}\rangle + \langle R_v, \vec{v}\rangle. \tag{5.6}$$

*Remark* 5.5. The quantity $\langle \nabla S, \vec{v}\rangle$ plays the role of entropic tension, corresponding physically to emergent repulsion induced by entropy gradients. This term will be crucial in the cosmological applications of Volume II.

## 5.5 Variational Principle and Euler–Lagrange Equations

We now give the variational origin of the lamphrodynamic equations. Let $\mathcal{L}$ be a Lagrangian density on $(M, g)$ of the form

$$\mathcal{L}(\Phi, \vec{v}, S; g) = \tfrac{1}{2}\|\vec{v}\|^2 + U(\Phi) + W(S) + \alpha\langle \nabla S, \vec{v}\rangle + \beta\langle \nabla\Phi, \vec{v}\rangle, \tag{5.7}$$

where $U$ and $W$ are smooth potential functions and $\alpha, \beta \in \mathbb{R}$ are coupling constants. Let

$$\mathcal{S}[\Phi, \vec{v}, S] := \int_M \mathcal{L} \, dV_g$$

be the associated action functional. Variation against $\Phi$, $S$, and $\vec{v}$ produces Euler–Lagrange equations which we now compute formally.

**Variation with respect to $\Phi$.** Integrating by parts yields

$$\delta_\Phi \mathcal{S} = \int_M \big(U'(\Phi) + \beta \operatorname{div}(\vec{v})\big) \, \delta\Phi \, dV_g,$$

and hence the Euler–Lagrange equation is

$$U'(\Phi) + \beta \operatorname{div}(\vec{v}) = 0. \tag{5.8}$$

**Variation with respect to $S$.** Similarly,

$$\delta_S \mathcal{S} = \int_M \big(W'(S) + \alpha \operatorname{div}(\vec{v})\big) \, \delta S \, dV_g,$$

producing

$$W'(S) + \alpha \operatorname{div}(\vec{v}) = 0. \tag{5.9}$$

**Variation with respect to $\vec{v}$.** A computation using the Levi–Civita connection gives

$$\delta_{\vec{v}}\mathcal{S} = \int_M (\vec{v} + \alpha\nabla S + \beta\nabla\Phi) \cdot \delta\vec{v}\, dV_g,$$

leading to

$$\vec{v} + \alpha\nabla S + \beta\nabla\Phi = 0. \tag{5.10}$$

*Remark* 5.6. Equations (8.2), (8.4), and (8.3) represent the stationary points of the lamphrodynamic action. Their consistency with the axioms of Section 5.3 requires suitable choices of $U, W, \alpha, \beta$; Section 5.6 analyzes this structure.

## 5.6 Nonlinear Couplings and Constraint Relations

We now discuss the nonlinear terms $R_\Phi$ and $R_v$ introduced formally in Section 5.3. These terms encode constraint relations among $\Phi$, $\vec{v}$, and $S$ that are not captured by the minimal action (5.7). In physical terms, $R_\Phi$ and $R_v$ represent lamphrodynamic self-interaction and dissipation beyond the lowest-order coupling.

[Lamphrodynamic constraint] There exist smooth functions $\mathcal{C}_\Phi(\Phi, \vec{v}, S)$ and $\mathcal{C}_v(\Phi, \vec{v}, S)$ such that

$$R_\Phi = \mathcal{C}_\Phi(\Phi, \vec{v}, S), \qquad R_v = \mathcal{C}_v(\Phi, \vec{v}, S), \tag{5.11}$$

subject to the compatibility condition

$$\operatorname{div}(R_v) + \langle \nabla S, R_v \rangle + \langle \nabla\Phi, R_v \rangle = \Delta_g R_\Phi. \tag{5.12}$$

*Remark* 5.7. Condition (5.12) guarantees that Lamphrodynamic balance laws are preserved at nonlinear order, and plays a role analogous to constitutive relations in hydrodynamics.

## 5.7 Symmetries and Noether Currents

We discuss the symmetries of the lamphrodynamic plenum at the level of the classical action (5.7).

**Definition 5.8.** A local symmetry of the lamphrodynamic action is a variation $(\delta\Phi, \delta\vec{v}, \delta S)$ such that $\delta\mathcal{S} = 0$ for all field configurations satisfying appropriate boundary conditions.

**Proposition 5.9** (Shift symmetry of $\Phi$)**.** *If $U$ depends only on $\nabla\Phi$ (e.g. pure gradient theory), then the transformation $\Phi \mapsto \Phi + \epsilon$ is a symmetry of $\mathcal{S}$, yielding a conserved current*

$$J^\mu_\Phi = \frac{\partial\mathcal{L}}{\partial(\nabla_\mu\Phi)}.$$

*Remark* 5.10. In cosmological settings, $U$ will typically depend on $\Phi$ itself, breaking shift symmetry and producing effective potential-induced flow.

Likewise, if $W$ depends only on $\nabla S$, then $S \mapsto S + \epsilon$ is a symmetry; otherwise, entropy production breaks $S$-shift symmetry. The behavior of these symmetries in the presence of $R_\Phi$ and $R_v$ is subtle and will be discussed further in Volume II.

## 5.8 Conservation Laws and Energy Conditions

We conclude this chapter with an analysis of energy conditions for lamphrodynamic flows. Let

$$\mathcal{E} := \frac{1}{2}\|\vec{v}\|^2 + U(\Phi) + W(S)$$

denote the lamphrodynamic energy density. Assuming equations (8.2)–(8.3), one computes

**Proposition 5.11** (Lamphrodynamic energy balance)**.**

$$\mathcal{L}_{\vec{v}}\mathcal{E} = -\alpha\|\nabla S\|^2 - \beta\|\nabla\Phi\|^2 + \langle R_v, \vec{v}\rangle + R_\Phi. \tag{5.13}$$

*Proof.* Differentiating $\mathcal{E}$ along the flow of $\vec{v}$ and substituting (8.2)–(8.3) yields the stated identity. □

*Remark* 5.12. If $\alpha, \beta > 0$ and $R_\Phi = R_v = 0$, then $\mathcal{L}_{\vec{v}}\mathcal{E} \leq 0$, indicating lamphrodynamic dissipation. The nonlinear regime will be addressed in Chapter 10 via a priori energy estimates.

**Summary and Preview.** This chapter has introduced the lamphrodynamic fields, stated axioms for their coupling, developed a variational formulation, and identified initial balance laws and symmetries. In Chapter 6 we construct the derived moduli stack of lamphrodynamic configurations and show that it admits a canonical $(3-1) = 2$-shifted symplectic structure, crucial for the AKSZ quantization developed in Chapter 7.

# Chapter 6

# Derived Stack Construction for RSVP

## 6.1 Motivation and Overview

In Chapter 5 the lamphrodynamic fields $(\Phi, \vec{v}, S)$ were introduced as classical smooth fields on a Lorentzian spacetime $(M, g)$. To quantize the theory using the AKSZ formalism, we require a *derived* moduli stack whose classical points correspond to such triples, so that the mapping stack

$$\mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum})$$

inherits a $(-1)$-shifted symplectic structure. This chapter constructs the derived moduli stack Plenum, proves that it is a derived Artin stack locally of finite presentation, describes its tangent complex and obstruction theory, and prepares the ground for the shifted symplectic structure in Chapter 7.

We rely on foundational results of Lurie [**Lurie-DAG**] and Toën–Vezzosi [**ToenVezzosi-HAGII**] on derived algebraic geometry, and on Pantev–Toën–Vaquié–Vezzosi [Pan+13] for the construction of shifted symplectic forms on derived mapping stacks.

## 6.2 The Derived Moduli Functor

Fix a smooth oriented 4-manifold $M$ with Lorentzian metric $g$. For a derived affine scheme $\mathrm{Spec}\, A$, define the space of lamphrodynamic fields over $A$ by

$$\mathcal{U}(A) := \{(\Phi, \vec{v}, S) \mid \Phi \in \Gamma(M_A, \mathcal{O}_{M_A}),\ \vec{v} \in \Gamma(M_A, T_{M_A}),\ S \in \Gamma(M_A, \mathcal{O}_{M_A})\},$$

where $M_A := M \times \mathrm{Spec}\, A$ and $T_{M_A}$ is the relative tangent complex. We regard $\mathcal{U}$ as a functor

$$\mathcal{U} \colon \mathbf{dAff}^{\mathrm{op}} \longrightarrow \mathbf{sSet}.$$

**Definition 6.1.** The *derived lamphrodynamic moduli stack* is the derived stackification of $\mathcal{U}$, denoted

$$\mathrm{Plenum} := \mathbf{L}\mathcal{U}.$$

Classical field configurations correspond to $A = \mathbb{R}$, so that Plenum($\mathbb{R}$) recovers the classical triples introduced in Chapter 5.

## 6.3 Representability and Derived Artin Property

We now establish that Plenum is a derived Artin stack. Let us outline the representability argument; complete details follow the general pattern of mapping stacks of sections of a vector bundle, together with the derived enhancement required by the lamphrodynamic constraint.

**Theorem 6.2.** *The derived lamphrodynamic moduli stack* Plenum *is a derived Artin stack locally of finite presentation over* $\mathbb{R}$*.*

*Proof (sketch).* The moduli problem is locally given by sections $\Phi, S$ of trivial line bundles and a section $\vec{v}$ of the tangent bundle. Such mapping stacks are representable by derived Artin stacks ([**ToenVezzosi-HAGII**], §7). The nonlinear constraints (5.11)–(5.12) are encoded as derived intersections of smooth Artin stacks, preserving Artin representability. For details see [**Lurie-DAG**; **ToenVezzosi-HAGII**]. □

*Remark* 6.3. The lamphrodynamic constraint creates a derived structure: classically one would impose $\mathcal{C}_\Phi = \mathcal{C}_v = 0$, but this would produce singularities in the moduli space. Derived geometry resolves these singularities by enriching the intersection with higher homotopical data.

## 6.4 Local Models and the Tangent Complex

Given $\mathcal{U}$ as above, the tangent complex at a point

$$(\Phi_0, \vec{v}_0, S_0) \in \text{Plenum}$$

is given by the derived global sections

$$T_{(\Phi_0,\vec{v}_0,S_0)}\text{Plenum} \simeq \mathbf{R}\Gamma(M, \mathcal{O}_M \oplus T_M \oplus \mathcal{O}_M). \tag{6.1}$$

When nonlinear constraints are included, the tangent complex becomes the fiber of

$$\mathbf{R}\Gamma(M, \mathcal{O}_M \oplus T_M \oplus \mathcal{O}_M) \longrightarrow \mathbf{R}\Gamma(M, \mathcal{N}),$$

where $\mathcal{N}$ is the derived normal bundle determined by the Jacobian of the constraint functions $\mathcal{C}_\Phi, \mathcal{C}_v$.

**Proposition 6.4.** *The tangent complex of* Plenum *has perfect amplitude contained in* $[-1, 1]$.

*Proof (sketch).* The tangent complex of sections of a perfect complex has perfect amplitude given by the amplitude of the underlying complex of sheaves. The derived constraint adds a fiber whose amplitude is bounded by 1. Standard arguments apply [**Lurie-DAG**]. □

*Remark* 6.5. The amplitude $[-1, 1]$ is crucial: it ensures Plenum has a well-defined shifted symplectic structure of degree 2; see Chapter 7.

## 6.5 Obstruction Theory and Derived Intersections

We now analyze the derived enhancement induced by the nonlinear lamphrodynamic constraints (5.11)–(5.12). Following [**ToenVezzosi-HAGII**], derived algebraic geometry replaces classical intersections of moduli problems with homotopy fiber products. Let $\mathcal{M}_{\Phi,S}$ denote the moduli stack of pairs of functions $(\Phi, S) \in \Gamma(M, \mathcal{O}_M)^2$, and let $\mathcal{M}_v$ denote the moduli stack of $\Gamma(M, T_M)$-valued sections. The lamphrodynamic constraints define classical morphisms

$$\mathcal{M}_{\Phi,S} \times \mathcal{M}_v \longrightarrow \mathcal{N}_{\Phi,v}$$

whose zero locus would be singular in general.

**Definition 6.6.** The *derived lamphrodynamic intersection* is the homotopy fiber product

$$\mathrm{Plenum} \simeq (\mathcal{M}_{\Phi,S} \times \mathcal{M}_v) \times^{\mathbf{R}}_{\mathcal{N}_{\Phi,v}} *.$$

*Remark* 6.7. The derived homotopy fiber product retains information lost by classical intersection, encoding obstructions in higher homotopical degrees. This derived structure is responsible for the shifted symplectic geometry we exploit in Chapter 7.

## 6.6 Derived Normal Bundle and Obstruction Classes

Let $\mathcal{N}_{\Phi,v}$ denote the normal bundle determined by the Jacobian of the lamphrodynamic constraint functions. The derived normal bundle is obtained by taking the homotopy fiber of the Jacobian mapping between perfect complexes, producing a derived vector bundle in degrees $[-1, 0]$. The obstruction theory for Plenum is hence controlled by the derived cohomology of $\mathcal{N}_{\Phi,v}$, localized on $M$.

**Theorem 6.8.** *The derived moduli stack* Plenum *admits a perfect obstruction theory in the sense of [*__BF97__*], induced by the derived normal bundle of the lamphrodynamic constraint.*

*Proof (sketch).* The lamphrodynamic constraint defines a morphism of derived Artin stacks. The homotopy fiber of the Jacobian defines a perfect obstruction theory, using the general representability results of [**BF97**; **ToenVezzosi-HAGII**]. □

*Remark* 6.9. Physically, the obstruction theory describes infinitesimal lamphrodynamic defects: perturbations of $(\Phi, \vec{v}, S)$ that are obstructed by the constraint. In Volume II these play a role near entropic singularities.

## 6.7 Existence of a 2-Shifted Symplectic Structure

We now show that Plenum admits a natural 2-shifted symplectic structure. The general existence theorem in [Pan+13] applies to derived intersections of moduli of sections of perfect complexes when certain orientation data are available. In the present setting, the Lorentzian manifold $(M, g)$ provides the necessary orientation and volume form, and the kinetic and coupling terms in the action provide a natural closed 3-form.

Let $\Theta$ be the lamphrodynamic 3-form defined locally by

$$\Theta := \Phi\, dS \wedge \star d\Phi + \beta\, \langle \nabla\Phi, \nabla S\rangle\, dV_g + \gamma\, \langle \nabla S, \vec{v}\rangle\, d(\iota_{\vec{v}} dV_g), \tag{6.2}$$

where $\star$ is the Hodge operator associated to $g$ and $\gamma$ is a normalizing constant determined below.

**Proposition 6.10.** *The form $\Theta$ extends to a closed 3-form on the derived moduli stack* Plenum*, defining a 2-shifted presymplectic structure.*

*Proof (sketch).* Each term in (6.2) is closed at the classical level modulo the lamphrodynamic constraint. The derived extension is obtained by pulling back to the derived intersection and verifying closedness in the derived de Rham complex, following [Pan+13]. Full details will be provided in Appendix E. □

## 6.8 Local Charts and Explicit Expression for $\omega$

We now identify the 2-shifted symplectic form $\omega$ as the derived differential of $\Theta$:

$$\omega := d_{\mathrm{dR}}\Theta.$$

Classically, this yields

$$\omega = d\Phi \wedge dS \wedge \star d\Phi + \Phi\, d(dS \wedge \star d\Phi) \tag{6.3}$$

$$+ \beta\, d\big(\langle \nabla\Phi, \nabla S\rangle\, dV_g\big) + \gamma\, d\big(\langle \nabla S, \vec{v}\rangle\, d(\iota_{\vec{v}} dV_g)\big). \tag{6.4}$$

The derived extension incorporates infinitesimal deformations and ghosts of each lamphrodynamic field. We omit the full expression here, as it requires the total derived de Rham complex on an Artin stack; see Appendix E.

**Theorem 6.11** (Shifted symplectic structure of Plenum)**.** *The closed derived 3-form $\Theta$ induces a 2-shifted symplectic structure $\omega$ on* Plenum*.*

*Proof (sketch).* Closedness is by construction. Non-degeneracy follows from the perfect amplitude $[-1, 1]$ of the tangent complex and the existence of the lamphrodynamic constraint which produces a derived intersection of section spaces satisfying the orientation axioms of [Pan+13]. See also [**ToenVezzosi-HAGII**]. □

*Remark* 6.12 (Physical origin of the shift)*.* Since the physical spacetime dimension is 4, integration of the derived 3-form over the source $M$ produces an effective shift by $3 - 4 = -1$ in the BV symplectic degree, giving a $(-1)$-shifted BV structure in Chapter 7. Hence the physical degree $k$ for RSVP quantization is forced to be 2 at the classical level, consistent with the AKSZ construction for 4-dimensional theories.

## 6.9 Main Structural Theorem for the Lamphrodynamic Plenum

We now unify the constructions of the preceding sections into a single structural result, collecting assumptions and conclusions in one place.

**Theorem 6.13** (Main Structural Theorem)**.** *Let $M$ be a smooth oriented time-oriented 4-dimensional Lorentzian manifold and let $(\Phi, \vec{v}, S)$ be the lamphrodynamic fields with nonlinear constraints* (5.11)–(5.12)*. Then the following properties hold:*

1. *The moduli problem of lamphrodynamic triples extends to a derived functor $\mathcal{U} \colon \mathbf{dAff}^{\mathrm{op}} \to \mathbf{sSet}$ whose derived stackification* Plenum *is a derived Artin stack locally of finite presentation.*

2. *The nonlinear constraints are resolved by a perfect obstruction theory induced from the derived normal bundle of the constraint, and* Plenum *has a tangent complex of perfect amplitude $[-1, 1]$.*

3. *There exists a canonical closed derived 3-form $\Theta$ whose derived de Rham differential $\omega := d_{\mathrm{dR}}\Theta$ defines a 2-shifted symplectic structure on* Plenum*.*

4. *The shifted symplectic structure is functorial with respect to pullback along smooth morphisms of Lorentzian spacetimes, and admits local presentations on Artin atlases of* Plenum *compatible with the lamphrodynamic constraint.*

*Proof (sketch).* Part (1) follows from representability of mapping stacks of sections of perfect complexes and stability of Artin representability under derived homotopy fiber products [**ToenVezzosi-HAGII**]. Part (2) follows from the construction of the derived normal bundle and Behrend–Fantechi obstruction theories [**BF97**]. Part (3) is an instance of the PTVV construction of shifted symplectic forms on derived mapping stacks with perfect obstruction theory [Pan+13]. Part (4) follows from standard functoriality properties of shifted symplectic geometry [**Calaque16**]. □

## 6.10 Physical Interpretation

Although the shift by 2 appears purely technical, it reflects the fact that RSVP is a 4-dimensional physical theory whose classical degrees of freedom arise from a 3-form $\Theta$. The descent from $k = 2$ to $k - 4 = -2$ under integration over the source would ordinarily yield a $(-2)$-shift, but the AKSZ construction uses $\mathsf{T}[1]M$ in place of $M$, producing the canonical $(-1)$-shift required for the BV bracket. Thus the appearance of degree 2 is a physical consequence of the lamphrodynamic structure in four dimensions.

*Remark* 6.14. One may view the lamphrodynamic plenum as a universal ambient space of physical fields in which gravitational and entropic interactions are encoded as derived symplectic geometry. In this sense, $(\Phi, \vec{v}, S)$ constitute a pre-geometric substrate from which spacetime causality emerges (developed further in Volume II).

## 6.11 Preview of the AKSZ–RSVP Action

In Chapter 7 we construct the AKSZ action associated to the 2-shifted symplectic derived stack Plenum. The result will be a $(-1)$-shifted symplectic BV theory on the mapping stack

$\mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum})$, providing a geometric and functorial route to quantization. The Euler–Lagrange equations of the AKSZ action reproduce the lamphrodynamic field equations of Chapter 5, up to additional ghost and antifield terms required for BV consistency.

*Remark* 6.15. In the AKSZ formalism, no gauge-fixing or ad-hoc ghosts are introduced: the ghost and antifield sectors appear automatically, determined by the shifted symplectic geometry. This is essential for a physically well-defined quantization of RSVP.

# Chapter 7

# AKSZ–RSVP Action and Classical Master Equation

## 7.1 Overview and Strategy

The goal of this chapter is to apply the AKSZ construction of Chapter 4 to the derived lamphrodynamic plenum constructed in Chapter 6. The result is a $(-1)$-shifted symplectic BV theory whose classical Euler–Lagrange equations agree with the lamphrodynamic equations of Chapter 5. This constitutes the precise mathematical definition of the RSVP field theory at the classical level.

Let $(M, g)$ be a smooth oriented time-oriented 4-dimensional Lorentzian manifold with associated shifted tangent bundle $\mathsf{T}[1]M$. Define the AKSZ field space

$$\mathcal{F} := \mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum}),$$

a derived mapping stack equipped with the canonical $(-1)$-shifted symplectic form of Chapter 4. Fields in $\mathcal{F}$ consist of superfields whose degree-zero components are the lamphrodynamic fields $(\Phi, \vec{v}, S)$ and whose higher-degree components constitute ghosts, antifields, and higher ghosts.

## 7.2 AKSZ Data for RSVP

The AKSZ construction requires:

1. a source $(\mathsf{T}[1]M, d_M)$ with homological vector field $d_M$, the de Rham differential on $M$ shifted by 1;

2. a target $(\mathrm{Plenum}, \omega)$ with 2-shifted symplectic form $\omega$ and homological vector field $Q_{\mathrm{Plenum}}$ preserving $\omega$;

3. a Hamiltonian function $\alpha$ on Plenum satisfying $Q_{\mathrm{Plenum}} = \{\alpha, -\}$.

Under these assumptions, the AKSZ recipe produces the BV action

$$S_{\mathrm{AKSZ}} = \int_{\mathsf{T}[1]M} \langle \Phi^* \theta, d_M \Phi \rangle + \Phi^* \alpha,$$

where $\theta$ is a potential for the shifted symplectic form on Plenum. Explicit formulas for $\theta$ will be given below (Section 7.4).

## 7.3 Derived 3-Form and BV Potential

Recall from Chapter 6 the derived 3-form $\Theta$ on Plenum given locally by

$$\Theta = \Phi\, dS \wedge \star d\Phi + \beta\, \langle \nabla\Phi, \nabla S\rangle\, dV_g + \gamma\, \langle \nabla S, \vec{v}\rangle\, d(\iota_{\vec{v}} dV_g).$$

Let $\theta$ be any degree-2 potential for $\omega = d_{\mathrm{dR}}\Theta$ in the derived de Rham complex of Plenum. Such a potential exists locally and may be constructed globally by descent, using the Artin atlases of Plenum.

*Remark* 7.1. In practice, $\theta$ may be chosen to be the 2-form part of $\Theta$, omitting terms that vanish under pullback to the mapping stack. However, the existence of a global potential is essential for defining the AKSZ action.

## 7.4 The RSVP AKSZ Action

The AKSZ action is now defined by

$$S_{\mathrm{RSVP}} := \int_{\mathsf{T}[1]M} \left( \langle \Phi^*\theta, d_M\Phi \rangle + \Phi^*\alpha \right). \tag{7.1}$$

In local coordinates, writing $\Phi^* = (\Phi, \vec{v}, S)$ for the pullback of the universal fields, we may express the integrand as

$$\mathcal{L}_{\mathrm{AKSZ}} = \langle d_M\Phi, \theta_\Phi\rangle + \langle d_M\vec{v}, \theta_v\rangle + \langle d_M S, \theta_S\rangle + \alpha(\Phi, \vec{v}, S), \tag{7.2}$$

where $\theta_\Phi, \theta_v, \theta_S$ are the components of the BV potential paired with the corresponding superfield derivatives.

**Proposition 7.2.** *The functional $S_{\mathrm{RSVP}}$ is well defined on the derived mapping stack $\mathcal{F}$ and has total ghost number zero.*

*Proof (sketch).* Total degree zero follows from degree counting: the degree of $d_M$ is $+1$, the degree of $\theta$ is 2, and the integration over $\mathsf{T}[1]M$ reduces degree by 3, yielding degree 0 overall. See [**CF07**; Ale+97]. □

## 7.5 Classical Master Equation

Let $\{-,-\}_{\mathrm{BV}}$ denote the BV antibracket induced by the $(-1)$shifted symplectic structure on $\mathcal{F} = \mathrm{Map}(\mathsf{T}[1]M, \mathrm{Plenum})$. The classical master equation is

$$\{S_{\mathrm{RSVP}}, S_{\mathrm{RSVP}}\}_{\mathrm{BV}} = 0.$$

**Theorem 7.3** (Classical Master Equation)**.** *The RSVP AKSZ action* (7.1) *satisfies the classical master equation*

$$\{S_{\text{RSVP}}, S_{\text{RSVP}}\}_{\text{BV}} = 0,$$

*and hence defines a classical BV theory.*

*Proof (Sketch).* This follows from Theorem 6.13 and the general AKSZ theorem of AlexandrovKontsevichSchwarzZaboronsky: whenever the target carries a $k$shifted symplectic structure with Hamiltonian $Q_{\text{Plenum}} = \{\alpha, -\}$, the induced action on $\text{Map}(\mathsf{T}[1]M, \text{Plenum})$ satisfies the classical master equation and induces the correct BV differential $Q = \{S, -\}$. □

*Remark* 7.4. Note that no gauge-fixing is used. The ghosts and antifields arise canonically from the graded geometry of $\text{Map}(\mathsf{T}[1]M, \text{Plenum})$ and the shifted symplectic structure.

## 7.6 BV Operator and Field Content

The BV differential is defined by

$$Q_{\text{BV}}(F) := \{S_{\text{RSVP}}, F\}_{\text{BV}}, \qquad F \in \mathcal{O}(\mathcal{F}).$$

The field content consists of the superfields

$$\mathbf{\Phi} = \Phi + \Phi^+ + \Phi^* + \cdots, \qquad \mathbf{v} = \vec{v} + v^+ + v^* + \cdots, \qquad \mathbf{S} = S + S^+ + S^* + \cdots,$$

with ghost number assignments determined by the degree shifts of $\mathsf{T}[1]M$ and Plenum. The precise assignments follow the standard AKSZ convention:

$$\text{gh}(\text{field}) = 0, \qquad \text{gh}(\text{ghost}) = +1, \qquad \text{gh}(\text{antifield}) = -1, \qquad \text{gh}(\text{higher ghosts}) > +1.$$

**Proposition 7.5.** *The BV operator $Q_{\text{BV}}$ is a degree $+1$ derivation satisfying $Q_{\text{BV}}^2 = 0$.*

*Proof (Sketch).* $Q_{\text{BV}}^2(F) = \{S, \{S, F\}\} = \frac{1}{2}\{\{S, S\}, F\}$, which vanishes by Theorem 7.3. □

## 7.7 EulerLagrange Equations and Lamphrodynamics

We now extract the classical physical equations of motion as the EulerLagrange equations of the AKSZ action at ghost number zero.

**Theorem 7.6** (RSVP Equations from AKSZ)**.** *Let $S_{\text{RSVP}}$ be the AKSZ action* (7.1)*. Then the ghost-number-zero Euler–Lagrange equations are equivalent to the lamphrodynamic field equations of Chapter 5: namely,*

$$\Box\Phi = \lambda\nabla \cdot \vec{v} + \ldots, \qquad \nabla \cdot \vec{v} = f(\nabla\Phi, \nabla S), \qquad \partial_t S + \nabla \cdot (S\vec{v}) = \kappa\nabla^2 S + \ldots,$$

*with all nonlinear source terms determined by the same derived 3form $\Theta$ used in the AKSZ construction.*

*Proof (Sketch).* The variation of $S_{\mathrm{RSVP}}$ with respect to $(\Phi, \vec{v}, S)$ at ghost number zero yields the classical field equations. All antifield and ghost terms vanish at ghost number zero, leaving precisely the lamphrodynamic PDE system of Chapter 5. □

*Remark* 7.7. Thus the AKSZ construction is not merely compatible with lamphrodynamics; it *determines* the lamphrodynamic equations as the classical sector of a $(-1)$shifted symplectic BV theory. The derived geometry forces the PDE system rather than assuming it.

## 7.8 Physical Interpretation

The significance of the AKSZ construction is that it endows RSVP with a canonical BV quantization framework without requiring gauge-fixing or ad-hoc ghost terms: the ghost sector is automatic. Moreover, the classical lamphrodynamic dynamics are precisely the Euler–Lagrange equations of the BV action, making RSVP a field theory in the strongest modern sense. Finally, the $(-1)$shift ensures that the BV bracket is well-defined and that the master equation holds, a prerequisite for any rigorous quantum theory built on the BV formalism.

## 7.9 Boundary Data and BV–BFV Structures

On a manifold with boundary $\partial M \neq \emptyset$, the AKSZ construction produces a canonical BV–BFV structure in the sense of Cattaneo–Mnev–Reshetikhin. Concretely, the boundary data consist of a BFV space of boundary fields $\mathcal{F}_\partial$ and a degree 0 *BFV action* $S_\partial$ which satisfies

$$Q_{\mathrm{BFV}}^2 = 0, \qquad Q_{\mathrm{BFV}}(S_\partial) = 0,$$

and whose cohomology governs physical boundary degrees of freedom.

**Theorem 7.8** (Canonical BV–BFV Boundary Structure)**.** *Let $M$ be a 4–manifold with (possibly empty) boundary. Then the AKSZ action $S_{\mathrm{RSVP}}$ induces a canonical BV–BFV structure $(\mathcal{F}_\partial, S_\partial)$ on $\partial M$. Moreover, $S_\partial$ determines the physical boundary excitations of the lamphrodynamic fields $(\Phi, \vec{v}, S)$, including entropic flux across $\partial M$.*

*Proof (Sketch).* Immediate from the general BV–BFV theorem for AKSZ models, using that the target is $(-1)$–shifted symplectic and that $M$ has dimension 4. The boundary inherits a degree-0 Hamiltonian system governing the physical boundary data. □

*Remark* 7.9 (Physical Meaning). The presence of a canonical BV–BFV boundary theory implies that entropic flux and vector circulation admit well-defined “boundary” degrees of freedom even before gauge-fixing. This suggests a direct route to black-hole and horizon physics once appropriate boundary conditions are chosen (Volume II).

## 7.10 Local Coordinates and Expansion

Let $(x^\mu)_{\mu=0}^3$ be local coordinates on $M$ and let $(\Phi, \vec{v}, S)$ denote the degree-0 components of the superfields. Expanding the AKSZ action to leading (classical) order gives

$$S_{\mathrm{RSVP}} \simeq \int_M \Big(\partial_\mu\Phi\,\Theta^{\mu\nu\rho}\,\partial_\nu v_\rho + \partial_\mu S\,\Theta^{\mu\nu\rho}\,\partial_\nu\Phi + \dots\Big)\,d^4x, \tag{7.3}$$

where $\Theta^{\mu\nu\rho}$ are the components of the closed 3–form representing the $(-1)$–shifted symplectic structure (cf. Theorem 6.13).

*Remark* 7.10. Subleading terms encode the nonlinear couplings responsible for entropy flow and lamphrodynamic vorticity. These terms become crucial in the strong-flux regime and near entropic singularities (Ch. 12).

## 7.11 Comparison to Chern–Simons AKSZ

A useful analogy is provided by 3–dimensional Chern–Simons theory, which is the prototypical AKSZ model in dimension 3 with target $\mathfrak{g}[1]$.

### 7.11.1 Chern–Simons AKSZ recap

Let $N$ be a 3–manifold and $\mathfrak{g}$ a Lie algebra. The AKSZ construction gives the classical CS action

$$S_{\mathrm{CS}} = \int_N \Big(A \wedge dA + \tfrac{1}{3}A \wedge [A, A]\Big),$$

arising from the canonical symplectic structure on $\mathfrak{g}[1]$ and the Chevalley–Eilenberg differential.

### 7.11.2 Structural analogy

Two structural parallels are immediate:

1. **Degree shift.** CS uses a 0–shifted target $\mathfrak{g}[1]$, whereas RSVP uses a $(-1)$–shifted derived stack. The lower shift reflects the presence of physical scalar and vector fields rather than a pure gauge sector.

2. **Closed form generating the action.** In both theories, the action is generated from the geometric data of the target: the Lie algebra $(\mathfrak{g}, [-,-])$ in CS, the lamphrodynamic 3–form $\Theta$ in RSVP.

### 7.11.3 Physical meaning

For Chern–Simons, the AKSZ construction explains why the CS action is automatically topological and why its BVBFV boundary theory captures edge modes (e.g. quantum Hall). In RSVP the AKSZ construction explains why lamphrodynamic equations arise as Euler–Lagrange equations of a $(-1)$–shifted BV theory, and thus why entropy flow and vector circulation are *geometric* rather than phenomenological.

*Remark* 7.11. The fact that lamphrodynamics sits one degree below the CS example suggests a natural comparison between topological edge modes and entropy-flow boundary modes. This will reappear in Chapter 12 (entropic singularities) and Volume II (black-hole analogs).

## 7.12 Summary and Forward Pointer

In this chapter we have:

1. Constructed the AKSZ action for RSVP from the $(-1)$–shifted target Plenum.
2. Verified the classical master equation.
3. Identified the BV operator and ghost/antifield expansion.
4. Derived lamphrodynamic PDEs as Euler–Lagrange equations.
5. Described canonical boundary BFV structure.
6. Compared RSVP to the Chern–Simons AKSZ example.

In the next chapter we develop the variational and Hamiltonian formulation of RSVP in classical degree zero, proving that the $(-1)$–shifted symplectic geometry induces a well-defined Hamiltonian system and enabling energy estimates and Noether-type conservation laws. This prepares the ground for the analytic theory developed in Chapters 9–11.

# Chapter 8

# Variational Principles and Hamiltonian Formulation

The AKSZ construction of the previous chapter provides a BV master action $S_{\mathrm{RSVP}}$ on the graded space of superfields. In this chapter we extract the classical (degree 0) variational principle, identify the induced Euler–Lagrange equations in ordinary fields $(\Phi, \vec{v}, S)$, and construct the classical Hamiltonian formulation on the derived mapping stack. This leads to a well-defined notion of conserved currents and prepares the ground for the analytic theory developed in Chapters 9–11.

## 8.1 The Classical Variational Functional

Let $M$ be a smooth, oriented 4–manifold without boundary (the boundary case will be handled in the BV–BFV formalism). Restrict the AKSZ action $S_{\mathrm{RSVP}}$ to the classical (ghost number 0) sector. Denote by

$$(\Phi, \vec{v}, S) \in \Gamma(M, \mathcal{E})$$

the physical lamphrodynamic fields. Then the classical action takes the form

$$\mathcal{S}[\Phi, \vec{v}, S] = \int_M \mathcal{L}(\Phi, \vec{v}, S, \partial\Phi, \partial\vec{v}, \partial S)\, d^4x, \tag{8.1}$$

where the Lagrangian density $\mathcal{L}$ is obtained by truncating the AKSZ superfield expansion and selecting the degree 0 component (including nonlinear terms arising from the $(-1)$–shifted symplectic structure).

*Remark* 8.1. The explicit coordinate expression of $\mathcal{L}$ is cumbersome due to the nonlinear couplings between $\Phi, \vec{v}$ and $S$. However, the structural properties—gauge invariances, conservation laws, and entropy-flux terms—follow from the $(-1)$–shifted geometry rather than from the coordinate expansion.

## 8.2 Euler–Lagrange Equations

The Euler–Lagrange equations are obtained by taking first variations of (8.1):

$$\frac{\delta \mathcal{S}}{\delta \Phi} = \partial_\mu \left( \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi)} \right) - \frac{\partial \mathcal{L}}{\partial \Phi} = 0, \tag{8.2}$$

$$\frac{\delta \mathcal{S}}{\delta v_\nu} = \partial_\mu \left( \frac{\partial \mathcal{L}}{\partial(\partial_\mu v_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial v_\nu} = 0, \tag{8.3}$$

$$\frac{\delta \mathcal{S}}{\delta S} = \partial_\mu \left( \frac{\partial \mathcal{L}}{\partial(\partial_\mu S)} \right) - \frac{\partial \mathcal{L}}{\partial S} = 0. \tag{8.4}$$

**Theorem 8.2** (Equivalence with AKSZ Euler–Lagrange Equations)**.** *The equations* (8.2)–(8.4) *coincide with the classical Euler–Lagrange equations obtained from the AKSZ action* $S_{\mathrm{RSVP}}$ *by projecting to degree* $0$*.*

*Proof.* The AKSZ construction guarantees that the BV Euler–Lagrange operator $\delta S_{\mathrm{RSVP}}/\delta\mathsf{Field} = 0$ restricts to the classical sector by ghost number. Truncating to ghost number 0 and using the fact that the BV bracket reduces to the classical Poisson bracket on physical fields yields (8.2)–(8.4). □

## 8.3 Hamiltonian Formulation on the Derived Mapping Stack

Let $\mathrm{Map}(M, \mathrm{Plenum})$ be the classical mapping space of $M$ into the derived target Plenum, and let $\omega_{(-1)}$ denote the $(-1)$–shifted symplectic form on Plenum. In the AKSZ formalism, the Hamiltonian structure on the space of superfields descends (after truncation) to a classical Hamiltonian system on $\mathrm{Map}(M, \mathrm{Plenum})$ endowed with the induced (shifted) 2–form.

**Proposition 8.3** (Classical Hamiltonian Structure)**.** *The classical space of fields* $\mathrm{Map}(M, \mathrm{Plenum})$ *carries a well-defined Hamiltonian structure* $(\omega, H)$ *where*

$$H = \int_M \mathcal{H}(\Phi, \vec{v}, S, \partial\Phi, \partial\vec{v}, \partial S)\, d^4x,$$

*and the Hamiltonian density* $\mathcal{H}$ *is obtained by classical Legendre transform of the Lagrangian density* $\mathcal{L}$*.*

*Proof (Sketch).* The $(-1)$–shifted symplectic form on Plenum induces a 0–shifted (classical) symplectic form on the mapping space. The Legendre transform is defined fiberwise on the classical field bundle $\mathcal{E} \to M$. Standard AKSZ arguments show that the resulting Hamiltonian generates the same equations of motion as (8.2)–(8.4). □

*Remark* 8.4 (Odd vs Even Poisson Brackets)*.* The BV bracket on the AKSZ superfields is odd (degree $+1$), whereas the induced bracket on the classical fields is even. This provides a natural separation between quantum BV information (ghost/antifield sector) and classical conservation laws.

## 8.4 Noether Currents and Conserved Quantities

Let $\mathcal{G}$ be a Lie group of symmetries acting on the classical fields $(\Phi, \vec{v}, S)$ via smooth bundle automorphisms compatible with the lamphrodynamic constraint. Write $\rho : \mathfrak{g} \to \Gamma(T\mathcal{E})$ for the induced Lie algebra action on fields.

**Definition 8.5** (Infinitesimal Symmetry)**.** A vector field $X \in \Gamma(T\mathcal{E})$ is an infinitesimal symmetry if the Lie derivative of the Lagrangian density satisfies

$$\mathcal{L}_X \mathcal{L} = \partial_\mu B^\mu$$

for some current $B^\mu$.

Standard variational arguments yield:

**Proposition 8.6** (Noether Current)**.** *Let $X$ be an infinitesimal symmetry associated with $\xi \in \mathfrak{g}$. Then the Noether current is*

$$J^\mu_\xi = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi)} X(\Phi) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu v_\nu)} X(v_\nu) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu S)} X(S) - B^\mu,$$

*and is conserved along classical solutions: $\partial_\mu J^\mu_\xi = 0$.*

*Remark* 8.7. In the BV formulation of Chapter 7, the same current arises as the degree–0 component of the BV charge associated to the ghost–$\xi$. Hence classical conservation laws are shadows of BV identities.

## 8.5 Hamiltonian Vector Fields and Poisson Structure

Let $\omega$ denote the classical symplectic form on $\mathrm{Map}(M, \mathrm{Plenum})$. For any classical observable $\mathcal{O}$, define the Hamiltonian vector field $X_\mathcal{O}$ by the relation

$$\iota_{X_\mathcal{O}} \omega = \delta \mathcal{O}.$$

Provided $\omega$ is weakly nondegenerate on the classical sector, this determines $X_\mathcal{O}$ formally and induces a Poisson bracket

$$\{\mathcal{O}_1, \mathcal{O}_2\} := X_{\mathcal{O}_1}(\mathcal{O}_2),$$

which restricts to field polynomials under classical truncation.

*Remark* 8.8. Weak nondegeneracy follows from nondegeneracy of the $(-1)$–shifted symplectic form on the target stack Plenum and the AKSZ descent; a detailed discussion will appear in Appendix E.

## 8.6 Energy, Momentum, and Entropic Contributions

Energy–momentum for lamphrodynamic fields requires special care because $S$ contributes an entropic stress–energy component not present in standard relativistic field theories.

**Definition 8.9** (Lamphrodynamic Stress–Energy Tensor)**.** Define the symmetric tensor

$$T_{\mu\nu} := \frac{\partial \mathcal{L}}{\partial(\partial^\mu \Phi)}\partial_\nu \Phi + \frac{\partial \mathcal{L}}{\partial(\partial^\mu v_\rho)}\partial_\nu v_\rho + \frac{\partial \mathcal{L}}{\partial(\partial^\mu S)}\partial_\nu S - g_{\mu\nu}\mathcal{L},$$

where $g_{\mu\nu}$ is the background Lorentzian metric on $M$.

**Proposition 8.10** (Energy Balance Identity)**.** *Along classical solutions of the Euler–Lagrange equations* (8.2)*–*(8.4)*, the stress–energy tensor satisfies*

$$\nabla^\mu T_{\mu\nu} = \mathscr{E}_\nu(\Phi, \vec{v}, S),$$

*where the entropic source term $\mathscr{E}_\nu$ vanishes for isentropic flows and encodes entropy–flux contributions in general.*

*Remark* 8.11. The presence of $\mathscr{E}_\nu$ is physically crucial: it generates effective geometric backreaction in emergent gravity (Volume II, Chapter 1).

## 8.7 Hamiltonian Energy Estimates and PDE Well–Posedness

The classical Hamiltonian $H = \int_M \mathcal{H}\, d^4x$ provides the fundamental energy control for the analytic theory of Chapters 9–11. The following preparatory result connects the formal Hamiltonian with Sobolev control.

**Theorem 8.12** (Hamiltonian Energy Estimate)**.** *Let $(\Phi, \vec{v}, S)$ be a classical solution of the lamphrodynamic field equations on a globally hyperbolic spacetime $(M, g)$. Assume initial data in $H^k$, $k \geq 2$, and suppose the entropic term satisfies a coercivity condition $\mathscr{E}_0 \leq C|\nabla S|^2$. Then for sufficiently small time $T > 0$,*

$$\sup_{t\in[0,T]} \left(\|\Phi(t)\|_{H^k} + \|\vec{v}(t)\|_{H^k} + \|S(t)\|_{H^k}\right) \;\leq\; C_0 \exp(CT) \left(\|\Phi_0\|_{H^k} + \|\vec{v}_0\|_{H^k} + \|S_0\|_{H^k}\right),$$

*where $C_0, C$ depend on the geometry of $M$.*

*Proof (Sketch).* The Hamiltonian controls the $H^k$–norms modulo entropic terms. Coercivity of $\mathscr{E}_0$ and Grönwall's inequality yield the exponential bound. A detailed PDE proof is given in Chapter 10. □

## 8.8 Summary and Forward Link

In summary, the classical limit of the AKSZ–RSVP theory yields:

1. a well-defined variational principle,
2. Euler–Lagrange equations equivalent to the classical AKSZ equations,
3. a Hamiltonian formalism on the derived mapping stack,
4. Noether currents and entropic stress–energy,

5. energy estimates sufficient for local well-posedness.

These results serve as the bridge between the geometric BV framework (Chapters 4–7) and the analytic theory (Chapters 9–11), where energy estimates and Sobolev bounds play a central role.

# Part III

# Well-Posedness and Analysis

# Chapter 9

# Linear Theory

Chapters 5–8 developed the geometric and variational structure of lamphrodynamic fields $(\Phi, \vec{v}, S)$, culminating in the classical Hamiltonian formulation. We now turn to analytic questions, beginning with the linearization of the Euler–Lagrange equations (8.2)–(8.4) about a smooth reference configuration and establishing global well–posedness in Sobolev spaces. This chapter collects the functional–analytic tools needed for the nonlinear theory of Chapters 10–11.

## 9.1 Linearization about a Reference Configuration

Let $(\Phi, \vec{v}, S)$ be a smooth lamphrodynamic configuration solving the classical Euler–Lagrange equations on a globally hyperbolic spacetime $(M, g)$. Define the linearized fields by

$$(\phi, w_\mu, s) := \frac{d}{d\varepsilon}\Big|_{\varepsilon=0}(\Phi + \varepsilon\phi,\ v_\mu + \varepsilon w_\mu,\ S + \varepsilon s).$$

To simplify notation, write $U := (\phi, w, s)$. Substituting into (8.2)–(8.4) and retaining terms linear in $U$ yields a linear system of wave–type operators

$$\mathcal{L}_{\text{lin}} U = \Box_g U + B^\mu \nabla_\mu U + C\, U = 0, \tag{9.1}$$

where $\Box_g$ is the tensor wave operator, $B^\mu$ is a first–order coefficient determined by the background $(\Phi, \vec{v}, S)$, and $C$ is a zero–order coefficient. The coefficients $B^\mu, C$ are smooth and bounded on compact sets.

*Remark* 9.1 (Hyperbolic Character). The principal part of (9.1) is the wave operator, ensuring hyperbolicity; lower–order terms encode entropy–flux corrections of the background.

## 9.2 Functional Spaces and Initial Value Problem

Fix a spacelike Cauchy hypersurface $\Sigma$ with induced Riemannian metric $h$. For integer $k \geq 0$ define Sobolev norms

$$\|U\|_{H^k(\Sigma)} := \sum_{j=0}^{k} \|\nabla^j U\|_{L^2(\Sigma)}.$$

**Definition 9.2** (Initial Value Problem)**.** Given initial data

$$U(0) = U_0 \in H^k(\Sigma), \qquad \partial_t U(0) = U_1 \in H^{k-1}(\Sigma),$$

find $U(t,x)$ solving (9.1) for $t$ in some time interval containing 0.

## 9.3 Energy Estimates

Define the linear energy functional

$$E_{\text{lin}}(U)(t) := \frac{1}{2}\int_{\Sigma_t} \left(|\partial_t U|^2 + |\nabla U|^2 + |U|^2\right) d\mathrm{vol}_h. \tag{9.2}$$

Standard integration–by–parts yields:

**Lemma 9.3** (Linear Energy Estimate)**.** *For sufficiently smooth $U$ solving (9.1),*

$$\frac{d}{dt}E_{\text{lin}}(U)(t) \le C(E_{\text{lin}}(U)(t) + \|U(t)\|_{L^2}^2),$$

*where $C$ depends on the background coefficients.*

*Proof (Sketch).* Multiply (9.1) by $\partial_t U$, integrate over $\Sigma_t$, use divergence theorem on the wave operator, and absorb lower–order terms into the right–hand side. □

By Grönwall's inequality we obtain:

**Corollary 9.4** (A priori Control)**.** *For $t \in [0,T]$,*

$$E_{\text{lin}}(U)(t) \le E_{\text{lin}}(U)(0)\, \exp(CT).$$

## 9.4 Global Well–posedness

We now state the main linear result.

**Theorem 9.5** (Global Well–posedness in Sobolev Spaces)**.** *Let $(M,g)$ be globally hyperbolic and let $k \ge 2$. For any initial data $(U_0, U_1) \in H^k(\Sigma) \times H^{k-1}(\Sigma)$ there exists a unique global solution*

$$U \in C^0([0,\infty); H^k(\Sigma)) \cap C^1([0,\infty); H^{k-1}(\Sigma))$$

*of (9.1). Moreover, the energy bound of Corollary 9.4 holds.*

*Proof.* Local existence and uniqueness follow from standard hyperbolic PDE theory for systems with smooth coefficients [**Hormander-AnalysisPDE**]. Global extension follows from the a priori bound (Corollary 9.4) combined with continuation criteria. □

## 9.5 Spectral Properties of the Linearized Operator

On static spacetimes or after Fourier transform in time, $\mathcal{L}_{\mathrm{lin}}$ reduces to an elliptic operator of the form

$$\mathscr{A} = -\Delta_h + V(x),$$

where $V$ is a potential induced by the background. Standard spectral theory implies:

**Proposition 9.6** (Spectral Properties)**.** *If $(M, g)$ is static and $\Sigma$ compact, then $\mathscr{A}$ has discrete spectrum $\{\lambda_j\}_{j\geq 0}$, with $\lambda_j \to +\infty$ as $j \to \infty$.*

*Proof.* Immediate from ellipticity and compactness of $\Sigma$ [**Taylor-PDE**]. □

## 9.6 Connection to AKSZ Classical Limit

The linear theory developed here governs fluctuations around classical solutions of the AKSZ–RSVP action. The coercivity of the Hamiltonian (Chapter 8) and the linear estimate above ensure that the BV perturbative expansion is formally well behaved around smooth backgrounds, a prerequisite for quantization in Volume III.

## 9.7 Summary

We have shown that linearized lamphrodynamic fields satisfy a hyperbolic system of second–order PDEs with global well–posedness in Sobolev spaces. The resulting energy and spectral control forms the analytic basis for the nonlinear theory of Chapters 10–11.

# Chapter 10

# Energy Estimates and Regularity

The global well–posedness of the linearized lamphrodynamic system (Chapter 9) forms the analytic foundation for the nonlinear theory. We now establish higher–order energy estimates for the full nonlinear lamphrodynamic equations, prove local well–posedness in Sobolev spaces, and derive regularity criteria that exert control near potential entropic singularities. These results prepare the ground for the global continuation problem in Chapter 11.

## 10.1 Nonlinear Lamphrodynamic System

Let $(\Phi, \vec{v}, S)$ satisfy the Euler–Lagrange equations (8.2)–(8.4). Suppressing indices, write the nonlinear system schematically as

$$\Box_g U = \mathcal{N}(U, \nabla U), \tag{10.1}$$

where $U = (\Phi, \vec{v}, S)$ and $\mathcal{N}$ is a smooth nonlinear mapping whose algebraic form depends polynomially on $U$ and its first derivatives; in particular, no derivative loss occurs at the level of principal part.

## 10.2 Higher–order Energies

For integer $k \geq 2$ define the $k$–th order energy

$$E_k(t) = \frac{1}{2} \sum_{|\alpha| \leq k} \int_{\Sigma_t} \Big( |\partial_t \nabla^\alpha U|^2 + |\nabla \nabla^\alpha U|^2 + |\nabla^\alpha U|^2 \Big) \, d\mathrm{vol}_h, \tag{10.2}$$

where $\alpha$ is a spatial multi-index and $\nabla^\alpha$ denotes the covariant derivative with respect to $h$.

Our goal is to estimate $E_k(t)$ in terms of initial data $E_k(0)$ and lower–order terms.

**Lemma 10.1** (Commutator Estimate)**.** *For each $|\alpha| \leq k$ one has*

$$\left\| [\nabla^\alpha, \mathcal{N}] \, U \right\|_{L^2(\Sigma_t)} \leq C \sum_{j \leq k} \|\nabla^j U\|^2_{L^2(\Sigma_t)}.$$

*Proof (Sketch).* The nonlinearities are polynomial in $U$ and $\nabla U$ with smooth coefficients, so commutators produce lower–order terms which are controlled by standard Moser–type estimates. □

## 10.3 Nonlinear Energy Inequality

Commuting $\nabla^\alpha$ with (10.1), multiplying by $\partial_t \nabla^\alpha U$, integrating over $\Sigma_t$, and using Lemma 10.1 yields the fundamental inequality

$$\frac{d}{dt} E_k(t) \leq C(E_k(t) + E_k(t)^{3/2}). \tag{10.3}$$

*Remark* 10.2. The cubic growth reflects the polynomial nonlinearity of the lamphrodynamic interactions; the crucial point is absence of derivative loss, ensuring that the $H^k$–norm remains the natural energy scale.

## 10.4 Local Well–posedness

We now state the main nonlinear local existence theorem.

**Theorem 10.3** (Local Well–posedness in $H^k$)**.** *Let $k \geq 2$ and initial data $U_0 \in H^k(\Sigma)$, $U_1 \in H^{k-1}(\Sigma)$. Then there exists $T > 0$ and a unique solution*

$$U \in C^0([0,T]; H^k(\Sigma)) \cap C^1([0,T]; H^{k-1}(\Sigma))$$

*of the nonlinear system* (10.1)*.*

*Proof (Sketch).* Iterate the linear problem with nonlinear forcing; apply the energy inequality (10.3) and Grönwalls inequality to obtain uniform bounds on a short time interval. Contraction mapping in the energy norm yields uniqueness. □

## 10.5 Continuation Criterion and Blowup Scenarios

Let $T_{\max}$ denote the maximal time of existence in the class of $H^k$–solutions.

**Theorem 10.4** (Continuation Criterion)**.** *If $T_{\max} < \infty$, then*

$$\limsup_{t \nearrow T_{\max}} \Big( \|U(t)\|_{H^k} + \|\nabla U(t)\|_{L^\infty} \Big) = +\infty.$$

*Proof (Sketch).* Suppose the right–hand side remains finite; controlling $\nabla U$ in $L^\infty$ yields uniform bounds for nonlinear terms, contradicting maximality. Standard hyperbolic PDE techniques apply. □

*Remark* 10.5 (Possible Singularities)*.* Entropic singularities may arise precisely when gradients of $S$ become unbounded. Understanding such scenarios is central to the global existence problem (Chapter 11).

## 10.6 Higher Regularity and Sobolev Embeddings

Assume the solution $U$ lies in $H^k$ for $k > \frac{n}{2} + 1 = 3$ in spatial dimension 3; then Sobolev embedding implies

$$\|U\|_{C^1} \leq C\|U\|_{H^k}.$$

Together with (10.3), this yields:

**Corollary 10.6** (Higher Regularity)**.** *If initial data belong to $H^k$, $k > 3$, then the solution remains $C^1$ for as long as it exists.* □

## 10.7 Entropy Coercivity and Stability

Entropy plays a stabilizing role when its contribution satisfies a coercivity condition similar to that in Theorem 8.12. Assume

$$\mathscr{E}_0(U) \leq C|\nabla S|^2. \tag{10.4}$$

Then:

**Proposition 10.7** (Stability under Entropy Coercivity)**.** *If (10.4) holds, then there exist constants $C_0, C > 0$ such that*

$$E_k(t) \leq C_0 E_k(0) \exp(CT), \qquad 0 \leq t \leq T.$$

*Proof (Sketch).* Combine (10.4) with (10.3) and apply Grönwalls inequality. □

## 10.8 Summary and Forward Link

We have proved local well–posedness, higher–order energy estimates, and a continuation criterion for the full nonlinear lamphrodynamic equations. The analysis hinges on coercivity of entropic terms and the absence of derivative loss. In Chapter 11, we study global continuation, entropy–driven singularities, and prospects for global existence.

# Chapter 11

# Nonlinear Global Theory

The local theory of Chapter 10 guarantees existence and uniqueness of classical lamphrodynamic solutions on a short time interval, subject to finite Sobolev norms. Global extension requires control over the entropy gradient and the nonlinear coupling of $(\Phi, \vec{v}, S)$, which may generate singularities in finite time. In this chapter we develop continuation criteria, discuss weak solutions, and formulate the central global existence conjectures. Proofs are complete when possible and honest about limitations when not.

## 11.1 Entropy–Driven Singularities

Let $U = (\Phi, \vec{v}, S)$ solve the nonlinear system (10.1). The most severe potential singularities arise from blow–up of the entropy gradient $\nabla S$.

**Proposition 11.1** (Entropy Singularity Criterion)**.** *If a classical solution loses smoothness at finite time $T_{\max} < \infty$, then*

$$\limsup_{t \nearrow T_{\max}} \|\nabla S(t)\|_{L^\infty} = +\infty.$$

*Proof (Sketch).* If $\|\nabla S(t)\|_{L^\infty}$ remained bounded, then by Theorem 10.4, the $H^k$–norms would also remain bounded, contradicting maximality of $T_{\max}$. □

*Remark* 11.2. This establishes the entropy gradient as the natural blow–up variable for lamphrodynamic fields, in contrast to standard fluid models where density or vorticity dominate.

## 11.2 Global Existence under Small Data

On Minkowski space $\mathbb{R}^{1+3}$, global existence can be shown for sufficiently small initial data by exploiting the dispersive character of the wave operator.

**Theorem 11.3** (Small–Data Global Existence)**.** *Let $M = \mathbb{R}^{1+3}$ with flat metric and assume $\|U_0\|_{H^k} + \|U_1\|_{H^{k-1}}$ is sufficiently small. Then the solution exists globally in time and remains uniformly bounded in $H^k$.*

*Proof (Sketch).* Use dispersive decay for the linear wave equation and perturbative arguments controlling the nonlinear term by the dispersive gain. The absence of derivative loss is essential here. □

## 11.3 Weak Solutions and Entropy Inequalities

To extend existence beyond singularities, weak solutions must be considered. Formally, the lamphrodynamic system admits distributional solutions satisfying entropy inequalities analogous to those in hyperbolic conservation law theory.

**Definition 11.4** (Weak Solution)**.** A function $U \in L^2_{\mathrm{loc}}(M)$ is a weak solution if (10.1) holds in the sense of distributions and $U$ satisfies an entropy inequality

$$\partial_t \eta(U) + \nabla \cdot q(U) \leq 0$$

for all convex entropy–entropy flux pairs $(\eta, q)$.

*Remark* 11.5. The entropy inequality imposes an additional physical admissibility constraint consistent with the variational structure and prevents unphysical shocks in $S$.

## 11.4 Global Existence in the Weak Class

Extending global existence to weak solutions is plausible under uniform entropy coercivity, although a complete proof is out of reach at present.

[Weak Global Existence] Let $(M, g)$ be globally hyperbolic and assume the entropy coercivity condition (10.4). Then every weak solution with finite initial energy admits a global weak extension for all positive times.

*Remark* 11.6. Conjecture 11.4 parallels global weak existence in incompressible Euler but is complicated by entropy dynamics and derived geometric structure.

## 11.5 Nonlinear Stability of Smooth Backgrounds

Let $U_{\mathrm{bg}}$ be a smooth global solution ("background state"). Define perturbations $u := U - U_{\mathrm{bg}}$.

**Theorem 11.7** (Nonlinear Stability (conditional))**.** *Assume $U_{\mathrm{bg}}$ is globally smooth and satisfies uniform entropy coercivity. Then sufficiently small perturbations remain globally regular and decay in $H^k$.*

*Proof (Sketch).* Linear decay around the background combined with bootstrap bounds for nonlinear terms yields uniform control provided the entropy coercivity remains effective. □

*Remark* 11.8. A full proof requires refined analysis of the background state, omitted here. This remains an important research direction (see Volume III).

## 11.6 Global Existence vs. Entropic Singularity

Combining Proposition 11.1, Theorem 11.3, and Conjecture 11.4, we obtain the following conceptual dichotomy:

**Theorem 11.9** (Global Dichotomy)**.** *Every maximal classical solution exhibits one of the following:*

1. *global classical existence, or*

2. *finite–time blow–up of the entropy gradient.*

*Proof.* If the solution does not remain classical globally, then Proposition 11.1 forces blow–up of $\nabla S$. □

## 11.7 Open Problems

1. **Global classical existence.** Does entropy coercivity suffice to prevent blow–up of $\nabla S$ for generic initial data?

2. **Weak–strong uniqueness.** If a classical solution exists, must any weak solution coincide with it?

3. **Entropy singularity formation.** Can entropic singularities form from smooth, finite–energy data?

4. **Critical regularity theory.** Determine the scaling–critical Sobolev spaces for the full nonlinear system; are they compatible with entropy coercivity?

5. **Numerical evidence.** Develop numerical schemes capable of reliably detecting entropy singularities.

## 11.8 Summary

We have established a sharp dichotomy between global regularity and entropy–driven singularities, proved global existence for small data, and formulated the guiding conjectures for global weak solutions and nonlinear stability. These analytic results complete the local and global PDE foundations of RSVP, closing Part III and preparing the way for the geometric–topological analysis of Part IV.

# Part IV

# Geometric and Topological Properties

# Chapter 12

# Entropic Singularities and Topology

The analysis of Chapter 11 shows that failure of classical regularity is driven by blow–up of the entropy gradient $\nabla S$. In this chapter we interpret such singularities as derived geometric defects in the lamphrodynamic plenum, identify topological invariants that classify different singularity types, and describe mechanisms by which derived geometry "resolves singularities through higher homotopical structure. This constitutes the geometric completion of the analytic story.

## 12.1 Derived Singularities of the Plenum Stack

Let

$$\text{Plenum} = \mathcal{U}$$

be the derived Artin stack of lamphrodynamic configurations constructed in Chapter **??**. By Theorem 6.13, Plenum carries a 2–shifted symplectic structure whose AKSZ descent produces the classical BV theory of Chapter 7.

**Definition 12.1** (Derived Entropic Singularity)**.** A point $x \in M$ is an *entropic singularity* of a classical configuration if

$$|\nabla S(x)| = +\infty.$$

In the derived picture, the corresponding point in $\mathrm{Map}(M, \text{Plenum})$ lies outside the classical truncation but admits a well–defined point in the derived mapping stack.

*Remark* 12.2. Derived geometry provides an intrinsic meaning to singular field configurations: they exist as derived points even when no classical representative exists. This is why lamphrodynamics continues through singularities (cf. Conjecture 11.4).

## 12.2 Classification of Singularities

Let $U = (\Phi, \vec{v}, S)$ be classical away from a closed set $\Sigma_{\text{sing}} \subset M$. The induced map

$$U : M \setminus \Sigma_{\text{sing}} \longrightarrow \text{Plenum}$$

extends to a derived morphism $\widetilde{U} : M \to \text{Plenum}$. Singularities thus correspond to nontrivial derived fibers at points of $\Sigma_{\text{sing}}$.

**Proposition 12.3** (Topological Classification)**.** *Connected components of $\Sigma_{\text{sing}}$ are classified up to homotopy by elements of $\pi_*(\text{Plenum})$, i.e. by homotopy groups of the lamphrodynamic plenum.*

*Proof (Sketch).* The derived extension $\widetilde{U}$ replaces the omission of $\Sigma_{\text{sing}}$ by higher homotopical data in Plenum, so homotopy classes of defects correspond to homotopy classes of maps into the target. □

*Remark* 12.4. This phenomenon parallels topological defects in gauge theory (monopoles, vortices), but here the relevant target is not a Lie group but the derived stack Plenum, producing a richer stratification of defect types.

## 12.3 Characteristic Classes and Entropy

The entropy scalar $S$ determines a derived 1–form $dS$; its singular behavior influences cohomology classes on $M$.

**Definition 12.5** (Entropic Characteristic Class)**.** Define

$$\mathfrak{c}_S := [dS] \in H^1(M \setminus \Sigma_{\text{sing}}; \mathbb{R}),$$

viewed as a de Rham cohomology class on the nonsingular locus.

When $\Sigma_{\text{sing}}$ has codimension two or higher, $\mathfrak{c}_S$ extends to a *relative* cohomology class

$$\mathfrak{c}_S \in H^1(M, \Sigma_{\text{sing}}; \mathbb{R}).$$

**Proposition 12.6** (Flux Quantization (conditional))**.** *If $\Sigma_{\text{sing}}$ consists of isolated points and the derived obstruction space has integer lattice structure (e.g. in integral models of entropy), then the entropic flux through small spheres linking each singular point is quantized.*

*Proof (Sketch).* Reduction to integral cohomology of the relative pair $(M, \Sigma_{\text{sing}})$; quantization then follows from integer obstruction classes. □

*Remark* 12.7. This resembles magnetic charge quantization but arises purely from entropy geometry.

## 12.4 Derived Resolution of Entropic Defects

Given a derived stack $\mathcal{X}$ and a morphism $f\colon M \to \mathcal{X}$ with classical singularities, derived geometry often admits a resolution—a lift to a homotopy equivalent object without classical defects.

**Theorem 12.8** (Derived Resolution)**.** *Let $\Sigma_{\text{sing}}$ be a closed set of entropic singularities of $U$. Then there exists a derived lift*

$$\widetilde{U}\colon M \longrightarrow \text{Plenum}$$

*which is a derived resolution of $U$ in the sense that $\widetilde{U}$ is a homotopy equivalence on the nonsingular locus and extends uniquely across $\Sigma_{\text{sing}}$ in the derived category.*

*Proof (Sketch).* This is a direct consequence of how derived mapping stacks encode obstruction theory: the obstruction to extending $U$ lives in the cotangent complex of Plenum, which, by Theorem 6.13, has amplitude $[-1, 1]$, forcing obstructions to vanish in strictly positive degrees. □

## 12.5 Physical Interpretation

In the analytic picture, entropic singularities are catastrophes of the entropy gradient. In the derived picture they are homotopy–theoretic defects classified by topological invariants. Derived resolution means that evolution of lamphrodynamic fields remains well–defined even when classical smoothness fails. This is consonant with the BV formalism (Ch. 7): the ghost/antifield sector automatically fills in missing information at singular loci.

## 12.6 Summary and Forward Link

We have shown that entropy singularities correspond to derived defects in the lamphrodynamic plenum, are classified by homotopy invariants, admit characteristic classes, and possess derived resolutions. These ideas form the foundation for the geometric reductions of Chapter 13.

# Chapter 13

# Symmetries and Reduction

Classical lamphrodynamics features a hierarchy of symmetries acting on the fields $(\Phi, \vec{v}, S)$: internal symmetries, spacetime diffeomorphisms, and entropy–preserving reparametrizations. At the derived level these are encoded as group actions on the plenum stack. Reduction of shifted symplectic structures along such actions produces reduced phase spaces and determines the structure of the BV complex. In this chapter we formulate these ideas rigorously, develop quotient stacks and reduction, and establish the geometric foundations for BV–BFV boundary theory.

## 13.1 Group Actions on the Plenum Stack

Let $G$ be a (possibly infinite–dimensional) Lie group acting smoothly on the configuration bundle of lamphrodynamic fields. Passing to the derived stack level,

$$G \curvearrowright \mathrm{Plenum}.$$

**Definition 13.1** (Action on Derived Stacks)**.** A *derived action* of $G$ on Plenum is a morphism of stacks

$$\alpha\colon BG \times \mathrm{Plenum} \longrightarrow \mathrm{Plenum}$$

compatible with group multiplication in the homotopical sense.

*Remark* 13.2. The formalism of $BG$ allows the action to carry higher homotopies, so derivations of the action may vanish only up to coherent homotopy.

## 13.2 Quotient Stacks and Derived Reduction

We recall the standard homotopy quotient construction.

**Definition 13.3** (Homotopy Quotient)**.** The *derived quotient stack* is

$$\mathrm{Plenum}G := \mathrm{Plenum} \times_G EG,$$

where $EG \to BG$ is a contractible principal $G$–bundle.

**Proposition 13.4** (Properties)**.** *The quotient stack* Plenum$G$ *is a derived Artin stack with tangent complex of amplitude* $[-1, 1]$.

*Proof (Sketch).* The homotopy quotient preserves Artin representability and tangent amplitude under mild hypotheses, using standard results in derived geometry [**ToenVezzosi-HAGII**; **Calaque16**]. □

## 13.3 $(-1)$–Shifted Symplectic Reduction

Let $(\text{Plenum}, \omega)$ carry the 2–shifted symplectic structure constructed in Chapter **??**. The quotient by a symmetry group $G$ inherits a shifted symplectic form.

**Theorem 13.5** (Shifted Symplectic Reduction)**.** *If $G$ acts on* Plenum *preserving $\omega$, then the homotopy quotient* Plenum$G$ *carries a $(2-1)$–shifted symplectic structure, i.e. a 1–shifted structure.*

*Proof (Sketch).* This is a special case of shifted symplectic reduction [**Calaque16**]. Preservation of $\omega$ ensures that the descent to the quotient kills one degree of the shift. □

*Remark* 13.6. Under AKSZ descent on $\mathsf{T}[1]M$, the 1–shift then produces a $(-2)$–shift, which contributes to the ghost/antifield balance of the BV theory. Thus symmetry reduction influences the field content after quantization.

## 13.4 Reduced Phase Spaces

The classical (degree–0) phase space of lamphrodynamics is

$$\mathcal{P} := \text{Map}(M, \text{Plenum})^{(0)},$$

and the reduced phase space is obtained by quotienting by the derived symmetry.

**Definition 13.7** (Reduced Classical Phase Space)**.**

$$\mathcal{P}_{\text{red}} := \text{Map}(M, \text{Plenum}G)^{(0)}.$$

**Proposition 13.8** (Reduced Symplectic Structure)**.** *$\mathcal{P}_{\text{red}}$ inherits a classical symplectic form which is the reduction of the classical form on $\mathcal{P}$.*

*Proof (Sketch).* Restrict the shifted reduction of Theorem 13.5 to degree 0. Standard arguments in AKSZ theory apply. □

## 13.5 Gauge Fixing and Derived Geometry

Gauge fixing can be regarded as selecting a section of the projection Plenum $\to$ Plenum$G$. Derived geometry provides a canonical method of choosing such sections via homotopy transfer.

**Theorem 13.9** (Homotopy Gauge Fixing)**.** *Under mild hypotheses on* Plenum*, there exists a homotopy splitting of the projection*

$$\text{Plenum} \longrightarrow \text{Plenum}G,$$

*so any gauge choice corresponds to a homotopy section.*

*Proof (Sketch).* The projection is a homotopy epimorphism in the derived category; a homotopy section exists by standard model category arguments. □

*Remark* 13.10*.* Homotopy gauge fixing means no physical content depends on gauge choices; ghost/antifield sectors encode the homotopy redundancies automatically.

## 13.6 BV–BFV Boundary Structure

Boundary terms of the AKSZ action produce a BV–BFV (Batalin–Vilkovisky / Batalin–Fradkin–Vilkovisky) theory on the boundary $\partial M$, encoding the symplectic structure of boundary degrees of freedom.

**Theorem 13.11** (BV–BFV Boundary Theorem)**.** *If $M$ has nonempty boundary, the AKSZ–RSVP action induces a* $(-1)$*–shifted symplectic form on the boundary field space and a BV–BFV structure consistent with the reduced phase space of the bulk theory.*

*Proof (Sketch).* This follows from the general AKSZ–BV–BFV correspondence [**CattaneoFelder2012**], applied to the 2–shifted plenum and its boundary reduction. □

## 13.7 Summary and Forward Link

We have developed derived symmetries, quotient stacks, shifted symplectic reduction, and reduced phase spaces for lamphrodynamic fields. Gauge fixing is homotopy–theoretic, and BV–BFV boundary structures appear automatically. These geometric foundations prepare for the detailed classical comparisons of the next chapter.

# Chapter 14

# Embedding Classical Field Theories into the Lamphrodynamic Plenum

### 14.0.1 Introduction

Classical field theory is usually formulated in terms of sections of bundles over a fixed spacetime manifold, together with variational principles determining the Euler–Lagrange equations. In the present volume we have recast field theory as arising from derived moduli spaces equipped with a $(-1)$–shifted symplectic structure. The goal of this chapter is to relate the ordinary formulation to the lamphrodynamic formalism by constructing a functor from a suitable category of classical field theories to the category representing RSVP field models.

We adopt the interpretative stance that classical theories embed *functorially* into RSVP. That is, there exists an $(\infty, 1)$–functor [ F : $\longrightarrow$, ]$whichisfullyfaithful(oratleastconservative), andwhoseessentiali$

*Remark* 14.1. This point of view corresponds to Option C of the discussion ending the previous chapter: RSVP is a categorical enlargement of classical theory, not merely a generalization nor an approximation scheme.

### 14.0.2 The category of classical field theories

Let $X$ be a smooth oriented spacetime manifold and let $E \to X$ be a smooth vector bundle (or principal bundle). A classical field configuration is a smooth section $\phi\colon X \to E$. A classical Lagrangian density $\mathcal{L}$ is a smooth function of the jet bundle $J^\infty(E)$, and the classical action is [ $\mathrm{S}_{\mathrm{class}}[\phi] = \int_X \mathcal{L}(\phi, \nabla\phi, \ldots), .$]

**Definition 14.2.** Define the $(\infty, 1)$–category  whose objects are triples $(X, E, \mathcal{L})$ and whose morphisms are field–bundle–variational morphisms preserving the Lagrangian structure.

This category includes scalar field theory, Yang–Mills theory, and Einstein gravity, among others. It constitutes the standard domain of classical theoretical physics.

### 14.0.3 Construction of the embedding functor

We now define a functor [ F : $\longrightarrow$ .]$Theideaisthataclassicalfield\phi\colon X \to E$ is assigned a lamphrodynamic triple $(\Phi, \mathbf{v}, S)$ via a canonical lift. The potential $\Phi$ encodes the classical field value, while $\mathbf{v}$ and $S$ are obtained from trivial (or canonical) background data determined by the geometry of $X$ and the structure of $E$. Formally, set [ F(X,E,L) = Map$(X, \mathsf{Plenum}_{\Phi,\mathbf{v},S})$, ]$where$Plenum$_{\Phi,\mathbf{v},S}$ is the lamphrodynamic moduli stack introduced previously, equipped with the $(-1)$–shifted symplectic structure constructed in Chapter 6.

**Proposition 14.3.** *The assignment $F$ extends to an $(\infty, 1)$–functor $F\colon \to$, preserving homotopy equivalences and sending classical Lagrangian morphisms to lamphrodynamic morphisms.*

*Sketch of proof.* A morphism of classical field theories induces a morphism of derived mapping stacks. The shifted symplectic forms are natural with respect to these maps by functoriality of the AKSZ construction. Hence the construction is functorial. The full proof uses the theory of tangent complexes discussed in Chapter 2. □

### 14.0.4 Examples

#### Scalar fields

Let $(X, E, \mathcal{L})$ be a scalar field theory with $E = X \times \mathbb{R}$ and [ L $= \frac{1}{2}(\partial\phi)^2 - V(\phi)$.]$Then[F(X, E, \mathcal{L}) =$ Map$(X, \mathsf{Plenum})$, ]$with$F$(\phi)$ = ($\Phi$,v,S)$givenby\Phi$(x) $= \phi$(x)$and$v,S$trivial.TheEuler--Lagrangeequationsbecom$ $componentofthelamphrodynamicequations.$

#### Yang–Mills fields

For Yang–Mills theory the classical gauge potential determines a component of $\Phi$ via a fixed embedding $\mathfrak{g} \hookrightarrow$ scalar sector, while $\mathbf{v}$ and $S$ remain trivial. The shifted symplectic form lifts from the moduli of gauge fields by AKSZ.

#### Einstein gravity

In the case of Einstein gravity, the Lorentzian metric determines $\Phi$ through a chosen geometric invariant such as curvature, and the components $\mathbf{v}$ and $S$ encode kinematic and entropic contributions from spacetime geometry. Details require the constructions of Chapter 12.

### 14.0.5 Conclusion

The functor $F$ embeds classical field theory into the RSVP formalism as a distinguished substructure. In this manner, the classical paradigm appears as a special case of a more general lamphrodynamic framework, and quantization can proceed categorically at the level of RSVP rather than theory by theory.

*Remark* 14.4. This approach allows one to regard classical field theories and RSVP models within a unified mathematical language, and it clarifies the relationship between classical physics and the derived geometric structures developed in the preceding chapters.

# Appendix A: Review of Differential Geometry

## .1 Manifolds and Smooth Maps

A *smooth manifold* of dimension $n$ is a second countable Hausdorff space $M$ together with an atlas of coordinate charts $(U_i, \varphi_i)$ where $\varphi_i \colon U_i \to \mathbb{R}^n$ are homeomorphisms onto their images and all transition maps $\varphi_j \circ \varphi_i^{-1}$ are smooth.

**Definition .5.** A map $f \colon M \to N$ between smooth manifolds is called smooth if for every pair of charts $\varphi$ on $M$ and $\psi$ on $N$, the coordinate representation $\psi \circ f \circ \varphi^{-1}$ is a smooth function between Euclidean spaces.

Note that in the categorical viewpoint, manifolds form a subcategory of smooth schemes, and ultimately of derived stacks. The lamphrodynamic formalism requires a broad generalization of these notions, but ordinary smooth manifolds remain the geometric base of physical spacetime throughout Volume I.

### .1.1 Tangent Bundle

For each $p \in M$, the tangent space $T_pM$ is defined as equivalence classes of smooth curves passing through $p$, or equivalently as derivations on germs of smooth functions at $p$. The union of all tangent spaces [ TM=$\bigsqcup_{p\in M} T_pM$]*isasmoothvectorbundleover*M.

*Remark* .6. Later, tangent complexes of derived stacks generalize these tangent spaces. In the derived setting, the tangent object is a chain complex rather than an ordinary vector space.

## .2 Vector Bundles and Differential Forms

A (real) vector bundle over $M$ is a smooth manifold $E$ together with a smooth projection $\pi \colon E \to M$ such that for each $p \in M$, the fiber $E_p = \pi^{-1}(p)$ is a real vector space and locally $E \simeq U \times \mathbb{R}^k$.

**Definition .7.** The space of smooth sections of $E$ is denoted $\Gamma(M, E)$.

In particular, the cotangent bundle $T^*M$ yields the spaces of differential $k$-forms [ $\Omega^k(M) = \Gamma(M, \wedge^k T^*M)$.]

## .3 Connections and Curvature

Let $E \to M$ be a vector bundle. A connection is a $\mathbb{R}$–linear operator [ $\nabla \colon \Gamma(M, E) \to \Gamma(M, T^*M \otimes E)$]*satisfyingtheLeibnizrule*∇(fs)=df⊗s + f,∇s*for*f∈C$^\infty(M)$ and $s \in \Gamma(M, E)$.

**Definition .8.** The curvature of a connection $\nabla$ is the bundle map [ R$_\nabla$ : $\Gamma(M, E) \to \Gamma(M, \wedge^2 T^*M \otimes E)$]*definedby*R$_\nabla(s) = \nabla^2 s$.

Connections appear in RSVP as part of geometric background data, but the lamphrodynamic fields themselves are defined intrinsically from the derived moduli stack, not imposed externally.

## .4 Riemannian and Lorentzian Geometry

A Riemannian metric $g$ on $M$ is a smooth section of $T^*M \otimes T^*M$ which is symmetric and positive definite. A Lorentzian metric has signature $(-,+,\dots,+)$ and is the standard setting for general relativity.

The Levi–Civita connection is the unique torsion–free connection compatible with a given metric. Its curvature tensor and associated invariants (Ricci curvature, scalar curvature) appear throughout classical field theory and reappear in the comparison sections of Chapter 14.

*Remark* .9. One conceptual theme of RSVP is that the lamphrodynamic metric need not be an externally fixed Lorentzian structure. Instead, the triple $(\Phi, \mathbf{v}, S)$ plays the role of the fundamental geometric data.

## .5 Exterior Calculus

The exterior derivative $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfies $d^2 = 0$, and the wedge product makes $\Omega^\bullet(M)$ into a graded commutative algebra. Many constructions in the $(-1)$–shifted symplectic setting rely on derived analogues of differential forms.

**Definition .10.** The de,Rham cohomology of $M$ is the cohomology of the complex $(\Omega^\bullet(M), d)$.

*Remark* .11. For derived stacks, the de,Rham complex is upgraded to a mixed complex whose cohomology yields the shifted symplectic form used in the AKSZ construction.

## .6 End of Appendix A

This appendix only summarizes the necessary smooth differential geometry. The derived generalizations in Chapters 2–4 systematically replace tangent spaces by tangent complexes, replace manifolds by derived Artin stacks, and replace ordinary forms by their graded, shifted, or mixed counterparts.

# Appendix B: Homological Algebra Essentials

## .7 Chain Complexes

Let **Vect** denote the category of real vector spaces. A chain complex $(C_\bullet, d)$ is a sequence of vector spaces [ $\cdots \longrightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \longrightarrow \cdots ]suchthat \mathrm{d}_n \circ d_{n+1} = 0$ for all $n \in \mathbb{Z}$. One defines the $n^{\text{th}}$ homology as [ $\mathrm{H}_n(C_\bullet) = \frac{\ker(d_n)}{\mathrm{im}(d_{n+1})}$.]

*Remark* .12. Tangent complexes of derived stacks are examples of chain complexes whose homology measures infinitesimal automorphisms and obstructions.

## .8 Double Complexes and Totalization

A double complex $C_{\bullet,\bullet}$ is a bi-graded collection $C_{p,q}$ with horizontal and vertical differentials. The total complex $\mathrm{Tot}(C)$ is defined by [ $\mathrm{Tot}(C)*n = \bigoplus *p + q = nC_{p,q}.]Totalization is crucial in defining the cotange$

$-BRST constructions.$

## .9 Tensor Products and Hom Functors

For chain complexes $C_\bullet$ and $D_\bullet$, the tensor product is the complex [ (C⊗D)*n = $\bigoplus *p + q = nC_p \otimes D_q,$ ]$withdifferentialdeterminedbytheLeibnizrule.Dually, theinternalHomcomplex$ Hom(C,D)$isgivenby$[Hom
$n = \prod_m \mathrm{Hom}(C_m, D*m + n),$ ]$withthestandarddifferential.Thisisthebackboneofenrichedhomotopicalstructu$

## .10 Exact Sequences and Derived Functors

A short exact sequence of complexes [ 0→A$_\bullet \to B_\bullet \to C_\bullet \to 0$]$inducesalongexacthomologysequence.Derivedfun$
and Tor arise by taking projective or injective resolutions of complexes.

*Remark* .13. For a derived stack $\mathcal{X}$, its cotangent complex $\mathbb{L} * \mathcal{X}$ and tangent complex $\mathbb{T} * \mathcal{X}$ are defined using homological algebra in suitable model categories.

## .11 Model Categories

A model category is a category equipped with three distinguished classes of morphisms (weak equivalences, fibrations, cofibrations) satisfying appropriate axioms. Homotopy categories and derived functors are constructed by formally inverting weak equivalences.

**Definition .14.** A Quillen adjunction between model categories induces an adjunction on homotopy categories, and a Quillen equivalence induces an equivalence of homotopy categories.

Model category language underlies much of the theory of derived algebraic geometry and is assumed in the main text. The reader seeking more detail will find standard references in the bibliography.

## .12 Spectral Sequences

Spectral sequences are computational tools associated to filtered chain complexes. They provide successive approximations to homology and cohomology and are indispensable for computing deformation and obstruction theories in derived geometry.

*Remark* .15. The obstruction theory used to construct the moduli stack Plenum depends on the vanishing of certain classes in higher derived Ext spaces, often computed via spectral sequence arguments.

## .13 End of Appendix B

We emphasize that the natural home for RSVP is the $(\infty, 1)$–categorical world introduced in Chapter 2, in which many of the above constructions are given a homotopical rather than merely algebraic interpretation.

# Appendix C: Functional Analysis for PDEs

## .14 Function Spaces

Let $\Omega \subset \mathbb{R}^n$ be an open set. The space $L^2(\Omega)$ consists of square–integrable functions with norm [ |u|*L$^{22}$ $= \int *\Omega |u(x)|^2, dx.]Moregenerally,$L$^p(\Omega)$ denotes the space of $p$–integrable functions. The Sobolev space $H^k(\Omega)$ is defined by [ H$^k(\Omega) = u \in L^2(\Omega) \mid \partial^\alpha u \in L^2(\Omega)$ for all $|\alpha| \leq k$.]

*Remark* .16. Sobolev spaces provide the natural framework for weak solutions of lamphrodynamic equations and for proving well–posedness theorems.

## .15 Weak Derivatives and Weak Solutions

Let $u \in L^1_{\text{loc}}(\Omega)$. A weak derivative $\partial_i u$ is a distribution satisfying [ $\int_\Omega u, \partial_i \varphi = -\int_\Omega (\partial_i u), \varphi]foralltestfunctions\varphi$
A weak solution of a PDE is a function satisfying the PDE in the sense of distributions.

**Definition .17.** A function $u \in H^1(\Omega)$ is called a weak solution to a differential operator $L$ if [ $\langle Lu, \varphi \rangle = 0]foralltestfunctions\varphi.$

## .16 Energy Methods

Consider an evolution equation [ $\partial_t u + Lu = N(u), ]where$L$isalinearoperatorand$N$anonlinearterm.Multiplying$
$\langle Lu, u \rangle = \langle N(u), u \rangle.]SuchestimatesarecentraltothelamphrodynamictheorydiscussedinChapters$ 9–
–11.

*Remark* .18. The entropy field $S$ introduces additional dissipation, strengthening the coercivity of energy estimates under suitable assumptions.

## .17 Semigroups and Linear Evolution

For linear equations $\partial_t u + Lu = 0$ on a Banach space $X$, well–posedness is often obtained by showing that $-L$ generates a strongly continuous semigroup $e^{-tL}{}_{t \geq 0}$ on $X$.

**Theorem .19** (Hille–Yosida)**.** *Let $X$ be a Banach space. A densely defined closed operator $-L$ generates a contraction semigroup on $X$ if and only if its resolvent satisfies suitable positivity and growth conditions.*

This theorem is used in the proof of global well–posedness for linearized lamphrodynamic fields.

## .18 Compactness and Regularity

Weak compactness and Rellich–Kondrachov compactness theorems allow one to extract convergent subsequences in Sobolev spaces. Elliptic regularity for elliptic operators $L$ improves the regularity of weak solutions: [ Lu=f∈H$^k \quad \Rightarrow \quad u \in H^{k+2}$.]

## .19 Nonlinear PDE Techniques

Local existence for nonlinear systems often follows from fixed point arguments (Picard iteration, contraction mapping principle) in suitable function spaces. Global existence usually requires energy estimates and a priori bounds.

*Remark* .20. Blow–up phenomena are typically associated with failure of energy control or growth of nonlinear terms. Entropy production in RSVP plays a role in mitigating such instabilities.

## .20 End of Appendix C

This appendix summarizes the analytic tools required for the proofs in Part III of this volume. The reader is referred to standard texts on PDEs and functional analysis for details beyond this short review.

# Appendix D: Conventions and Notation

## .21 Indices and Summation

Throughout this monograph, lower case Latin indices $i, j, k, \ldots$ range over spatial components $1, \ldots, n$, while Greek indices $\mu, \nu, \rho, \sigma$ range over spacetime indices $0, \ldots, n$. The Einstein summation convention over repeated indices is always assumed unless stated otherwise.

Boldface symbols such as $\mathbf{v}$ denote spatial vector fields, whereas non-bold symbols $v^\mu$ denote spacetime components.

## .22 Metrics and Signatures

Spacetime metrics $g_{\mu\nu}$ have Lorentzian signature $(-, +, \ldots, +)$ unless otherwise stated. Raising and lowering of indices is performed with $g_{\mu\nu}$ and its inverse $g^{\mu\nu}$.

The metric used in classical background comparisons may differ from the effective geometric structures arising in lamphrodynamics; the latter are derived from the triple $(\Phi, \mathbf{v}, S)$ rather than postulated *a priori.*

## .23 Differential Operators

The gradient operator $\nabla$ denotes the Levi–Civita covariant derivative unless the context dictates a gauge or derived connection. The d'Alembertian operator is [ $\Box = \nabla^\mu \nabla_\mu$.]

The divergence of a vector field is $\nabla \cdot \mathbf{v}$, and the Laplacian on scalar functions is $\Delta = \nabla^i \nabla_i$. In derived settings, the operator $\Delta$ refers to the BV Laplacian unless explicitly disambiguated.

## .24 Lamphrodynamic Fields

The symbol $\Phi$ denotes the scalar potential field. The bold symbol $\mathbf{v}$ denotes the spatial vector field of lamphrodynamic flow, and $S$ denotes the entropy density. The triple $(\Phi, \mathbf{v}, S)$ is the fundamental object of RSVP field theory.

*Remark* .21. It is important not to confuse the entropy field $S$ with the classical thermodynamic entropy, although there are interpretative analogies. In RSVP, $S$ is a geometric quantity arising from derived symplectic structures.

## .25 Derived Geometry

The symbols $\mathbb{L} * \mathcal{X}$ and $\mathbb{T} * \mathcal{X}$ denote the cotangent and tangent complexes of a derived stack $\mathcal{X}$. The shifted symplectic structure on $\mathsf{Plenum}$ is denoted by $\omega_{[-1]}$.

## .26 Variational Notation

The classical BV action is denoted $\mathcal{S}$, and the classical master equation is $\mathcal{S}, \mathcal{S} = 0$. The BV bracket is written $-, -_{BV}$ or simply $-, -$ when unambiguous.

## .27 Constants and Parameters

Physical constants such as $c$ and $G$ may be set to unity except in comparative formulas with observational cosmology. Dimensional analysis is included in Volume II as needed.

## .28 End of Appendix D

These conventions are used uniformly throughout all volumes of the monograph. Any deviation in later chapters will be indicated explicitly.

# Appendix E: Proof of Technical Lemmas

## .29 Introduction

This appendix collects proofs of auxiliary lemmas whose statements appear in earlier chapters but whose proofs were deferred in the interest of readability. We assume the notation and background from Chapters 2–11 and Appendices A–D.

Where possible, we refer to standard theorems in the literature (for example, PTVV for shifted symplectic structures or Hovey for model categories). Nevertheless, we provide explicit argumentation whenever those references require substantial adaptation to the lamphrodynamic setting.

## .30 Lemma 3.4: Exactness of the Shifted de,Rham Differential

**Lemma .22** (Lemma 3.4). *Let $\omega_{[-1]}$ be the $(-1)$–shifted presymplectic form on a derived Artin stack $\mathcal{X}$. Then the de,Rham differential $d_{\mathrm{dR}}\omega_{[-1]}$ vanishes in degree 2 of the truncated mixed complex.*

*Proof.* By PTVV, a $k$–shifted symplectic structure is a closed 2–form in the truncated de,Rham complex of degree $k$. In the case $k = -1$, we have a degree 1 form in cohomological grading, which corresponds to a degree 2 form in total degree after the shift. The closure condition then asserts $d_{\mathrm{dR}}\omega_{[-1]} = 0$. Since the truncation functor commutes with $d_{\mathrm{dR}}$ on derived Artin stacks, the vanishing holds in the truncated complex as required. □

## .31 Lemma 6.2: Smoothness Condition for the Moduli Stack Plenum

**Lemma .23** (Lemma 6.2)**.** *Let* Plenum *denote the lamphrodynamic moduli stack defined in Chapter 6. Then* Plenum *is a derived Artin stack locally of finite presentation.*

*Sketch of proof.* Recall that Plenum is constructed as a derived mapping stack $\mathrm{Map}(X, \mathcal{F})$, where $\mathcal{F}$ is a derived stack parameterizing the lamphrodynamic triple $(\Phi, \mathbf{v}, S)$. Since $X$ is smooth and $\mathcal{F}$ is locally of finite presentation, standard results on derived mapping stacks imply that $\mathrm{Map}(X, \mathcal{F})$ is a derived Artin stack locally of finite presentation. The details rely on the representability of mapping stacks by Toën–Vezzosi and Lurie, together with local finite presentation properties of $\mathcal{F}$. □

## .32 Lemma 9.7: Sobolev Inequality in Lamphrodynamic Variables

**Lemma .24** (Lemma 9.7)**.** *Let $\Phi, \mathbf{v}, S$ satisfy the hypotheses of Chapter 9. Then there exists a constant $C > 0$ such that [ /Φ/*$L^\infty(\Omega) \le C|\Phi| * H^k(\Omega)]foranyopenboundeddomain\Omega \subset R^n$ and integer $k > \frac{n}{2}$.*

*Proof.* This is the classical Sobolev embedding theorem. The presence of $\mathbf{v}$ and $S$ does not affect the embedding since the estimate is purely functional-analytic. The required regularity conditions are satisfied by the hypotheses of linearization in Chapter 9. □

## .33 Lemma 10.5: A Priori Energy Estimate

**Lemma .25** (Lemma 10.5)**.** *Under the assumptions of Theorem 10.3, the a priori energy estimate [ /(Φ,v,S)(t)/*$H^{12} \le C\exp(CT)\,|(\Phi, \mathbf{v}, S)(0)| * H^{12}]holdsforallt$∈[0,T].*

*Proof.* Multiply the linearized lamphrodynamic equations by $(\Phi, \mathbf{v}, S)$ and integrate in space. Integration by parts eliminates first–order derivatives. The entropy term contributes a negative definite form, and Grönwall's inequality yields the stated estimate. □

## .34 Lemma 11.8: Local Existence for the Nonlinear System

**Lemma .26** (Lemma 11.8)**.** *Assume the nonlinear lamphrodynamic system satisfies the hypotheses of Chapter 11. Then there exists a time $T > 0$ such that a unique solution exists on $[0, T]$ in $H^k$ for sufficiently large $k$.*

*Proof.* Local existence follows from a fixed–point argument (Picard iteration) in $H^k$ spaces. The nonlinear terms satisfy a Lipschitz condition in $H^k$ due to Sobolev embedding and Moser estimates. Uniqueness follows from a Grönwall argument applied to the difference of two solutions. □

## .35 End of Appendix E

*This concludes Volume I: Mathematical Foundations of the Relativistic Scalar–Vector Plenum.*

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# Relativistic Scalar–Vector Plenum: Physical Applications: Cosmology and Gravity

## Emergent Spacetime, Entropic Gravity, and Observational Predictions

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# Preface

s second volume develops the physical and cosmological consequences of the Relativistic Scalar–Vector Plenum (RSVP) introduced and constructed in Volume I. Whereas the foundational volume focused on the derived geometric and analytic underpinnings of the lamphrodynamic field theory, the present work applies these structures to a broad set of physical phenomena ranging from entropy-driven modifications of general relativity, to cosmic redshift without metric expansion, to observational tests against contemporary cosmological datasets.

The central thesis of this volume is that gravitational behavior and cosmic structure can be derived directly from entropy flow and lamphrodynamic coupling, without assuming classical spacetime expansion. In this sense, gravity emerges not as a fundamental force but as an effective description of entropic relaxation in the plenum. The resulting picture is partially aligned with other emergent gravity programs—most notably the thermodynamic perspectives of Jacobson and Padmanabhan, and the entropic interpretation of Verlinde—but differs in essential respects: gravity in RSVP is not postulated as an entropic force on a background spacetime, but is obtained from the variational structure of the lamphrodynamic fields themselves.

We emphasize at the outset that Volume II is not primarily a work of phenomenology. All derivations proceed from the rigorous mathematical framework developed in Volume I, especially the shifted symplectic construction and the AKSZ action functional. The modified Einstein equations derived here follow from lamphrodynamic variation, not from heuristic analogy. Likewise, the redshift function is obtained by solving null geodesic transport equations in the lamphrodynamic background, rather than postulating a phenomenological dilation factor.

From a methodological perspective, this volume weaves together three strands:

1. **Theoretical Development.** We formulate the entropic corrections to Einstein geometry, derive modified Friedmann-like equations, and analyze structure formation and cosmic microwave background anisotropies within the lamphrodynamic regime.

2. **Emergent Interpretation.** We develop a conceptual bridge between thermodynamic and gravitational descriptions, articulating in what precise sense gravity and metric expansion emerge from lamphrodynamic entropic descent rather than from a background metric dynamics.

3. **Empirical Confrontation.** Where feasible, we present comparisons with observational cosmology, including Type Ia supernovae, large-scale structure, weak lensing, and CMB data. Although many computations remain at a prototype stage, we include explicit statistical comparisons and outline a roadmap toward full falsifiability.

The philosophy of this volume is conservative: calculations are carried out explicitly, observational comparisons acknowledge uncertainties and limitations, and speculative ideas are clearly distinguished from derived results. While certain claims will remain conjectural pending further development, we have attempted to provide the strongest possible mathematical justification for each step, and to identify precisely where empirical confirmation or refutation is possible.

The reader is assumed to have internalized the core results of Volume I, especially the construction of the lamphrodynamic stack and the associated AKSZ/BV machinery. We will not repeat these foundational arguments, but will refer to specific theorems and propositions where needed.

Finally, we note that the structure of Volume II is intentionally modular: the chapters on emergent gravity, cosmology, and observational tests can be read semi-independently, provided the reader accepts the lamphrodynamic field equations as derived in Volume I. Readers primarily interested in phenomenology may safely skim the foundational derivations, whereas readers focused on mathematical rigor may be more interested in the structural chapters.

As with Volume I, every effort has been made to maintain a clear distinction between proven results, conjectural claims, and purely speculative proposals. The author welcomes any and all criticism, both mathematical and empirical, that sharpens or challenges the framework developed here.

# Contents

# Notation and Conventions

Throughout this volume we adopt the following conventions, which extend and specialize those of Volume I. The reader should consult the Notation and Conventions of Volume I for background definitions and general geometric conventions; only new or cosmologyspecific notation is introduced here.

- $(M, g)$ continues to denote a smooth Lorentzian fourmanifold with signature $(-,+,+,+)$, which we interpret as a lamphrodynamic spacetime. In cosmological settings we restrict to homogeneous and isotropic configurations, in which case $(M, g)$ admits a foliation by spatial hypersurfaces $\Sigma_t$.

- Greek indices $\mu, \nu, \ldots$ range over spacetime coordinates; Latin indices $i, j, \ldots$ range over spatial coordinates on $\Sigma_t$. The abstractindex notation of Penrose is used whenever tensorial structure is being emphasized independently of components.

- The core lamphrodynamic fields are denoted $\Psi = (\Phi, \boldsymbol{v}, S)$, where $\Phi$ is a scalar potential, $\boldsymbol{v}$ is a spatial vector flow field, and $S$ is the entropy density. Unless otherwise stated, we regard $(\Phi, \boldsymbol{v})$ as fields on $\Sigma_t$ evolving in $t$ via the lamphrodynamic equations derived in Volume I.

- The modified Einstein tensor is denoted $\widetilde{G}_{\mu\nu}$ and is defined by adding the entropyinduced correction tensor $\Theta_{\mu\nu}$ to the usual Einstein tensor $G_{\mu\nu}$. Explicitly,

$$\widetilde{G}_{\mu\nu} := G_{\mu\nu} + \Theta_{\mu\nu},$$

  where $\Theta_{\mu\nu}$ is computed from $(\Phi, \boldsymbol{v}, S)$ via the lamphrodynamic variation principle (see Chapter 1 and **??**).

- The cosmological metric used for homogeneous and isotropic solutions is written in the lamphrodynamic gauge as

$$g = -\mathrm{d}t^2 + a^2(t)\,\mathrm{d}\Sigma^2,$$

  where $a(t)$ is the effective scale factor induced by entropy flow and $\mathrm{d}\Sigma^2$ is the constantcurvature spatial metric. We emphasize that $a(t)$ is *not* assumed to arise from geometric expansion, but appears as an effective quantity derived from the lamphrodynamic fields.

- Entropy gradients are denoted by $\nabla_\mu S$; the associated entropy flux vector is denoted by $J_S^\mu$. In comoving coordinates we write $J_S^0 := \partial_t S$ and $J_S^i := v^i S$.

- The redshift function is denoted by $z = f(S, \boldsymbol{v})$ and arises from integrating null geodesics in the lamphrodynamic background. In many computations it is convenient to write

$$1 + z = \exp\Big(\int_\gamma \omega\Big),$$

where $\omega$ is the lamphrodynamic optical form along a null geodesic $\gamma$.

- Observational quantities follow standard cosmological notation: $H_0$ denotes the presentday Hubble parameter, $\Omega_m$, $\Omega_\Lambda$, and $\Omega_k$ denote matter, dark energy, and curvature density fractions, respectively (used only for comparison with $\Lambda$CDM). Distance measures include luminosity distance $d_L(z)$ and angular diameter distance $d_A(z)$, both defined with respect to the lamphrodynamic redshift $z$.

- Throughout this volume we adopt units in which $c = 1$, unless otherwise specified. Newtons constant $G$ is retained in analytic expressions where comparison with general relativity is explicit.

- Observational data sources (Planck, Pantheon, DES, ...) are cited by abbreviated names; full references appear in the bibliography and in Chapter C. Statistical conventions (posterior distributions, likelihood functions, $\chi^2$) follow standard cosmological and particle physics usage.

Unless otherwise indicated, all results in this volume rely on the field equations and shifted symplectic construction developed in Volume I. For ease of reading, we frequently refer to specific theorems by number (for example, "Theorem I.7.12"), meaning Theorem 12 in Chapter 7 of Volume I.

We recommend that readers with a primary interest in observational cosmology first consult Chapters 4 and 7 after reviewing the modified Einstein equations in Chapter 1, while readers primarily interested in emergentgravity theory may focus on Chapter 1 and **??**.

# Part I

# Emergent Gravity

# Chapter 1

# Gravity from Entropy Gradients

## 1.1 Motivation: Why Emergence?

Classical general relativity models the dynamics of spacetime geometry by relating curvature to stress–energy via the Einstein field equations. Lamphrodynamics modifies this paradigm by introducing an entropy–driven correction term arising from the lamphrodynamic fields $\Psi := (\Phi, \boldsymbol{v}, S)$ defined in Volume I.

Informally, the guiding principle of RSVP is that gravitational attraction is not a fundamental interaction mediated by a metric tensor, but is an *entropic phenomenon* induced by the dynamics of $\Psi$. In this view spacetime geometry emerges as an effective description of deeper lamphrodynamic degrees of freedom. The aim of this chapter is to explain, derive, and justify this perspective in a mathematically precise manner.

From a physical standpoint, this interpretation is motivated by the observation that regions of high entropy density tend to induce lamphrodynamic flows which pull field configurations into lower–energy, higher–coherence states. The associated effective stress–energy contributes an additional curvature term in the field equations, one which reduces to the usual Einstein tensor in the limit where entropy gradients vanish.

## 1.2 Review of Lamphrodynamic Variational Structure

Recall from Volume I that lamphrodynamic field configurations are described by the action functional

$$\mathcal{S}_{\mathrm{AKSZ}}[\Psi] = \int_M \omega_{-1}(\Psi, \mathrm{d}\Psi) + \mathcal{V}[\Psi],$$

constructed via the AKSZ procedure with target the $(-1)$shifted symplectic stack $\mathcal{P}$ introduced in **??**. The Euler–Lagrange equations take the schematic form

$$E_\Psi(\Phi, \boldsymbol{v}, S) = 0,$$

coupled to the metric variation arising from the underlying AKSZ action. We now recall the basic structure needed for the gravitational sector.

**Theorem 1.1** (Metric Variation of the AKSZ–RSVP Action)**.** *Let* $(M, g)$ *be a lamphrodynamic* *spacetime and* $\Psi = (\Phi, \boldsymbol{v}, S)$ *a smooth configuration. Then the variational derivative of the AKSZ–RSVP action with respect to* $g$ *is given by*

$$\frac{\delta \mathcal{S}_{\mathrm{AKSZ}}}{\delta g^{\mu\nu}} = -\frac{1}{2}\sqrt{|g|}\,(T^{\mathrm{matter}}_{\mu\nu} + \Theta_{\mu\nu}),$$

*where* $\Theta_{\mu\nu}$ *is an entropy–induced tensor determined by* $(\Phi, \boldsymbol{v}, S)$ *and the* $(-1)$*shifted symplectic form* $\omega_{-1}$*.*

*Sketch of proof.* The variation follows from the AKSZ construction and the explicit form of the target $(-1)$shifted symplectic structure. The term $\Theta_{\mu\nu}$ arises from the nontrivial dependence of $\omega_{-1}$ and $\mathcal{V}$ on the metric through entropy gradients and lamphrodynamic flow. The full derivation is given in **??**. □

## 1.3 Derivation of Modified Einstein Equations

Combining Theorem 1.1 with the standard variation of the Einstein–Hilbert action leads to the modified Einstein field equations

$$G_{\mu\nu} + \Theta_{\mu\nu} = 8\pi G\, T^{\mathrm{matter}}_{\mu\nu}, \tag{1.1}$$

where $\Theta_{\mu\nu}$ depends on $(\Phi, \boldsymbol{v}, S)$ and vanishes when the lamphrodynamic fields are trivial. By construction, $\Theta_{\mu\nu}$ is symmetric and covariantly conserved:

$$\nabla^{\mu}\Theta_{\mu\nu} = 0,$$

which follows directly from the $(-1)$shifted symplectic structure and the AKSZ master equation.

Equation (1.1) should be interpreted as a set of effective gravitational field equations, where curvature is driven not only by ordinary matter but also by entropy dynamics. The geometry is therefore emergent in the sense that it is determined by lamphrodynamic evolution rather than postulated a priori.

## 1.4 Explicit Computation of the Entropic Correction Tensor

We now give an explicit formula for the tensor $\Theta_{\mu\nu}$ introduced in Theorem 1.1. As in Volume I, let $\Psi = (\Phi, \boldsymbol{v}, S)$ denote the lamphrodynamic fields, and write $J^{\mu}_{S}$ for the entropy–flux vector. The $(-1)$shifted symplectic structure supplies a natural bilinear pairing

$$\omega_{-1} : T_{-1}\mathcal{P} \times T_{-1}\mathcal{P} \longrightarrow \mathbb{R}[-1],$$

whose local expression, in the gauge chosen in **??**, takes the schematic form

$$\omega_{-1}(\Psi, \mathrm{d}\Psi) = S\, g_{\mu\nu}\, \mathrm{d}\Phi \wedge \mathrm{d}v^{\nu} + \cdots .$$

Varying this with respect to the metric and extracting the $\delta g^{\mu\nu}$ component produces the following.

**Proposition 1.2.** *The entropic correction tensor associated to the lamphrodynamic fields is given in local coordinates by*

$$\Theta_{\mu\nu} := \alpha\left(\nabla_\mu S\,\nabla_\nu\Phi + \nabla_\mu\Phi\,\nabla_\nu S - g_{\mu\nu}\,\nabla_\rho S\,\nabla^\rho\Phi\right) + \beta\left(J^S_\mu\,J^S_\nu - \tfrac{1}{2}g_{\mu\nu}J^S_\rho J^\rho_S\right), \tag{1.2}$$

*where $\alpha, \beta$ are universal coupling constants determined by the AKSZ target and the normalization of $\omega_{-1}$.*

*Proof.* The term proportional to $\alpha$ arises from metric variations of the $(-1)$shifted symplectic pairing between $\Phi$ and $S$; the term proportional to $\beta$ arises from the kinetic part of the entropy–flux sector. The full derivation follows the standard variation procedure for AKSZ actions, using the explicit form of $\omega_{-1}$ in a local coordinate chart and the master equation to guarantee symmetry and conservation. Details appear in Appendix **??**. $\square$

*Remark* 1.3. The tensor $\Theta_{\mu\nu}$ plays the role of an "entropic stress–energy tensor," but unlike the usual perfectfluid form, it captures both entropy gradients (through $\alpha$) and entropy fluxes (through $\beta$).

### 1.4.1 Weak–Field Limit and Newtonian Approximation

To understand the physical content of (1.1) and (1.2), we linearize the metric around Minkowski space. Write $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$. Retaining only linear terms in $h_{\mu\nu}$ and the lamphrodynamic fields yields the approximate field equations

$$\Box h_{\mu\nu} = -16\pi G\left(T^{\text{matter}}_{\mu\nu} + \Theta^{(1)}_{\mu\nu}\right), \tag{1.3}$$

where $\Theta^{(1)}_{\mu\nu}$ denotes the linear part of (1.2). In the Newtonian gauge and for slowly varying fields, this reduces to the effective Poisson equation

$$\Delta\Phi_{\text{grav}} = 4\pi G(\rho_{\text{matter}} + \rho_{\text{entropy}}), \tag{1.4}$$

with $\rho_{\text{entropy}}$ obtained from the time–time component of $\Theta^{(1)}_{\mu\nu}$.

This shows that entropy gradients and entropy fluxes contribute additional "massdensity" terms, leading to enhanced gravitational attraction in regions with strong lamphrodynamic activity.

### 1.4.2 Comparison with General Relativity

In the limit $\alpha, \beta \to 0$ (or equivalently when $S$ and $J_S$ vanish), equation (1.4) reduces to the classical Newtonian potential equation with source $\rho_{\text{matter}}$ only. Hence general relativity is recovered as a degenerate limit of the lamphrodynamic theory.

By contrast, when $S$ and $J_S$ are non–trivial, the right–hand side of (1.4) includes additional contributions, providing a potential explanation for phenomena traditionally attributed to dark matter or modified gravity. Subsequent chapters will analyze these possibilities in detail.

### 1.4.3 Phenomenological Interpretation of $\rho_{\text{entropy}}$

Equation (1.4) suggests interpreting

$$\rho_{\text{entropy}} := \frac{1}{4\pi G}\,\Theta^{(1)}_{00}$$

as an effective mass–density associated to lamphrodynamic degrees of freedom. Two qualitatively distinct mechanisms appear:

1. *Static entropic gradients*: regions with $\nabla_i S \neq 0$ produce a contribution to $\rho_{\text{entropy}}$ similar in form to scalar field sources, but with reversed sign depending on the parameters $(\alpha, \beta)$. In typical lamphrodynamic regimes, $\alpha > 0$ yields additional attractive potential.

2. *Dynamical entropy flux*: regions with $J_i^S \neq 0$ contribute via the flux–flux term in (1.2), producing attractive gravitational effects even when $S$ is spatially homogeneous.

These two effects are conceptually different: entropy density gradients correspond to spatial variation in the thermodynamic state of the plenum, whereas entropy flux corresponds to ongoing irreversible processes or negentropic flows. Their coexistence in $\Theta_{\mu\nu}$ is a distinctive signature of the lamphrodynamic theory absent in general relativity.

### 1.4.4 Observational Consequences

In the weak–field, quasi–stationary regime, the effective density $\rho_{\text{entropy}}$ acts as an additional gravitational source. This leads to several immediate observational consequences:

1. *Galaxy rotation curves.* A spatially varying $S$ can modify the radial gravitational potential without invoking dark matter halos. Preliminary rotation–curve fits are discussed in **??**.

2. *Cluster binding.* Enhanced effective mass at large scales may stabilize galaxy clusters consistent with observed velocity dispersions.

3. *Gravitational lensing.* The entropy–flux term modifies the deflection angle for light rays, potentially explaining lensing anomalies in clusters or strong–lensing systems.

These consequences will be quantified and compared with observational data in **??**.

### 1.4.5 Dimensional Analysis and Coupling Constants

The constants $\alpha, \beta$ appearing in (1.2) are dimensionful and depend on the normalization of the AKSZ target, as well as the units in which entropy and potential are measured. A convenient choice is

$$[\Phi] = \text{energy per unit mass}, \qquad [S] = k_B \, (\text{dimensionless}),$$

so that $(\alpha, \beta)$ carry dimensions of length$^2$ in geometrized units. In the classical limit $\hbar \to 0$ (where the AKSZ structure becomes purely classical), one may treat $\alpha, \beta$ as phenomenological constants to be fit by observation; in the quantum regime, they are determined by the target geometry of the BV theory.

### 1.4.6 Recovery of Newtonian Gravity

Setting $S = 0$ and $J_S = 0$ exactly, the tensor $\Theta_{\mu\nu}$ vanishes identically, and (1.4) reduces to

$$\Delta\Phi_{\text{grav}} = 4\pi G\rho_{\text{matter}},$$

the usual Newtonian gravity. Thus, the lamphrodynamic theory strictly contains general relativity as a special case, without introducing additional fields beyond $(\Phi, \boldsymbol{v}, S)$.

### 1.4.7 Relation to Scalar–Tensor Theories

At first sight, $\Theta_{\mu\nu}$ resembles the stress–energy tensor of a scalar field $\varphi$ in scalar–tensor gravity. However, the presence of $\boldsymbol{v}$ and the flux sector $J_S$ distinguishes the lamphrodynamic case: there is no single scalar field but a coupled system of scalar potential, vector flow, and entropy density. Moreover, $S$ is thermodynamic rather than fundamental, encoding local entropy rather than a new fundamental scalar. This qualitative difference will be crucial in subsequent chapters, especially when comparing with scalar–tensor screening mechanisms.

### 1.4.8 Summary of the Weak–Field Analysis

The main conclusions of this chapter are:

- the lamphrodynamic correction tensor $\Theta_{\mu\nu}$ provides an effective gravitational source proportional to entropy gradients and fluxes;
- these corrections produce additional gravitational attraction in situations with nontrivial lamphrodynamic activity;
- Newtonian gravity is recovered when $S$ and $J_S$ vanish or are negligible;
- the resulting phenomenology can potentially address astrophysical anomalies without invoking nonbaryonic dark matter.

In the next chapter we compare these predictions with other emergent–gravity frameworks, especially Jacobson's thermodynamic derivation of Einstein's equations and Verlinde's entropic gravity proposals, highlighting both conceptual and quantitative differences.

# Chapter 2

# Comparison with Other Emergent Gravity Theories

## 2.1 Overview

A central claim of the lamphrodynamic framework is that gravitational phenomena emerge from entropy gradients and thermodynamic fluxes in the plenum. This places RSVP in conceptual proximity to several other emergent–gravity proposals, most prominently those of Jacobson, Verlinde, and Padmanabhan, as well as more recent quantum–information approaches (notably Carney). The goal of this chapter is to compare these theories in detail, highlighting overlaps, divergences, and potential experimental discriminants.

Throughout, we work at the level of qualitative mechanism, then translate to quantitative predictions where possible. A comparison table summarizing assumptions and key predictions appears in Section 2.6.

## 2.2 Jacobson's Thermodynamic Gravity

### 2.2.1 Local Rindler Horizons and Clausius Relation

Jacobson's 1995 derivation of Einstein's equations interprets gravity as a thermodynamic equation of state. Consider a local Rindler horizon $H$ generated by accelerated observers. The Clausius relation

$$\delta Q = T\,\mathrm{d}S$$

applied to local horizons yields, under certain assumptions,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\,T_{\mu\nu},$$

where $G_{\mu\nu}$ is the Einstein tensor and $T_{\mu\nu}$ the stress–energy tensor of matter fields. Crucially, this derivation presupposes a specific relation between area and entropy ($\mathrm{d}S \propto \mathrm{d}A$), borrowed from black–hole thermodynamics.

### 2.2.2 Comparison with Lamphrodynamic Entropy

RSVP shares the theme that entropy plays a gravitational role, but differs in three fundamental ways:

1. Entropy in RSVP is *volume–based* rather than horizon–based: $S$ is defined locally throughout the plenum, without reference to causal horizons.

2. Gravity arises from *gradients and fluxes* of $S$ rather than equilibrium relations at horizons.

3. The field equations are modified Einstein equations with an explicit lamphrodynamic contribution $\Theta_{\mu\nu}$ derived from the AKSZ action, not Einstein's equations recovered from thermodynamic identities.

Thus, whereas Jacobson infers Einstein's equations from thermodynamic assumptions, RSVP derives *modified* Einstein equations from a microscopic lamphrodynamic field theory.

## 2.3 Verlinde's Entropic Gravity

### 2.3.1 Holographic Screens and Entropic Force

Verlinde proposes that gravity is an entropic force arising from changes of entropy associated with holographic screens. A test mass approaching a screen experiences an entropic force

$$F = T \frac{\partial S}{\partial x},$$

leading, in heuristic derivations, to Newtonian gravity. Later developments attempt to explain dark–matter–like phenomena through entropic effects associated with de Sitter entropy.

### 2.3.2 Differences with RSVP

Although both frameworks employ entropy gradients, several sharp differences exist:

1. Verlinde's entropy is associated with *holographic screens* and fundamentally tied to information storage capacity; RSVP entropy is a local thermodynamic field defined in the bulk.

2. Verlinde's derivation is essentially Newtonian or weak–field, whereas RSVP yields fully covariant modified Einstein equations.

3. Verlinde's approach postulates emergent space from holography; RSVP assumes a lamphrodynamic plenum with intrinsic geometry, not constructed from entanglement entropy.

Consequently, RSVP predicts distinct strong–field and cosmological deviations from general relativity, while Verlinde's modifications primarily affect galaxy–scale dynamics.

## 2.4 Padmanabhan's Emergent Paradigm

### 2.4.1 Degrees of Freedom Balance

Padmanabhan argues that spacetime dynamics emerge from the thermodynamic equipartition of degrees of freedom between bulk and boundary, encoded in holographic relations such as

$$N_{\text{sur}} - N_{\text{bulk}} \sim \text{cosmological acceleration.}$$

Acceleration of cosmic expansion results from a mismatch between horizon and bulk degrees of freedom.

### 2.4.2 Comparison with RSVP

The lamphrodynamic theory shares with Padmanabhan the idea that cosmological acceleration is thermodynamic in origin. However:

1. RSVP attributes acceleration to entropy *flux* and gradient terms in $\Theta_{\mu\nu}$, not mismatch of horizon degrees of freedom.

2. Cosmology in RSVP does not rely on de Sitter entropy or horizon thermodynamics.

3. RSVP does not posit holographic equipartition as a fundamental principle; instead, AKSZ variations yield modified field equations directly.

## 2.5 Quantum Information Approaches (Carney)

### 2.5.1 Gravity from Decoherence and Entanglement

Carney and collaborators propose that gravity may emerge from quantum information dynamics such as decoherence, entanglement structure, or information flow in quantum systems. These approaches focus on microscopic quantum mechanisms rather than macroscopic thermodynamics.

### 2.5.2 Relevance to RSVP

Lamphrodynamic entropy $S$ is classical and thermodynamic rather than quantum informational. However, a deep connection may exist if $S$ coarse–grains microscopic quantum correlations. If so, lamphrodynamic fluxes could represent emergent semiclassical remnants of underlying quantum information flows. This is speculative but suggests potential connections between AKSZ quantization and holographic quantum information geometry.

## 2.6 Comparison Table

| **Property** | **RSVP** | **Jacobson** | **Verlinde** | **Padmanabha** |
|---|---|---|---|---|
| Entropy type | volume, local | horizon | holographic screen | horizon/bulk bala |
| Mechanism | gradients, flux | Clausius relation | entropic force | DOF mismatc |
| Field equations | modified Einstein | Einstein | Newtonian/heuristic | cosmological dynami |
| Dark–matter effects | entropy flux | not primary | entropic de Sitter | holographic acceler |
| Quantum origin | AKSZ, classical limit | thermo heuristic | holography heuristic | holographic DO |

## 2.7 Summary and Outlook

While RSVP is thermodynamic in spirit, it differs from other emergent–gravity proposals in mechanism, ontology, and mathematical structure. The lamphrodynamic corrections arise from explicit field equations, not from thermodynamic identities or holographic assumptions. In the following chapter, we turn to observable consequences for geodesic motion and test particles, including perihelion precession and light bending, which provide immediate empirical tests of the lamphrodynamic corrections.

# Chapter 3

# Geodesic Equations and Test Particles

## 3.1 Overview

In standard general relativity, freely falling test particles follow geodesics of the spacetime metric $g_{\mu\nu}$. In RSVP, the presence of lamphrodynamic entropy flux modifies the connection and thus the notion of free fall. The purpose of this chapter is to derive the modified geodesic equation, analyze its consequences for classical tests of general relativity (perihelion precession, light bending), and discuss gravitational time dilation in lamphrodynamic backgrounds.

Throughout, we assume the modified Einstein equations from **??**:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \Theta_{\mu\nu} = 8\pi G\, T_{\mu\nu},$$

where $\Theta_{\mu\nu}$ encodes lamphrodynamic contributions from $(\Phi, \boldsymbol{v}, S)$ as derived in **??**.

## 3.2 Modified Connection and Effective Metric

### 3.2.1 Lamphrodynamic Correction

We model the lamphrodynamic influence on geodesic motion by defining an effective connection

$$\tilde{\Gamma}^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\mu\nu} + \Delta^{\lambda}_{\mu\nu},$$

where $\Gamma$ is the Levi–Civita connection of $g$ and $\Delta^{\lambda}_{\mu\nu}$ is a small correction depending on $\Theta_{\mu\nu}$ and derivatives of $\Phi$, $\boldsymbol{v}$, and $S$. For definiteness, we adopt the model ansatz

$$\Delta^{\lambda}_{\mu\nu} := \alpha\, g^{\lambda\rho} \nabla_{(\mu}(S\, \boldsymbol{v}_{\nu)}), \tag{3.1}$$

where $\alpha$ is a coupling constant with dimensions of length, and $(\cdot)_{(\mu\nu)}$ denotes symmetrization. More sophisticated forms of $\Delta$ could be derived from the AKSZ action, but (3.1) captures the leading lamphrodynamic contribution in weak fields.

### 3.2.2 Effective Geodesic Equation

The modified geodesic equation becomes

$$\frac{d^2x^\lambda}{d\tau^2} + \Gamma^\lambda_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} + \Delta^\lambda_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0. \tag{3.2}$$

When $\Delta \equiv 0$, we recover the usual geodesic equation.

*Remark* 3.1 (Interpretation). The correction term encodes the effect of entropy flow and lamphrodynamic gradients on inertial motion. In heuristic terms, test particles “feel” an additional force resulting from entropy flux, analogous to thermodynamic drag or buoyancy effects in continuum media, but in a fully covariant setting.

## 3.3 Modified Raychaudhuri Equation

### 3.3.1 Classical Form

For a congruence of timelike geodesics with tangent vector $u^\mu$, the classical Raychaudhuri equation is

$$\frac{d\theta}{d\tau} = -\frac{1}{3}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu,$$

where $\theta$ is expansion, $\sigma$ shear, and $\omega$ vorticity.

### 3.3.2 Lamphrodynamic Correction

In RSVP, the Ricci tensor $R_{\mu\nu}$ is replaced by an effective Ricci tensor $\tilde{R}_{\mu\nu}$ defined by

$$\tilde{R}_{\mu\nu} := R_{\mu\nu} + \Theta_{\mu\nu}.$$

Thus, the modified Raychaudhuri equation becomes

$$\frac{d\theta}{d\tau} = -\frac{1}{3}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - (R_{\mu\nu} + \Theta_{\mu\nu})u^\mu u^\nu. \tag{3.3}$$

The extra term $-\Theta_{\mu\nu}u^\mu u^\nu$ may either enhance or reduce focusing of geodesics depending on the sign and magnitude of entropy gradients and lamphrodynamic flux.

## 3.4 Perihelion Precession

### 3.4.1 Standard GR Result

In general relativity, the perihelion of Mercury advances by approximately $43''$ per century due to relativistic corrections. Observational agreement is excellent.

### 3.4.2 Lamphrodynamic Correction

To leading order in the weak–field, slow–motion limit, the correction to the effective potential due to $\Delta^\lambda_{\mu\nu}$ yields a modified perihelion advance of the form

$$\Delta\phi_{\mathrm{RSVP}} \approx \frac{6\pi GM}{a(1-e^2)c^2}(1+\epsilon_{\mathrm{RSVP}}),$$

where $a$ and $e$ are the semi–major axis and eccentricity, and $\epsilon_{\mathrm{RSVP}}$ is a small dimensionless parameter proportional to $\alpha\,\nabla S\cdot\boldsymbol{v}$ evaluated along the orbit.

Empirical bounds on $\epsilon_{\mathrm{RSVP}}$ arise from detailed fits to planetary ephemerides; current observational precision suggests $\epsilon_{\mathrm{RSVP}} \lesssim 10^{-5}$, depending on the specific model of $S$ and $\boldsymbol{v}$.

## 3.5 Light Bending

### 3.5.1 Null Geodesics

For null geodesics ($ds^2 = 0$), the same modified geodesic equation (3.2) applies. To leading order, the light–bending angle near a massive body is

$$\delta\phi_{\mathrm{RSVP}} \approx \frac{4GM}{c^2 b}(1+\epsilon_{\mathrm{RSVP}}),$$

where $b$ is the impact parameter. Current measurements of gravitational lensing by the Sun impose tight constraints on $\epsilon_{\mathrm{RSVP}}$, potentially at the level of $10^{-4}$ or better.

## 3.6 Gravitational Time Dilation

Lamphrodynamic corrections modify redshift and time dilation through $\Delta^\lambda_{\mu\nu}$ and $\Theta_{\mu\nu}$. The proper time ratio between two stationary observers at different potentials receives a multiplicative correction of the form

$$\frac{d\tau_1}{d\tau_2} = \sqrt{\frac{g_{00}(x_1)}{g_{00}(x_2)}}(1+\epsilon_{\mathrm{RSVP}}),$$

where again $\epsilon_{\mathrm{RSVP}}$ is controlled by entropy gradients and lamphrodynamic fluxes. Existing experimental tests of gravitational redshift are sufficiently precise that bounds on $\epsilon_{\mathrm{RSVP}}$ are likely stringent, though a systematic analysis awaits detailed phenomenological modeling.

## 3.7 Summary

Modifications to geodesic motion provide one of the clearest empirical handles on lamphrodynamic gravity. Classical tests of relativity — perihelion precession, light bending, gravitational time dilation — place strong constraints on the magnitude of lamphrodynamic corrections. Nonetheless, small deviations remain possible and could be detected by higher–precision experiments, providing a direct test of the RSVP framework.

In the next chapter (**??**), we turn to cosmological implications, deriving Friedmann–like equations from homogeneous and isotropic lamphrodynamic configurations and exploring the entropic contribution to cosmic expansion.

# Part II

# RSVP Cosmology

# Chapter 4

# Friedmann-Like Equations from Lamphrodynamics

## 4.1 Overview

Classical cosmology relies on the FriedmannLemaîtreRobertsonWalker (FLRW) class of metrics, representing homogeneous and isotropic universes with scale factor $a(t)$. The standard Friedmann equations follow from Einsteins field equations applied to the FLRW ansatz. In RSVP, the modified Einstein equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \Theta_{\mu\nu} = 8\pi G\, T_{\mu\nu}$$

lead, under homogeneity and isotropy assumptions, to *modified* Friedmannlike equations with explicit entropic contributions. This chapter derives these equations in detail and discusses physical interpretation, including the effective equation of state and critical densities.

## 4.2 Lamphrodynamic FLRW Ansatz

We adopt the standard FLRW metric

$$ds^2 = -c^2 dt^2 + a(t)^2 \left( \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right),$$

with curvature parameter $k \in \{-1, 0, 1\}$. Homogeneity and isotropy impose that the lamphrodynamic fields $(\Phi, \boldsymbol{v}, S)$ respect FLRW symmetry. We assume

$$\boldsymbol{v}(t, \mathbf{x}) = 0, \qquad S(t, \mathbf{x}) = S(t), \qquad \Phi(t, \mathbf{x}) = \Phi(t),$$

so that only time dependence remains. (More general anisotropic configurations are possible but beyond the scope of this chapter.)

*Remark* 4.1 (Physical interpretation). In the FLRW context, lamphrodynamic entropy becomes a *cosmic entropy density* varying only with cosmic time. Entropy production and lamphrodynamic couplings generate effective pressure and energydensity terms modifying the cosmological expansion.

## 4.3 Modified Einstein Tensor

In the FLRW metric, the Einstein tensor components are

$$G_{00} = 3\left(\frac{\dot{a}}{a}\right)^2 + 3\frac{k}{a^2}, \qquad G_{ij} = -\left[2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2}\right] g_{ij},$$

with dots denoting time derivatives.

## 4.4 Lamphrodynamic Stress Tensor $\Theta_{\mu\nu}$

From the AKSZ derivation (see Volume I, **??**), the lamphrodynamic stress tensor for homogeneous configurations takes the form

$$\Theta_{00} = \rho_{\mathrm{lam}}(t), \qquad \Theta_{ij} = p_{\mathrm{lam}}(t)\, g_{ij},$$

where

$$\rho_{\mathrm{lam}} := \alpha_1 \dot{S}^2 + \alpha_2 S^2 + \alpha_3 \dot{\Phi}^2, \tag{4.1}$$

$$p_{\mathrm{lam}} := \beta_1 \dot{S}^2 + \beta_2 S^2 + \beta_3 \dot{\Phi}^2 \tag{4.2}$$

for certain coupling constants $\alpha_i, \beta_i$ determined by the AKSZ action and variational principle. (Explicit formulas may be modeldependent and appear in Appendix I of this volume.)

*Remark* 4.2. Entropy production $\dot{S} > 0$ contributes positively to $\rho_{\mathrm{lam}}$, potentially driving accelerated expansion even in the absence of a cosmological constant $\Lambda$.

## 4.5 Modified Friedmann Equations

Inserting $G_{\mu\nu}$ and $\Theta_{\mu\nu}$ into

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \Theta_{\mu\nu} = 8\pi G\, T_{\mu\nu},$$

with perfectfluid matter stress tensor

$$T_{00} = \rho, \qquad T_{ij} = p\, g_{ij},$$

we obtain

$$3\left(\frac{\dot{a}}{a}\right)^2 + 3\frac{k}{a^2} = 8\pi G\,\rho + \Lambda + \rho_{\mathrm{lam}}, \tag{4.3}$$

$$-2\frac{\ddot{a}}{a} - \left(\frac{\dot{a}}{a}\right)^2 - \frac{k}{a^2} = 8\pi G\, p - \Lambda + p_{\mathrm{lam}}. \tag{4.4}$$

## 4.6 Effective Equation of State

Define an effective lamphrodynamic equationofstate parameter

$$w_{\mathrm{lam}} := \frac{p_{\mathrm{lam}}}{\rho_{\mathrm{lam}}}.$$

Depending on the signs and magnitudes of $\alpha_i, \beta_i$, one may have

$$-1 \leq w_{\mathrm{lam}} \leq +1,$$

with $w_{\mathrm{lam}} \approx -1$ corresponding to darkenergylike behavior and $w_{\mathrm{lam}} \approx 0$ mimicking cold dark matter. In this sense, lamphrodynamic entropy can *unify* dark energy and dark matter behavior in a single effective field.

## 4.7 Critical Densities

Define the critical lamphrodynamic density

$$\rho_{\mathrm{crit,lam}} := \frac{3}{8\pi G}\left(\frac{\dot{a}}{a}\right)^2 - \rho - \frac{\Lambda}{8\pi G}.$$

This characterizes the lamphrodynamic contribution needed to close the universe ($k = +1$) or accelerate expansion in the flat case ($k = 0$). Unlike in ΛCDM, critical density may be timedependent due to entropy production.

## 4.8 Discussion

Lamphrodynamic fields generically modify the cosmic expansion history, potentially explaining accelerated expansion without a cosmological constant and altering the interpretation of dark matter. Subsequent chapters explore these consequences in detail, focusing first on redshift without metric expansion (Chapter 4), then structure formation and darksector phenomenology.

# Chapter 5

# Redshift Without Metric Expansion

## 5.1 Overview

A distinctive feature of the RSVP cosmology is the possibility of explaining observed redshift entirely through lamphrodynamic effects rather than metric expansion. Whereas standard cosmology attributes the redshift of light to the stretching of spacetime (increase of the scale factor $a(t)$), RSVP allows for redshift generated by entropy gradients and the interaction of null geodesics with the lamphrodynamic fields $(\Phi, \boldsymbol{v}, S)$.

In this chapter we provide a complete derivation of the redshift formula

$$z = \int_\gamma f(S, \boldsymbol{v})\, d\tau,$$

starting from the modified Einstein equations and the null geodesic equations from Chapter 3. We then compare the resulting distance–redshift relation with the standard $z \approx H_0 d/c$ law and derive higher–order corrections.

## 5.2 Null Geodesics in Lamphrodynamic Backgrounds

Consider a null curve $\gamma(\tau)$ parameterized by affine parameter $\tau$ satisfying the modified geodesic equation

$$\frac{d^2 x^\lambda}{d\tau^2} + \Gamma^\lambda_{\mu\nu} \frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} + \Delta^\lambda_{\mu\nu} \frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0, \tag{5.1}$$

with $\Gamma$ the Levi–Civita connection and $\Delta$ given by (3.1). For null curves,

$$g_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0.$$

Let $k^\mu = dx^\mu/d\tau$ denote the null tangent vector. Contracting (4.1) with $g_{\lambda\rho}$ yields

$$k^\nu \nabla_\nu k_\rho = -\Delta_{\rho\mu\nu} k^\mu k^\nu,$$

where $\Delta_{\rho\mu\nu} := g_{\rho\lambda}\Delta^\lambda_{\mu\nu}$. Thus, lamphrodynamics introduces an effective *drag* or *frequency–shift* term along null rays.

## 5.3 Frequency Transport Equation

Let $\omega(\tau)$ denote the photon frequency measured by a comoving observer with 4–velocity $u^\mu$. Then

$$\omega = -g_{\mu\nu} k^\mu u^\nu.$$

Differentiating along $\gamma$, we obtain

$$\frac{d\omega}{d\tau} = -u^\nu k^\mu k^\rho \nabla_\rho g_{\mu\nu} - g_{\mu\nu} k^\mu k^\rho \nabla_\rho u^\nu - g_{\mu\nu} k^\mu k^\rho \Delta^\nu_{\rho\sigma} u^\sigma. \tag{5.2}$$

For a comoving observer in an FLRW background with lamphrodynamics, the first two terms reproduce the standard cosmic redshift associated with the scale factor, while the last term introduces new entropy–flux effects.

Using the homogeneous ansatz from **??** and the expression for $\Delta^\nu_{\rho\sigma}$ from (3.1), (4.2) simplifies to

$$\frac{1}{\omega}\frac{d\omega}{d\tau} = -\frac{\dot{a}}{a} - \alpha\, k^\mu k^\rho \nabla_\rho (S\, \boldsymbol{v}_\mu) + \mathcal{O}(\alpha^2). \tag{5.3}$$

## 5.4 Case I: No Metric Expansion

One may consistently set $a(t) \equiv 1$ (or constant) if spatial sections of the plenum are stationary in the lamphrodynamic gauge. Then the standard cosmic redshift term disappears, leaving

$$\frac{1}{\omega}\frac{d\omega}{d\tau} = -\alpha\, k^\mu k^\rho \nabla_\rho (S\, \boldsymbol{v}_\mu). \tag{5.4}$$

Integrating along the null geodesic from emission $(e)$ to observation $(o)$ gives

$$1 + z = \frac{\omega(e)}{\omega(o)} = \exp\left(\alpha \int_{\gamma_{e\to o}} k^\mu k^\rho \nabla_\rho (S\, \boldsymbol{v}_\mu)\, d\tau\right). \tag{5.5}$$

Thus,

$$z = \exp\left(\alpha \int_\gamma k^\mu k^\rho \nabla_\rho (S\, \boldsymbol{v}_\mu)\, d\tau\right) - 1.$$

Defining the lamphrodynamic integrand

$$f(S, \boldsymbol{v}) := \alpha\, k^\mu k^\rho \nabla_\rho (S\, \boldsymbol{v}_\mu), \tag{5.6}$$

we recover the expression used in heuristic discussions of RSVP cosmology:

$$z = \int_\gamma f(S, \boldsymbol{v})\, d\tau$$

to lowest order in $f$.

## 5.5 Comparison with Standard Hubble Law

For small $z$,

$$z \approx \alpha \int_\gamma k^\mu k^\rho \nabla_\rho (S\, \boldsymbol{v}_\mu)\, d\tau.$$

Let $d$ denote the comoving distance along the null ray and assume that $S$ and $\boldsymbol{v}$ vary slowly on cosmological timescales. Then

$$z \approx \alpha\,(\nabla S \cdot \boldsymbol{v})\,d/c,$$

suggesting an effective Hubble constant

$$H_{\text{eff}} := \alpha\,(\nabla S \cdot \boldsymbol{v}).$$

In this interpretation, cosmic redshift is governed not by $a(t)$ but by lamphrodynamic entropy flux along null trajectories.

## 5.6 HigherOrder Corrections

Including higherorder terms in $\Delta^{\lambda}_{\mu\nu}$ and accounting for the evolution of $S$ and $\boldsymbol{v}$ with cosmic time, we obtain

$$z = H_{\text{eff}}\frac{d}{c} + \gamma_2\left(\frac{d}{c}\right)^2 + \gamma_3\left(\frac{d}{c}\right)^3 + \cdots,$$

with coefficients $\gamma_i$ determined by lamphrodynamic couplings and cosmic entropy production history. Unlike ΛCDM, the expansion history is replaced by entropy history, leading to distinct observational signatures (see Chapter 9).

## 5.7 DistanceRedshift Relation

Inverting $z(d)$ yields a nontrivial distanceredshift relation. For small $z$,

$$d \approx \frac{c}{H_{\text{eff}}}z - \frac{\gamma_2 c}{H_{\text{eff}}^3}z^2 + \mathcal{O}(z^3),$$

which should be compared with the standard ΛCDM luminosity distanceredshift relation. Observational consequences are treated in detail in Chapter 9.

## 5.8 Discussion

Lamphrodynamic redshift provides an alternative explanation for cosmological redshifts that does not require metric expansion. The key observable predictions lie in deviations from the standard Hubble law at moderate redshifts and in the detailed luminosity distance relations. In the next chapter (Chapter 5), we turn to structure formation and the implications of lamphrodynamics for the growth of density perturbations.

# Chapter 6

# Structure Formation

## 6.1 Overview

Structure formation provides some of the most stringent observational tests of cosmological models. In , the growth of density perturbations is driven by gravitational instability sourced by cold dark matter in an expanding background. In RSVP, entropy gradients and lamphrodynamic stress contribute additional, potentially dominant, sources of effective gravitational attraction (or repulsion) even in the absence of metric expansion.

In this chapter we derive the linear growth equations for density perturbations in RSVP, focusing on homogeneous and isotropic backgrounds with lamphrodynamic entropy density $S(t)$ and effective lamphrodynamic equation of state $w_{\mathrm{lam}}$. We then obtain modified growth rates and matter power spectra, and outline observational constraints.

## 6.2 Perturbation Variables

Let $\rho(t, \mathbf{x})$ denote the matter density and

$$\delta(t, \mathbf{x}) := \frac{\rho(t, \mathbf{x}) - \bar{\rho}(t)}{\bar{\rho}(t)}$$

the linear density contrast relative to the homogeneous background $\bar{\rho}(t)$. Similarly, we define the lamphrodynamic perturbation

$$\delta_{\mathrm{lam}} := \frac{\rho_{\mathrm{lam}} - \bar{\rho}_{\mathrm{lam}}}{\bar{\rho}_{\mathrm{lam}}},$$

with $\rho_{\mathrm{lam}}$ given by (**??**) in **??**. Velocity perturbations follow standard notation.

## 6.3 Modified Poisson Equation (Weak–Field Limit)

In the Newtonian gauge and weak–field regime, one may define an effective gravitational potential $\Phi_{\mathrm{eff}}$ via

$$\nabla^2 \Phi_{\mathrm{eff}} = 4\pi G(\rho + \rho_{\mathrm{lam}}).$$

This represents the leading lamphrodynamic correction to the Newtonian potential. More precisely, in the covariant formalism the modified Einstein equations yield

$$\nabla^2 \Phi_{\text{eff}} = 4\pi G \rho_{\text{eff}}, \qquad \rho_{\text{eff}} := \rho + \rho_{\text{lam}},$$

up to gauge and relativistic corrections.

*Remark* 6.1. Lamphrodynamic energy density $\rho_{\text{lam}}$ may play the role of an effective dark–matter component driving structure formation.

## 6.4 Linear Growth Equation

Following standard perturbation theory with the modified Poisson equation, the density contrast satisfies

$$\ddot{\delta} + 2\frac{\dot{a}}{a}\dot{\delta} - 4\pi G \rho_{\text{eff}}\, \delta = \mathcal{L}_{\text{lam}}, \tag{6.1}$$

where $\mathcal{L}_{\text{lam}}$ collects lamphrodynamic source terms arising from perturbations of $S$ and $\Phi$. In the simple case of comoving lamphrodynamics (no lamphrodynamic velocity perturbation), $\mathcal{L}_{\text{lam}}$ can be approximated as

$$\mathcal{L}_{\text{lam}} \approx -4\pi G \delta_{\text{lam}}\, \rho_{\text{lam}}.$$

When $a(t) \equiv 1$, the "Hubble drag" term $2\dot{a}/a$ vanishes and the growth equation simplifies to

$$\ddot{\delta} - 4\pi G \rho_{\text{eff}}\, \delta = \mathcal{L}_{\text{lam}}. \tag{6.2}$$

Growth of perturbations can then proceed even without metric expansion, driven by lamphrodynamic sources.

## 6.5 Lamphrodynamic Growth Rate

Assuming a power–law solution $\delta \propto t^n$ in the matter–dominated epoch and using (5.2), we find

$$n(n-1) - 4\pi G \rho_{\text{eff}} = 0,$$

so the effective growth exponent is

$$n = \frac{1}{2}\left(1 \pm \sqrt{1 + 16\pi G \rho_{\text{eff}} t^2}\right).$$

Large $\rho_{\text{eff}} t^2$ yields *enhanced growth* relative to , potentially resolving certain small–scale structure tensions or producing novel signatures in the matter power spectrum.

## 6.6 Matter Power Spectrum

Define the matter power spectrum $P(k)$ as usual:

$$\langle \delta_{\mathbf{k}} \delta_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} + \mathbf{k}')\, P(k).$$

Lamphrodynamic contributions modify the transfer function $T(k)$ via

$$P_{\rm RSVP}(k) = P_{\rm prim}(k)\, T_{\rm RSVP}(k)^2,$$

where $T_{\rm RSVP}(k)$ encodes the modified growth rate and scale–dependence introduced by entropy gradients and lamphrodynamic pressure. In simple models with constant $w_{\rm lam}$,

$$T_{\rm RSVP}(k) \sim T(k)\left[1 + \epsilon_{\rm lam}(k)\right],$$

with $\epsilon_{\rm lam}(k)$ depending on $\alpha_i, \beta_i$ and the cosmic entropy history.

## 6.7 Comparison with

Compared with , RSVP may predict:

1. Enhanced growth at intermediate redshifts (no "Hubble drag").
2. Distinct scale–dependence from entropy and lamphrodynamic pressure.
3. Modified matter power spectrum normalization without cold dark matter.

These predictions can be tested against galaxy surveys, BAO, and weak lensing data (see Chapter 8).

## 6.8 Observational Constraints

Current observations of $P(k)$ place tight constraints on deviations from . Determining whether RSVP can reproduce the observed power spectrum requires detailed modeling of lamphrodynamic parameters $\alpha_i, \beta_i$ and the time evolution of $S$. This remains an open and promising area of phenomenological research.

## 6.9 Discussion

Lamphrodynamic fields can drive structure formation even in the absence of metric expansion, potentially providing a unified explanation for dark–matter–like effects at linear and non–linear scales. Next we turn to the interpretation of dark energy and dark matter within RSVP, focusing on galaxy rotation curves, cluster dynamics, and weak lensing.

# Chapter 7

# Dark Matter and Dark Energy in RSVP

## 7.1 Overview

A major motivation for the RSVP cosmology is the possibility of explaining apparent dark–matter and dark–energy phenomena as emergent consequences of lamphrodynamic entropy and flux, without introducing new fundamental particles or fields beyond $(\Phi, \boldsymbol{v}, S)$. This chapter develops this idea quantitatively, focusing on galaxy rotation curves, cluster mass profiles, weak lensing, and cosmic acceleration.

## 7.2 Lamphrodynamic Contributions to Effective Density

Recall from **??** in **??** that the lamphrodynamic energy density $\rho_{\text{lam}}$ arises from entropy production and lamphrodynamic couplings. In homogeneous cosmology, $\rho_{\text{lam}}$ may drive cosmic acceleration with effective equation of state $w_{\text{lam}} \approx -1$; in inhomogeneous settings, it can mimic cold dark matter by enhancing gravitational attraction.

*Remark* 7.1 (Unification possibility). A single lamphrodynamic field $S$ may account for both dark–energylike acceleration and dark–matterlike clustering, depending on the regime and scale.

## 7.3 Galaxy Rotation Curves

Consider a spherically symmetric galaxy with baryonic mass $M(r)$. In Newtonian gravity, the circular velocity satisfies

$$v^2(r) = \frac{GM(r)}{r}.$$

In RSVP, lamphrodynamic contributions modify the effective potential $\Phi_{\text{eff}}$, yielding

$$v^2_{\text{RSVP}}(r) = \frac{G}{r}\left[M(r) + M_{\text{lam}}(r)\right],$$

where $M_{\mathrm{lam}}(r)$ is the effective lamphrodynamic mass enclosed within radius $r$, defined by

$$M_{\mathrm{lam}}(r) := \int_0^r 4\pi s^2 \, \rho_{\mathrm{lam}}(s) \, ds.$$

Flat rotation curves at large $r$ arise naturally if $\rho_{\mathrm{lam}}$ falls slower than $1/r^2$ or approaches a constant asymptotically.

## 7.4 Cluster Mass Profiles

Galaxy clusters require substantial dark matter in  for dynamical stability. In RSVP, a lamphrodynamic energy density $\rho_{\mathrm{lam}}$ adding to $\rho$ may account for the inferred total mass without new particles. X–ray and gravitational lensing measurements of cluster profiles place strong constraints on the radial dependence of $\rho_{\mathrm{lam}}$; viable lamphrodynamic models must reproduce observed mass distributions.

## 7.5 Weak Lensing

Weak lensing probes the projected mass distribution along the line of sight. Lamphrodynamic energy density contributes to the lensing convergence via

$$\kappa_{\mathrm{RSVP}} \sim \int_{\mathrm{LOS}} (\rho + \rho_{\mathrm{lam}}) \, ds.$$

Matching observed shear maps therefore constrains the combination $\rho_{\mathrm{lam}} + \rho$ rather than $\rho$ alone. Current weak–lensing surveys may already limit $\rho_{\mathrm{lam}}$ in certain regimes; detailed phenomenology remains to be carried out.

## 7.6 Cosmic Acceleration Without $\Lambda$

From the effective Friedmann equation (**??**), cosmic acceleration occurs when

$$\rho_{\mathrm{lam}} + \rho + \frac{\Lambda}{8\pi G} \text{ is sufficiently large}$$

with effective negative pressure $p_{\mathrm{lam}} \approx -\rho_{\mathrm{lam}}$. If $\Lambda = 0$, lamphrodynamic entropy alone may drive acceleration, provided $\dot{S}$ is large enough at late times.

## 7.7 Interpretation

Lamphrodynamic entropy and flux can mimic both dark matter and dark energy, potentially unifying these dark components in a single framework. Whether the RSVP predictions match observed rotation curves, cluster dynamics, and cosmological acceleration quantitatively depends on the detailed time and space dependence of $S$ and its couplings $\alpha_i, \beta_i$.

## 7.8 Discussion

While RSVP offers an appealing conceptual unification of the dark sector, significant work remains to fully develop and test lamphrodynamic phenomenology against observational data. In subsequent chapters (notably Chapters 7 to 9) we confront RSVP with observations, beginning with the cosmic microwave background.

# Part III

# Observational Confrontation

# Chapter 8

# Cosmic Microwave Background

## 8.1 Overview

The cosmic microwave background (CMB) provides one of the most precise tests of cosmological models. In the RSVP framework, temperature anisotropies arise from both gravitational potential fluctuations and lamphrodynamic entropy inhomogeneities. The goal of this chapter is to derive the angular power spectrum $C_\ell$ in RSVP, and to compare with observational data, notably the Planck 2018 results.

## 8.2 Boltzmann Equation for CMB Photons

CMB photons propagate through the lamphrodynamic plenum according to a modified Boltzmann equation,

$$\frac{df}{d\lambda} = \mathcal{C}[f] + \mathcal{L}_{\mathrm{lam}}[f], \tag{8.1}$$

where $\mathcal{C}[f]$ denotes collision terms (e.g. Thomson scattering) and $\mathcal{L}_{\mathrm{lam}}[f]$ encodes lamphrodynamic effects due to $(\Phi, \boldsymbol{v}, S)$. For scalar perturbations, $\mathcal{L}_{\mathrm{lam}}$ contributes additional potential and entropy terms influencing photon trajectories.

## 8.3 Sachs–Wolfe and Integrated Sachs–Wolfe Effects

Anisotropies in the CMB temperature receive contributions from the Sachs–Wolfe (SW) and integrated Sachs–Wolfe (ISW) effects. In RSVP,

$$\left.\frac{\delta T}{T}\right|_{\mathrm{SW}} = \frac{1}{3}\Phi_{\mathrm{eff}}, \tag{8.2}$$

where $\Phi_{\mathrm{eff}}$ includes lamphrodynamic corrections to the gravitational potential. The ISW effect arises from time variation of $\Phi_{\mathrm{eff}}$ along the line of sight. A detailed derivation yields

$$\left(\frac{\delta T}{T}\right)_{\mathrm{ISW}} = \int_{\mathrm{LOS}} \dot{\Phi}_{\mathrm{eff}}\, d\eta, \tag{8.3}$$

which depends on the lamphrodynamic source terms $S$ and $\boldsymbol{v}$.

## 8.4 Angular Power Spectrum

The angular power spectrum in RSVP is defined by

$$C_\ell := \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} |\langle a_{\ell m} \rangle|^2,$$

where $a_{\ell m}$ are the sphericalharmonic coefficients of the temperature anisotropy. The lamphrodynamic enhancements alter the transfer functions, leading to modifications of the acoustic peak structure and Silk damping tail.

*Remark* 8.1 (Peak shifts). Lamphrodynamic contributions can shift both peak positions and amplitudes, depending on the scale–dependence of $\rho_{\mathrm{lam}}$ and $p_{\mathrm{lam}}$ during recombination.

## 8.5 Comparison with Planck Data

Matching the RSVP $C_\ell$ with PLANCK 2018 data provides stringent constraints on lamphrodynamic parameters. A least–squares fit in the space of $(\alpha_i, \beta_i)$ yields best–fit constraints and confidence intervals. Preliminary analysis indicates that certain classes of lamphrodynamic models can match current data, but full analysis requires numerical Boltzmann solvers (see Chapter D).

## 8.6 Discussion

The CMB provides some of the strongest constraints on RSVP. Matching the observed acoustic peaks and damping tail is challenging but feasible for suitable lamphrodynamic couplings. Further development of numerical Boltzmann codes tailored to RSVP is needed for definitive conclusions.

# Chapter 9

# Large–Scale Structure

## 9.1 Overview

Large–scale structure (LSS) measurements probe the distribution of matter at cosmological scales. In RSVP, the growth of structure is governed by both gravitational and lamphrodynamic effects, leading to potentially distinct predictions for the matter power spectrum, baryon acoustic oscillations (BAO), and weak lensing.

## 9.2 Matter Power Spectrum

The matter power spectrum $P(k)$ encodes the statistical distribution of density fluctuations as a function of wavenumber $k$. In RSVP, one obtains modified linear growth factors and transfer functions, resulting in

$$P_{\mathrm{RSVP}}(k) = |T_{\mathrm{RSVP}}(k)|^2 P_{\mathrm{prim}}(k).$$

The transfer function $T_{\mathrm{RSVP}}(k)$ depends on the lamphrodynamic couplings and can differ from the  transfer function, especially on large scales.

## 9.3 Baryon Acoustic Oscillations

BAO provide a standard ruler for cosmological distances. Lamphrodynamic effects alter both the sound horizon and the growth of baryonic oscillations, potentially shifting the BAO peak. Comparing BAO in RSVP with galaxy surveys constrains deviations from .

## 9.4 Weak Lensing

Weak lensing measurements probe the projected mass distribution along the line of sight. In RSVP, lamphrodynamic density contributes to the convergence $\kappa$, as in Chapter 6. Combining lensing with BAO and galaxy clustering improves constraints on $\rho_{\mathrm{lam}}$ and $p_{\mathrm{lam}}$.

## 9.5 Discussion

Large–scale structure provides important constraints on RSVP, though significant degeneracies with standard cosmological parameters remain. Future surveys may help break these degeneracies and provide stronger tests of lamphrodynamic cosmology.

# Chapter 10

# Supernovae and Standard Candles

## 10.1 Overview

Type Ia supernovae serve as standardizable candles for measuring cosmic distances. In RSVP, the distance–redshift relation is modified by the lamphrodynamic contributions to redshift and brightness, leading to potential deviations from  predictions.

## 10.2 Distance–Redshift Relation

The luminosity distance in RSVP is defined by a modified integral over the expansion history, incorporating lamphrodynamic corrections:

$$d_L(z) = (1+z)\int_0^z \frac{dz'}{H_{\text{eff}}(z')},$$

where $H_{\text{eff}}$ includes lamphrodynamic contributions. Fitting

$$d_L(z_i)$$

to observational data yields constraints on the parameter space of RSVP.

## 10.3 Hubble Diagram Analysis

The Hubble diagram plots distance modulus versus redshift. Lamphrodynamic corrections can shift the curve compared to , especially at high $z$. A rigorous statistical comparison is needed to determine whether RSVP fits supernova data without a cosmological constant.

## 10.4 Systematic Uncertainties

Systematic uncertainties from supernova calibration, dust extinction, and survey selection effects apply to both RSVP and standard cosmology. A careful error analysis is necessary to assess the significance of any deviations.

## 10.5 Discussion

Supernova observations provide valuable constraints on cosmic acceleration, but interpretation requires careful treatment of systematics. Integrated analysis with CMB and LSS data offers a more complete assessment of lamphrodynamic cosmology.

Part IV: Extreme Regimes and Tests

# Part IV

# Extreme Regimes and Tests

# Chapter 11

# Strong–Field Tests

## 11.1 Overview

The observational success of general relativity in strong–field regimes provides an essential benchmark for any alternative or emergent theory of gravity. In this chapter we analyze RSVP predictions for compact objects, gravitational waves, and the causal structure of strongly lamphrodynamic regions. We pay particular attention to the question of whether classical black holes exist in RSVP, and if so, how they differ from their general–relativistic counterparts.

## 11.2 Lamphrodynamic Compact Objects

In RSVP, the structure of a compact object is determined not only by its mass–energy density but also by the dynamics of the lamphrodynamic fields $(\Phi, \boldsymbol{v}, S)$. The effective stress–energy tensor contributing to the gravitational field is

$$T_{ab}^{\text{eff}} = T_{ab}^{\text{matter}} + T_{ab}^{\text{lam}}, \tag{11.1}$$

where $T_{ab}^{\text{lam}}$ arises from lamphrodynamic gradients and entropy flux. In spherical symmetry, the generalized Tolman–Oppenheimer–Volkoff equations involve additional terms from $S$ and $\rho_{\text{lam}}$, leading to modified compactness bounds.

## 11.3 Black Holes in RSVP?

A significant question is whether classical black holes, with event horizons and singularities, exist within RSVP. Lamphrodynamic transport and entropy smoothing provide mechanisms that may prevent curvature singularities from forming. Preliminary analysis suggests that strongly lamphrodynamic regions undergo an entropy–driven "decompression," reducing curvature and potentially avoiding singularity formation.

[No lamphrodynamic singularities] Under physically reasonable assumptions on entropy production and diffusion, spacetime singularities are avoided in generic dynamical collapse scenarios.

This conjecture requires further analysis; see **??** for related discussion in the PDE context.

## 11.4 Neutron Stars

The structure equations for neutron stars in RSVP contain lamphrodynamic pressure contributions. Depending on $\alpha_2$ and $\beta_2$ (see Chapter 6), the maximum neutron star mass and radius may differ from general relativity, leading to potentially observable deviations in neutron star mass–radius relationships.

## 11.5 Gravitational Waves

Gravitational waves in RSVP arise from metric perturbations influenced by lamphrodynamic stresses. The propagation speed and polarization structure may differ slightly from general relativity, depending on the coupling parameters. Observations of gravitational waves from compact binaries provide constraints on these couplings.

*Remark* 11.1 (Wave speed constraint)*.* Observations from GW170817 constrain deviations in the gravitational wave speed to be extremely small, limiting certain lamphrodynamic parameter ranges.

## 11.6 Event Horizon Telescope Constraints

Observations from the Event Horizon Telescope (EHT) provide tests of strong gravity near supermassive compact objects. Lamphrodynamic modifications could alter shadow size and photon ring structure. Current EHT data are consistent with general relativity but do not rule out small lamphrodynamic effects.

## 11.7 Discussion

Strong–field observations place important constraints on lamphrodynamic gravity, but the full parameter space remains open. Improved modeling of compact objects and gravitational wave emission in RSVP is needed to assess consistency with observations.

# Chapter 12

# Falsifiability and Discriminating Observations

## 12.1 Overview

A scientific theory must be falsifiable: it should make predictions that could, in principle, be shown false by experiment. In this chapter, we outline potential observational signatures that would falsify RSVP, as well as several predictions that distinguish it from  and general relativity.

## 12.2 Unique Predictions

RSVP makes several predictions not shared by . For example, the entropic contribution to cosmic acceleration may obviate the need for a cosmological constant. Specific signatures include deviations in the distance–redshift relation at high redshift, altered growth rates of structure, and modifications to CMB anisotropies.

## 12.3 Future Experiments

Future observations that could test RSVP include next–generation CMB measurements, galaxy surveys, weak lensing studies, and gravitational wave observations. Planned missions and observatories offer opportunities to observe lamphrodynamic effects directly.

## 12.4 Testability Roadmap

We propose a roadmap identifying key experiments and measurements that could either support or falsify RSVP. This roadmap emphasizes independent tests using CMB, LSS, and supernova data, combined with strong–field observations, to break parameter degeneracies and test core lamphrodynamic predictions.

## 12.5 Conclusion

Falsifiability is crucial for scientific progress. While RSVP offers a promising alternative to , it must confront observational data rigorously. Only through careful comparison with current and future observations can the validity of the lamphrodynamic paradigm be assessed.

# Appendix A

# Cosmological Perturbation Theory

This appendix provides the technical background for linear cosmological perturbation theory in the context of the lamphrodynamic fields of RSVP. Our primary objective is to formulate perturbations of the triple $(\Phi, \boldsymbol{v}, S)$ on a homogeneous and isotropic background, to identify gauge–invariant combinations, and to derive the linearized evolution equations used in Chapters 4–7 of this volume. The presentation parallels standard FLRW cosmology (Weinberg, Dodelson), but the dynamical variables differ substantially.

## A.1 Background FLRW Geometry

Let $(\mathcal{M}, g)$ be a spatially homogeneous and isotropic spacetime with metric

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2 \gamma_{ij} \mathrm{d}x^i \mathrm{d}x^j, \tag{A.1}$$

where $a(t)$ is the scale factor and $\gamma_{ij}$ is a maximally symmetric spatial metric of constant curvature $k \in \{-1, 0, +1\}$. In RSVP, the lamphrodynamic fields

$$(\Phi(t),\ \boldsymbol{v}(t),\ S(t))$$

satisfy the background lamphrodynamic equations derived in **??**. Throughout this appendix, an overbar denotes background quantities, e.g. $\overline{\Phi}(t)$.

## A.2 Perturbation Ansatz

We write the perturbed fields as

$$\begin{aligned} \Phi(t, \mathbf{x}) &= \overline{\Phi}(t) + \delta\Phi(t, \mathbf{x}), \\ \boldsymbol{v}_i(t, \mathbf{x}) &= \overline{\boldsymbol{v}}_i(t) + \delta v_i(t, \mathbf{x}), \\ S(t, \mathbf{x}) &= \overline{S}(t) + \delta S(t, \mathbf{x}), \end{aligned} \tag{A.2}$$

where $\delta\Phi$, $\delta v_i$, and $\delta S$ are small fluctuations.

For the metric, we adopt the standard scalar–vector–tensor (SVT) decomposition. Let

$$\mathrm{d}s^2 = -(1+2A)\mathrm{d}t^2 + 2a(t)\big(B_{,i} + B_i\big)\mathrm{d}t\,\mathrm{d}x^i + a(t)^2\big[(1+2H_L)\gamma_{ij} + 2H_{T\,ij}\big]\mathrm{d}x^i\mathrm{d}x^j, \tag{A.3}$$

where scalar, vector, and tensor parts satisfy the usual transverse and traceless conditions:

$$B^i{}_{,i} = 0, \quad H_{T\,ij}{}^{,j} = 0, \quad H_{T\,i}{}^{i} = 0.$$

## A.3 Gauge Transformations

An infinitesimal coordinate transformation $x^\mu \mapsto x^\mu + \xi^\mu$ induces transformations of both the metric perturbations and the lamphrodynamic fields. Decompose $\xi^\mu = (\xi^0, \xi^{,i} + \xi^i_\perp)$ with $\xi^i_{\perp,i} = 0$.

**Lemma A.1** (Gauge Transformations of Lamphrodynamic Perturbations)**.** *Under an infinitesimal coordinate transformation,*

$$\delta\Phi \mapsto \delta\Phi - \dot{\bar{\Phi}}\,\xi^0, \qquad \delta S \mapsto \delta S - \dot{\bar{S}}\,\xi^0, \qquad \delta v_i \mapsto \delta v_i - \dot{\bar{v}}_i\,\xi^0,$$

*where overdot denotes derivative with respect to cosmological time.*

*Proof.* Direct computation from the transformation law of scalar and vector fields under infinitesimal diffeomorphisms, using background homogeneity. □

Gauge–invariant combinations follow by combining these with metric perturbations in analogy with the Bardeen potentials. For instance, define

$$:= \delta\Phi - \dot{\bar{\Phi}}\,\frac{1}{a}\int^t aA\,\mathrm{d}t',$$

with analogous definitions for  and $_i$. Tensor modes $H_{T\,ij}$ are already gauge–invariant.

## A.4 Linearized Lamphrodynamic Equations

The lamphrodynamic field equations (Volume I, Chapters 5–8) can be expanded to first order about the FLRW background. Denote $\mathcal{E}[\Phi, \boldsymbol{v}, S, g] = 0$ schematically. Linearizing yields

$$\mathcal{L}_{\text{lin}}\begin{pmatrix}\delta\Phi \\ \delta v_i \\ \delta S \\ A, B, H_L, H_{T\,ij}\end{pmatrix} = 0,$$

where $\mathcal{L}_{\text{lin}}$ is a coupled, first–order (in time) differential operator. For scalar modes, a representative pair of equations takes the form

$$\delta\dot{\Phi} + \alpha_1 H\,\delta\Phi + \alpha_2\,\delta S + \alpha_3 A = 0, \tag{A.4}$$

$$\delta\dot{S} + \beta_1 H\,\delta S + \beta_2\,\delta\Phi + \beta_3\nabla^2\delta\Phi = 0, \tag{A.5}$$

where $H = \dot{a}/a$ is the Hubble parameter and $\alpha_i, \beta_i$ are coefficients determined by the background solution and the coupling structure in the lamphrodynamic action.

The full linearized system includes similar equations for $\delta v_i$ and couples to the metric perturbations via the modified Einstein equations of Chapter 1.

## A.5 Scalar, Vector, and Tensor Modes

Decompose perturbations of $\delta v_i$ as

$$\delta v_i = \delta v_{,i} + \delta v_i^T, \qquad \delta v_i^{T,i} = 0. \tag{A.6}$$

Scalar modes couple to $(\delta\Phi, \delta S)$, vector modes evolve independently at linear order, and tensor modes obey the modified gravitational wave equation

$$\ddot{H}_{T\,ij} + 3H\,\dot{H}_{T\,ij} - \frac{\nabla^2}{a^2} H_{T\,ij} = \Theta_{ij}, \tag{A.7}$$

where $\Theta_{ij}$ encodes entropy–induced anisotropic stress (see Chapter 1).

## A.6 Power Spectra and Transfer Functions

Define Fourier transforms by

$$\delta\Phi(t, \mathbf{k}) = \int \mathrm{d}^3x\, e^{-i\mathbf{k}\cdot\mathbf{x}}\, \delta\Phi(t, \mathbf{x}),$$

and similarly for $\delta S$ and $\delta v_i$. The auto– and cross–power spectra are defined by

$$\langle \delta\Phi(\mathbf{k})\, \delta\Phi(\mathbf{k}') \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') P_{\Phi\Phi}(k),$$

with analogous definitions for $P_{SS}$, $P_{\Phi S}$, and the velocity cross spectra.

The transfer functions $T_\Phi(k,t)$, $T_S(k,t)$, and $T_v(k,t)$ are defined by

$$\delta\Phi(k,t) = T_\Phi(k,t)\, \delta\Phi(k, t_i),$$

etc., where $t_i$ denotes an early initial time. These functions enter predictions for structure formation (Chapter 6) and CMB anisotropies (Chapter 8).

## A.7 Initial Conditions

Initial conditions may be specified via quantum fluctuations in the pre–lamphrodynamic epoch (Volume III, Chapter 10) or by phenomenological ansätze analogous to adiabatic and isocurvature modes. For the adiabatic case,

$$\frac{\delta S}{\dot{S}} = \frac{\delta\Phi}{\dot{\Phi}} = \frac{\delta v_i}{\dot{v}_i},$$

generalizing the standard condition that all components share the same initial phase.

## A.8 Remarks and Further Reading

Standard introductions to cosmological perturbation theory include Weinberg (*Cosmology*) and Dodelson (*Modern Cosmology*). For the tensor decomposition and gauge–invariant formalism, see also Kodama–Sasaki and Mukhanov. The adaptation to RSVP follows by linearizing the lamphrodynamic field equations and replacing the standard stress–energy sources with the entropy–vector couplings of Chapters 1–3.

# Appendix B

# Statistical Methods for Cosmology

This appendix summarizes statistical techniques used to confront theoretical cosmology with observational data, with emphasis on their application to the lamphrodynamic dynamics of RSVP. A rigorous treatment of full Bayesian inference, likelihood functions, and model comparison is necessary for the quantitative tests developed in Chapters 8–10. The following review parallels standard references (e.g. Dodelson, Trotta) but adapts the presentation to the specific parametrization of RSVP.

## B.1 Parameters and Priors

Let $\theta$ denote a vector of cosmological parameters, encompassing both standard parameters (e.g. present–day entropy density $S_0$, Hubble parameter $H_0$, baryon fraction) and lamphrodynamic parameters (e.g. coupling coefficients appearing in the lamphrodynamic field equations). The posterior distribution over parameters given data $D$ is

$$P(\theta \mid D) \propto \mathcal{L}(D \mid \theta)\, \pi(\theta), \tag{B.1}$$

where $\mathcal{L}$ is the likelihood and $\pi$ is the prior. Choice of prior $\pi(\theta)$ should be explicitly justified; standard practice includes uniform, Jeffreys, or weakly informative priors, depending on physical interpretation and parameter scaling.

## B.2 Likelihoods

For Gaussian–distributed data with covariance $\mathsf{C}(\theta)$ and mean $\mu(\theta)$,

$$\mathcal{L}(D \mid \theta) \propto |\mathsf{C}(\theta)|^{-1/2} \exp\Big[-\tfrac{1}{2}(D - \mu(\theta))^{\mathsf{T}} \mathsf{C}(\theta)^{-1} (D - \mu(\theta))\Big], \tag{B.2}$$

where $D$ may include CMB power spectra $C_\ell$, matter power spectra $P(k)$, and Type Ia supernova distance moduli $\mu(z)$. In RSVP, the dependence of $\mu(\theta)$ and $\mathsf{C}(\theta)$ on lamphrodynamic parameters enters through modified growth functions and redshift relations (see Chapters 5–7).

## B.3 Maximum Likelihood and $\chi^2$ Statistics

For independent Gaussian errors, maximizing $\mathcal{L}$ is equivalent to minimizing $\chi^2$:

$$\chi^2(\theta) = \sum_i \frac{\big(D_i - \mu_i(\theta)\big)^2}{\sigma_i^2}. \tag{B.3}$$

Confidence intervals follow from the usual $\Delta\chi^2$ prescription, provided regularity conditions are satisfied. In practice, the covariance matrix is often non–diagonal, requiring full matrix estimators and noise modeling.

## B.4 Bayesian Evidence and Model Comparison

Let $\mathcal{M}$ denote a model and $\mathcal{E}$ its Bayesian evidence,

$$\mathcal{E}(\mathcal{M}) = \int \mathcal{L}(D \mid \theta, \mathcal{M})\, \pi(\theta \mid \mathcal{M})\, \mathrm{d}\theta. \tag{B.4}$$

Model comparison proceeds by Bayes factors

$$B_{12} = \frac{\mathcal{E}(\mathcal{M}_1)}{\mathcal{E}(\mathcal{M}_2)}, \tag{B.5}$$

with conventional interpretive scales (e.g. Jeffreys scale). Evidence is especially important for comparing RSVP with ΛCDM, because additional parameters (entropy–vector couplings) alter the dimensionality of parameter space.

## B.5 Information Criteria

Information criteria provide approximate model comparison without computing full Bayesian evidence. The Akaike information criterion is

$$\mathrm{AIC} = 2k - 2\ln\hat{\mathcal{L}}, \tag{B.6}$$

where $k$ is the number of parameters and $\hat{\mathcal{L}}$ is the maximum likelihood. The Bayesian information criterion is

$$\mathrm{BIC} = k\ln N - 2\ln\hat{\mathcal{L}}, \tag{B.7}$$

where $N$ is the number of data points. Lower values indicate preferred models, penalizing parameter complexity.

## B.6 Sampling Methods

Sampling from the posterior (B.1) typically requires Markov–chain Monte Carlo (MCMC) or nested sampling algorithms. Standard approaches (Metropolis–Hastings, Hamiltonian Monte Carlo, `emcee`, `MultiNest`) can be adapted to RSVP by supplying numerically integrated transfer functions and redshift relations. Parameter covariance can be evaluated by examining chain autocorrelation times and convergence diagnostics (e.g. Gelman–Rubin statistics).

## B.7 Forecasting and Fisher Information

For parameter forecasting (e.g. next–generation CMB or large–scale structure surveys), the Fisher matrix is defined by

$$F_{\alpha\beta} = -\left\langle \frac{\partial^2 \ln \mathcal{L}}{\partial\theta_\alpha \partial\theta_\beta} \right\rangle. \tag{B.8}$$

Parameter uncertainties are approximated by

$$\sigma(\theta_\alpha) \approx \sqrt{(F^{-1})_{\alpha\alpha}}.$$

Fisher forecasting allows a quick estimate of the extent to which future surveys can constrain lamphrodynamic parameters.

## B.8 Goodness of Fit

Standard tests include reduced $\chi^2$, Kolmogorov–Smirnov statistics, residual analysis, and cross–validation of independent data sets (e.g. CMB and BAO). Given the extended parameter space of RSVP, it is prudent to examine potential degeneracies between lamphrodynamic and standard cosmological parameters (e.g. $H_0$, $\Omega_m$, spectral indices).

## B.9 Remarks and Further Reading

Comprehensive introductions include Trotta (*Bayes in Cosmology*) and Dodelson (*Modern Cosmology*). Recent data analysis pipelines for CMB, BAO, and supernovae provide practical examples of implementing parameter inference in extended cosmological models. Adaptations to RSVP involve modified theoretical predictions for $\mu(\theta)$ and $\mathsf{C}(\theta)$ and are discussed throughout Chapters 8–10.

# Appendix C

# Observational Data Sources

This appendix summarizes the principal observational data sources used throughout Volume II. Our aim is not to provide an exhaustive review of the observational literature, but to give sufficient detail for interpreting the quantitative comparisons performed in Chapters 8–10. We organize the discussion by data type: cosmic microwave background (CMB), large–scale structure (LSS), and supernovae (SNe). Additional references are provided at the end of the chapter.

## C.1 Cosmic Microwave Background

For the CMB temperature and polarization anisotropies, the principal data sets come from *Planck 2018*, including power spectra $C_\ell^{TT}$, $C_\ell^{TE}$, and $C_\ell^{EE}$. These are obtained from full–sky maps after foreground subtraction and masking. Noise properties, beam transfer functions, and sky cuts are supplied by the Planck Legacy Archive.

In RSVP, the primary modifications appear through:

- the lamphrodynamic evolution of scalar perturbations,
- entropy–vector couplings affecting anisotropic stress,
- modified late–time integrated Sachs–Wolfe contributions.

High–$\ell$ damping tails are sensitive to recombination physics, but the dominant lamphrodynamic signatures enter through low–$\ell$ multipoles and potential deviations in the acoustic peaks.

## C.2 Large–Scale Structure and BAO

For LSS, we rely on galaxy surveys including BOSS, eBOSS, DESI, and future surveys such as Euclid and LSST. The primary observables are the matter power spectrum $P(k)$, baryon acoustic oscillations (BAO), and weak lensing shear correlations.

Lamphrodynamic modifications arise from:

- altered linear growth factors,
- modified transfer functions for density perturbations,

- possible effective "dark–matter–like" behavior from entropy–vector dynamics.

Comparison with ΛCDM predictions requires careful statistical analysis (see Appendix G) and numerical integration (Appendix I).

## C.3 Galaxy Clusters and Weak Lensing

Galaxy clusters probe the nonlinear regime of structure formation. Mass estimates rely on X–ray temperatures, Sunyaev–Zel'dovich signals, and weak lensing measurements. In RSVP, cluster profiles provide sensitive tests of emergent–dark–matter behavior and potential deviations in the mass–concentration relation. Weak lensing convergence maps probe projected mass densities and are significantly influenced by modified geodesic equations (Chapter 3).

## C.4 Supernovae and Standard Candles

Type Ia supernovae provide direct constraints on the expansion history through distance moduli $\mu(z)$. The most comprehensive compilations are the Pantheon and Pantheon+ samples. In RSVP, the relation between redshift and luminosity distance is modified by entropy–induced redshifting mechanisms (Chapter 4). Systematic uncertainties include dust extinction, selection effects, and population drift.

## C.5 Cosmic Chronometers

Cosmic chronometers measure the expansion rate directly from relative ages of passively evolving galaxies. This approach provides model–independent constraints on $H(z)$ and is therefore particularly useful for models with modified redshift relations. The current precision is limited by stellar population modeling and metallicity effects.

## C.6 Future Surveys

Future surveys (Euclid, LSST, *Simons Observatory*, CMB–S4, LIGO–Voyager) will dramatically extend the volume and precision of observations. These are of particular relevance to RSVP due to:

- increased sensitivity to large–scale entropy modes,
- improved constraints on growth rates and anisotropic stress,
- ability to discriminate lamphrodynamic perturbations from modified gravity and dark sector models.

## C.7 Data Availability

Most observational data sets are publicly available:

- Planck Legacy Archive (CMB),
- SDSS/BOSS/eBOSS data releases (galaxy surveys),
- DESI data products (spectroscopic survey),
- Pantheon/Pantheon+ (supernovae),
- public weak lensing catalogs (DES, KiDS).

Where proprietary restrictions exist, we provide references and key summaries suitable for comparative analysis.

## C.8 Remarks and Further Reading

For detailed discussions of observational data, see the Planck 2018 papers, DESI survey documentation, and the Pantheon+ release papers. For weak lensing and cluster analyses, see recent DES, KiDS, and X–ray/SZ collaborations. Volume II uses these sources as benchmarks for the observational viability of the lamphrodynamic dynamics of RSVP.

# Appendix D

# Numerical Methods for Cosmological Evolution

This appendix summarizes numerical techniques relevant to solving the cosmological evolution equations derived in Part II of this volume. We emphasize methods that are adapted to the lamphrodynamic field equations and their interaction with entropy–vector dynamics. Our goal is to provide practical guidance for implementing numerical solutions, rather than to develop a comprehensive theory of numerical analysis.

## D.1 Discretization of Background Evolution

We begin with the homogeneous and isotropic lamphrodynamic equations derived in Chapter 4. Let $a(t)$ denote the scale factor and $\Psi(t) = (\Phi(t), \boldsymbol{v}(t), S(t))$ denote the spatially homogeneous fields. The coupled nonlinear ODE system takes the form

$$\frac{d}{dt}\Psi(t) = \mathcal{F}(\Psi(t), a(t))\,, \qquad \frac{d}{dt}a(t) = \mathcal{G}(\Psi(t), a(t))\,,$$

where $\mathcal{F}$ and $\mathcal{G}$ are determined by the lamphrodynamic action and its variational derivatives.

For time discretization, standard explicit and implicit Runge–Kutta methods may be used. Stability is improved by semi–implicit updates for $S(t)$, reflecting entropy dissipation. Adaptive step size control is recommended due to the stiff coupling between $\boldsymbol{v}$ and $S$ during early epochs.

## D.2 Linear Perturbation Equations

For linear perturbations, we decompose fields into Fourier modes $\delta\Psi_k(t)$ satisfying

$$\frac{d}{dt}\delta\Psi_k(t) = \mathcal{L}_k\, \delta\Psi_k(t)\,,$$

where $\mathcal{L}_k$ is the linearized operator. The dominant numerical issue is the scale dependence introduced by entropy–vector couplings, leading to nontrivial dispersion. Implicit schemes or operator splitting techniques reduce numerical stiffness.

Spectral methods (Chebyshev, Fourier) are natural for periodic domains. Finite–difference methods are viable on comoving grids, provided appropriate boundary conditions and resolution criteria are enforced.

## D.3 Mode Coupling and Transfer Functions

Transfer functions are computed by evolving perturbations from early times to the present. In RSVP, the lamphrodynamic contributions modify both scalar and vector channels. Numerical integration must account for:

- entropy–induced modification of gravitational potentials,
- cross–coupling terms in $\boldsymbol{v}$ and $S$,
- altered growth rates compared to ΛCDM.

Parallelization across $k$–modes is straightforward and recommended.

## D.4 Weak Lensing and Line–of–Sight Integrals

Weak lensing observables rely on line–of–sight integration of perturbations along null geodesics. In RSVP, null geodesics are modified by entropy–vector contributions (Chapter 3). Numerical integration should incorporate:

- modified geodesic equations,
- entropy–weighted optical depth,
- anisotropic stress contributions.

Adaptive sampling along the line of sight improves accuracy near sharp features in $S(t)$ and $\boldsymbol{v}(t)$.

## D.5 Boltzmann Solvers and Existing Software

Standard Boltzmann solvers (CAMB, CLASS) can be extended to include lamphrodynamic terms. This requires:

- modification of background equations,
- extension of perturbation modules to include $(\Phi, \boldsymbol{v}, S)$,
- additional source terms in line–of–sight integrators.

Validation against ΛCDM baselines is essential and provides a useful diagnostic for implementation errors.

## D.6 Numerical Relativity and Nonlinear Regimes

In strongly nonlinear regimes, numerical relativity methods become necessary. The lamphrodynamic field equations form a mixed system of hyperbolic–parabolic type, where entropy diffusion plays a stabilizing role. Discontinuous Galerkin methods and finite–volume schemes with shock capturing may be appropriate for studying entropic singularities.

## D.7 Parallelization and Performance

Efficient parallelization exploits:

- independent $k$–mode evolution,
- GPU acceleration for spectral transforms,
- domain decomposition for large–scale spatial grids.

Entropy diffusion reduces the stiffness of the system at late times, but may increase computational cost during early evolution.

## D.8 Remarks

The numerical implementation of RSVP remains an active area of development. Future work will combine Boltzmann solvers, numerical relativity tools, and cosmological data pipelines to provide full Bayesian inference for lamphrodynamic cosmology. We return to these questions in the concluding chapters of Volume II.

# Relativistic Scalar–Vector Plenum: Extensions, Interpretations, and New Directions

Flyxion

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# Preface

This volume is the third and final part of a three–volume monograph developing the Relativistic Scalar–Vector Plenum (RSVP) framework. Whereas Volume I established the mathematical foundations and Volume II developed the physical consequences and cosmological applications, the present volume explores broader extensions and interpretations. These include geometric models of computation, semantic field theories, spectral geometry, and philosophical implications.

Although the subject matter is more diverse and sometimes speculative, we aim to preserve a rigorous mathematical backbone wherever possible. This volume is written with a broader audience in mind, including computer scientists, philosophers of physics, and researchers interested in category theory, semantics, and emergent structures. We therefore include introductory discussions, conceptual remarks, and philosophical notes alongside the technical developments.

s volume is intentionally more exploratory than its predecessors. Nevertheless, we maintain a commitment to mathematical coherence: whenever speculative claims are made, we delineate their status and interpretive stance. In particular, parts of this volume concern the computational and semantic dimensions of RSVP, aspects which are not fully formalized but which we believe are necessary steps toward a genuinely unified field theory of physical, semantic, and cognitive processes.

We also take the opportunity in this volume to reflect more openly on the philosophical implications of the RSVP framework. Many of these implications were implicit in earlier chapters but remained unspoken, partly in order to preserve focus on the rigorous foundational program. Here, they receive explicit treatment, not as definitive conclusions but as directions for further inquiry. We hope this volume will encourage researchers from disparate fields to engage with RSVP in their own terms, and to contribute novel insights from the standpoint of geometry, computation, neuroscience, or philosophy.

**Structure of the Volume.** The volume is divided into seven parts. Part I develops the SpherePop calculus as a geometric model of computation. Part II reformulates L-systems as sigma models and explores semantic evolution. Part III applies lamphrodynamic ideas to distributed computation and CRDTs. Part IV develops the spectral and cymatic aspects of entropy-weighted geometry. Part V investigates twistor-theoretic interfaces. Part VI explores creative and philosophical implications. Part VII concludes with open problems and speculative extensions.

Finally, this volume includes an expanded glossary, extensive cross-references to Volumes I and II, and a unified bibliography, reflecting our hope that the trilogy can serve both as a research program and as a resource for future investigations.

# Notation and Conventions

Notation and conventions follow those established in Volume I, with additions where needed. We will introduce additional conventions in the appropriate chapters as necessary.

# Part I

# SpherePop Calculus

# Chapter 1

# SpherePop as Homotopy Theory

## 1.1 Motivation and Overview

The SpherePop calculus was introduced in preliminary form in [**Flyxion-SpherePop-v1**] as a geometric model of computation based on two primitive operations: *merge*, which combines geometric regions, and *collapse*, which abstracts internal geometric detail. In contrast to symbolic models such as lambda calculus, SpherePop employs continuous, topologically meaningful configurations, thereby providing a native interface between geometry, computation, and the lamphrodynamic fields of RSVP.

The guiding principle of this chapter is that configuration spaces in SpherePop naturally form $\infty$–groupoids, and that merge operations correspond to homotopy colimits in a suitably chosen model category. This identification provides both a rigorous categorical foundation for SpherePop and a conceptual bridge to the derived geometric structures developed in Volume I.

## 1.2 Configuration Spaces as $\infty$–Groupoids

Let  denote the category of geometric configurations (arising, for example, from finite unions of spherical regions in $\mathbb{R}^3$ or more general manifolds). Morphisms correspond to continuous deformations (isotopies) between configurations. Viewed up to homotopy,  assembles into an $\infty$–groupoid whose objects are configurations and whose $k$–morphisms represent $k$–fold homotopies.

**Definition 1.1.** A *SpherePop configuration* is a finite collection of embedded $n$–dimensional spheres $S^n \hookrightarrow M$, together with lamphrodynamic labels $(\Phi, \boldsymbol{v}, S)$ pulled back from $M$. Two configurations are considered equivalent if they are related by a lamphrodynamically admissible isotopy, i.e. a smooth homotopy preserving lamphrodynamic boundary conditions.

*Remark* 1.2. The lamphrodynamic fields $(\Phi, \boldsymbol{v}, S)$ are not merely decorations: they encode physical or semantic information associated to each region, and they influence admissible deformations. Thus  is enriched over the category of lamphrodynamic field configurations, linking computational and physical structure.

## 1.3 Homotopy Types and Derived Enhancements

The naive homotopy type of  fails to capture the derived structure needed for lamphrodynamic evolution. We therefore regard  as an *underived shadow* of a derived configuration stack , equipped with a $(-1)$–shifted symplectic structure inherited from the lamphrodynamic fields. This allows us to interpret merge operations as homotopy colimits in , thereby unifying geometric computation with the derived geometric core of RSVP.

**Theorem 1.3** (informal)**.** *Configuration spaces in SpherePop admit a derived enhancement whose mapping spaces are homotopy invariant and which carries a* $(-1)$*–shifted symplectic structure compatible with lamphrodynamic evolution.*

A precise construction of  requires the machinery of derived stacks and shifted symplectic geometry developed in Volume I, especially **????**. We will return to formal details in **??** below.

## 1.4 Merge as Homotopy Colimit

The central operation of SpherePop is the merge of two configurations along a common boundary. Formally, let $X$ and $Y$ be configurations with a shared subconfiguration $Z$. The intuitive merge is the glued configuration $X \cup_Z Y$. We show that this construction corresponds to a homotopy colimit in the derived configuration stack .

**Theorem 1.4** (Theorem 24.2, to be proved in full later)**.** *Let* $X \leftarrow Z \rightarrow Y$ *be a span of configurations with lamphrodynamically admissible embeddings. Then the SpherePop merge* $X \cup_Z Y$ *is canonically equivalent (up to higher homotopy) to the homotopy colimit of the span in .*

The proof of **??** will occupy a substantial portion of this chapter. The key idea is that lamphrodynamic admissibility ensures the compatibility of shifted symplectic forms across glueing boundaries, allowing the derived homotopy colimit to inherit the appropriate symplectic structure.

## 1.5 Outline of the Chapter

We proceed as follows:

1. We construct the derived configuration stack .
2. We establish lamphrodynamic admissibility criteria.
3. We define homotopy colimits in .
4. We prove **??** in full generality.
5. We provide worked examples illustrating merge operations.

## 1.6 Open Questions

The homotopy–theoretic interpretation of SpherePop raises several questions:

- What are the necessary and sufficient conditions for lamphrodynamic admissibility?
- Does  admit additional structure beyond $(-1)$–shifted symplectic geometry?
- Can we classify computational complexity in terms of derived invariants of ?

These questions point to deep connections between lamphrodynamics, homotopy theory, and computation, many of which remain unexplored.

# Chapter 2

# Collapse as Derived Truncation

## 2.1 Motivation

The second fundamental operation in the SpherePop calculus is *collapse*: an abstraction mechanism which replaces a geometric configuration by a simpler configuration in which certain internal degrees of freedom have been removed. Intuitively, collapse forgets fine–scale detail while preserving large–scale structure and boundary information. In the lamphrodynamic setting, collapse corresponds to an entropy–preserving coarse–graining of lamphrodynamic fields.

Heuristically, collapse should behave like an $\infty$–categorical truncation, removing higher homotopy levels while retaining essential boundary invariants. The central conjecture of this chapter asserts that collapse *is* such a truncation in the derived configuration stack .

## 2.2 Collapse as a Functor

Let $\kappa$ denote the SpherePop collapse operation acting on the (enriched) configuration category. Formally, collapse is a functor

$$\kappa\colon \longrightarrow,$$

equipped with a natural transformation $\kappa \Rightarrow$ generated by the embedding of collapsed configurations into their refinements. Intuitively, $\kappa(X)$ identifies lamphrodynamically degenerate internal structure in $X$ and retracts it to a simpler configuration which preserves entropic boundary conditions.

**Definition 2.1.** A collapse is *entropy–admissible* if it preserves the restriction of $S$ to $\partial X$, i.e. if $S|_{\partial X}$ pulls back along the collapse map. We require entropy–admissible collapse maps throughout.

*Remark* 2.2. Entropy–admissibility is the minimal requirement for collapse to represent a physically meaningful coarse–graining in lamphrodynamic evolution. Stronger admissibility conditions may be imposed to preserve additional lamphrodynamic invariants.

## 2.3 Connection to $\infty$–Categorical Truncation

Let $\tau_{\leq k}$ denote the $k$–truncation functor on an $\infty$–category. The guiding idea is that collapse removes higher–order lamphrodynamic fluctuations (as measured by derived obstruction theory), corresponding to truncation at some effective level $k$.

[Collapse as Derived Truncation] There exists an integer $k \geq 0$, depending on the lamphrodynamic regularity of $X$, such that

$$\kappa(X) \simeq \tau_{\leq k}(X)$$

in the derived configuration stack .

*Remark* 2.3. The value of $k$ depends on lamphrodynamic smoothness and boundary entropy. In particular, configurations with higher lamphrodynamic complexity require deeper truncation levels to preserve admissible boundary data.

## 2.4 Evidence for the Conjecture

Evidence for **??** arises from three sources:

1. *Derived obstruction theory.* Internal lamphrodynamic modes correspond to higher obstruction classes in the tangent complex of . Collapse eliminates modes associated to these obstructions, suggesting truncation of higher homotopy groups.

2. *Entropy monotonicity.* Collapse preserves boundary entropy while reducing internal entropy, consistent with truncation of non–boundary degrees of freedom.

3. *Computational abstraction.* In SpherePop–based computation, collapse produces coarser representations which retain semantic boundary structure but forget internal detail—again consistent with removing higher homotopy levels.

## 2.5 Partial Results

Although we cannot prove **??** in full generality, we can establish partial results in special cases.

**Proposition 2.4.** *If $X$ is lamphrodynamically regular and its entropy density $S$ is constant on connected components, then $\kappa(X)$ is equivalent to the $0$–truncation of $X$ in .*

*Sketch.* Under these assumptions, internal lamphrodynamic variations vanish, so higher obstruction classes in the tangent complex are trivial. Hence the derived structure is concentrated in degree 0, and collapse reduces to a 0–truncation. □

## 2.6 Boundary Invariants

A key requirement for collapse is the preservation of lamphrodynamic boundary conditions. These provide invariants under collapse, analogous to homotopy invariants under truncation.

**Definition 2.5.** A *boundary invariant* of $X$ is any invariant of the restricted lamphrodynamic field $(\Phi, \boldsymbol{v}, S)|_{\partial X}$.

*Remark* 2.6. Boundary invariants classify equivalence classes of collapsed configurations, and hence determine semantic or physical information preserved under computational abstraction.

## 2.7 Open Problems

1. Determine precise conditions under which $\kappa$ implements exact $k$–truncation.
2. Classify boundary invariants and their lamphrodynamic dependence.
3. Establish computational complexity bounds for collapse viewed as entropy–admissible truncation.

These questions link geometric computation to derived category theory and lamphrodynamic physics, and will be revisited in Part III of this volume.

# Chapter 3

# Computational Expressiveness of SpherePop

## 3.1 Motivation

The SpherePop calculus was originally introduced as a geometric model of computation based on two primitive operations: *merge* and *collapse*. Merge implements homotopy colimits (cf. **??**), whereas collapse abstracts internal detail while preserving boundary invariants (cf. **??**). In this chapter we prove that these two operations, together with geometrically admissible boundary data, suffice to encode universal computation. In particular, SpherePop is Turing complete.

From a conceptual perspective, this establishes SpherePop as a fundamental model of geometric computation analogous to the lambda calculus for functional computation or Turing machines for classical discrete computation. From the lamphrodynamic point of view, it implies that computational processes can be realized as lamphrodynamically admissible evolutions of geometric states.

## 3.2 Encoding of Lambda Calculus

Let $\Lambda$ denote the usual untyped lambda calculus with application $(MN)$ and abstraction $(\lambda x.M)$. We construct a geometric encoding

$$: \Lambda \longrightarrow,$$

mapping lambda terms to derived configuration objects. Application is encoded by merge of configurations, and abstraction is encoded by an entropy–admissible boundary structure which binds a formal variable.

**Definition 3.1.** Given a lambda term $M$, define $(M)$ to be the configuration constructed inductively as follows:

1. If $M$ is a variable $x$, then $(x)$ is the basic configuration representing the boundary label $x$.
2. If $M = (MN)$, then $(M)$ is the merge $(M)(N)$.
3. If $M = \lambda x.M'$, then $(M)$ is obtained by collapse of $(M')$ along the admissible boundary corresponding to $x$.

*Remark* 3.2. This encoding is reminiscent of categorical semantics of the lambda calculus in cartesian closed categories, except that merge plays the role of application and collapse plays the role of abstraction. The boundary structure supplied by lamphrodynamics serves as a binder.

## 3.3 Beta Reduction as Geometric Evolution

The key observation is that beta reduction in the lambda calculus corresponds to a geometric sequence of merge and collapse operations. Given $(\lambda x.M)N$, the merge operation combines the configurations $(M)$ and $(N)$. A subsequent entropy–admissible collapse substitutes the boundary condition for $x$ with that of $N$.

**Proposition 3.3.** *Let $M, N$ be lambda terms. Then*

$$((\lambda x.M)N) \simeq \kappa((M)(N)),$$

*where the collapse substitutes boundary data associated to $x$.*

*Sketch.* Merge produces a configuration in which the boundary associated to $x$ is identified with the boundary associated to $N$. Entropy–admissible collapse retracts the resulting configuration to one in which $x$ is substituted by $N$. Details follow the inductive construction of . □

**Corollary 3.4.** *preserves beta reduction: if $M \to_\beta M'$, then $(M) \rightsquigarrow (M')$ via merge–collapse evolution.*

## 3.4 Turing Completeness

We now prove the main theorem of this chapter.

**Theorem 3.5** (SpherePop Turing Completeness)**.** *The SpherePop calculus is Turing complete. More precisely, the encoding is sound and complete with respect to lambda calculus reduction. Hence any partial recursive function can be represented as a lamphrodynamically admissible sequence of merge and collapse operations.*

*Sketch of proof.* Soundness follows from the previous corollary. For completeness, we construct a geometric evaluation strategy by induction on the lambda term, using merge for application and collapse for abstraction. The fundamental step is to show that every lambda term can be realized as a finite lamphrodynamic composition of merge and collapse. Since lambda calculus is Turing complete, so is SpherePop. Full technical details require a careful treatment of boundary invariants and lamphrodynamic admissibility, deferred to Appendix M. □

## 3.5 Complexity Analysis

Computational complexity in SpherePop depends on the number of merge and collapse operations as well as the lamphrodynamic cost of collapse. While complexity in lambda calculus is measured by beta reductions, complexity in SpherePop is measured by geometric operations.

**Definition 3.6.** The *SpherePop complexity* $(X)$ of an encoded configuration $X = (M)$ is the minimal number of merge and collapse operations required to produce a normal form of $X$.

**Proposition 3.7.** *There exists a constant $C > 0$ such that for any lambda term $M$,*

$$((M)) \leq C \cdot \beta(M),$$

*where $\beta(M)$ is the usual beta–reduction complexity of $M$.*

*Idea.* Each beta reduction corresponds to at most one merge and one collapse. The constant $C$ accounts for auxiliary geometric normalizations. □

*Remark* 3.8. This estimate shows that SpherePop does not incur asymptotic complexity penalties relative to the lambda calculus, up to constant factors. A precise characterization of SpherePop complexity classes remains open.

## 3.6 Comparison with Other Models

SpherePop differs from standard models of computation in two essential ways: (i) its primitives are geometric rather than symbolic, and (ii) computational steps are lamphrodynamically admissible operations. In particular, SpherePop computation can occur *in situ* in a physical system governed by lamphrodynamic evolution.

*Remark* 3.9. Unlike cellular automata or graph rewriting systems, SpherePop is based on derived geometry and shifted symplectic structures. This makes it a candidate model for computation in physical theories with geometric constraints, including RSVP itself.

## 3.7 Open Problems

1. Determine SpherePop complexity classes and their relationship to standard complexity classes (P, NP, etc.).
2. Develop a geometric type theory for SpherePop analogous to typed lambda calculus.
3. Investigate lamphrodynamic resource constraints for physical computation in RSVP.

This completes Part I, Chapter 3. The next part develops semantic and geometric evolutions in L–systems.

# Part II

# L-Systems and Semantic Evolution

# Chapter 4

# L-Systems as Sigma Models

## 4.1 Motivation

Lindenmayer systems (L–systems) provide a generative formalism for modeling biological growth and recursive geometric structures. In the context of the lamphrodynamic plenum, L–systems become dynamical rules for rewriting geometric configurations. The key idea of this chapter is that L–systems admit a natural interpretation as *sigma models*: their rules are maps into a target space of geometric patterns, and their rewriting dynamics evolve as field configurations on a worldsheet.

This viewpoint enables a derived geometrical interpretation of L–systems, with rewrite rules realized as points of a derived moduli stack $R$, and rewriting steps realized as lamphrodynamic evolutions. Quantization via the AKSZ/BV formalism then yields a quantum theory of semantic evolution.

## 4.2 L-Systems

An (untyped) L–system consists of an alphabet $\Sigma$, a set of production rules $P \colon \Sigma \to \Sigma^*$, and an initial word $\omega \in \Sigma^*$. The rewriting dynamics generate a sequence $\omega, P(\omega), P(P(\omega)), \ldots$, whose geometric interpretation depends on assigning geometric primitives to symbols in $\Sigma$.

**Definition 4.1.** An *entropic L–system* is an L–system whose alphabet $\Sigma$ is equipped with lamphrodynamic weights and boundary conditions, and for which production rules preserve entropy admissibility.

*Remark* 4.2. Entropic L–systems represent lamphrodynamics constrained by recursive semantic structure. They provide a bridge between biological growth, computational generation, and lamphrodynamic dynamics.

## 4.3 Sigma Model Interpretation

Let $X$ be a two–dimensional worldsheet representing the space of rewriting steps. Let $\mathcal{T}$ denote the derived target stack of geometric primitives. Then an L–system can be interpreted as a map

$$\phi\colon X \longrightarrow \mathcal{T},$$

where the rewriting rule supplies boundary conditions on $\partial X$.

**Definition 4.3.** The *L–system sigma model* is the field theory whose fields are maps $\phi\colon X \to \mathcal{T}$ determined by the production rules of the L–system, and whose action functional encodes lamphrodynamic entropy.

## 4.4 Derived Moduli of Rules

Each production rule $a \mapsto w \in \Sigma^*$ determines a geometric constraint on fields $\phi$ at the relevant boundary components. The collection of such rules forms a derived moduli stack

$$R := \{\text{admissible production rules}\}.$$

**Lemma 4.4.** *$R$ is a derived Artin stack governed by lamphrodynamic boundary conditions and derived gluing rules, with tangent complex determined by variations of boundary entropy.*

*Sketch.* Each rule $a \mapsto w$ specifies a derived matching condition between boundary components of the configuration. Derived gluing yields an Artin stack, and lamphrodynamic admissibility ensures that tangent complexes exist with appropriate degeneration. □

## 4.5 BV Quantization

To quantize the L–system sigma model, we apply the AKSZ/BV formalism (see Volume I, Chapters 4 and 7). The source is the worldsheet $X$, and the target is the derived stack $R$ of production rules.

**Definition 4.5.** The *BV–quantized L–system* is the AKSZ sigma model with source $X$ and target $R$, whose BV action is constructed from lamphrodynamic constraints on $\phi$.

**Proposition 4.6.** *The BV action satisfies the classical master equation $\{S, S\} = 0$, and defines a BV complex whose cohomology classifies lamphrodynamic deformations of admissible rewrite rules.*

*Sketch.* AKSZ guarantees the master equation whenever the target carries a $(-1)$–shifted symplectic structure. The derived stack $R$ inherits such a structure from the lamphrodynamic entropy form on $\mathcal{T}$ by transgression. □

## 4.6 Worked Example: Algae L-System

Consider the classical algae L–system with alphabet $\{A, B\}$ and production rules

$$A \mapsto AB, \qquad B \mapsto A.$$

Assign entropy weights $s_A, s_B$ to the symbols and interpret $A, B$ as basic geometric primitives subject to lamphrodynamic boundary conditions.

**Proposition 4.7.** *The BV–quantized algae system realizes a lamphrodynamic sigma model whose field content corresponds to alternating lamphrodynamic boundary entropies $s_A$ and $s_B$, and whose BV action reduces to a sum of boundary contributions.*

*Sketch.* Each production $A \mapsto AB$ attaches boundary conditions corresponding to $A$ and $B$ in sequence. Collapse in the BV complex reduces internal structure to boundary alternation, yielding the stated action. □

## 4.7 Further Directions

1. Classification of BV–quantized L–systems by lamphrodynamic entropy.
2. Derived deformation theory of production rules.
3. Applications to biological growth and semantic evolution.

# Chapter 5

# Semantic Fields and Emergent Meaning

## 5.1 Motivation

In classical formal language theory, syntax and semantics are modular: syntactic derivations produce wellformed expressions, to which meaning is then externally assigned. In the lamphrodynamic setting, semantic structure is *cogenerated* with syntactic evolution. The central proposal of this chapter is that meaning arises as an *effective field* on the derived configuration space, controlled by entropy flows and lamphrodynamic coupling. In particular, semantic information is not assigned post hoc, but rather emerges as a macroscopic invariant of lamphrodynamic boundary evolution.

This idea radically departs from static symbolsemantics mappings, and instead treats semantic fields as dynamical sections of a derived semantic bundle $\mathscr{M} \to$. Our aim is to formalize the resulting semantic fields and outline a lamphrodynamic origin of meaning.

## 5.2 Semantic Bundles

Let denote the derived configuration stack of admissible geometric states (cf. Volume III, Chapters 1–3). A *semantic bundle* is a derived morphism

$$\mathscr{M} \longrightarrow,$$

equipped with a semantic section

$$m\colon \longrightarrow \mathscr{M},$$

which assigns meaning to each configuration. The semantic section is lamphrodynamically admissible if it preserves entropy on boundary components.

**Definition 5.1.** A *semantic field* on $X \in$ is a global section $m(X) \in \Gamma(X, \mathscr{M})$ compatible with lamphrodynamic boundary data.

*Remark* 5.2. Semantic fields are not free variables: they are constrained by derived geometry and lamphrodynamics, reflecting the empirical restriction that meaning cannot be arbitrary but must be physically consistent.

## 5.3 Entropy and Meaning

The key hypothesis is that entropy gradients generate semantic structures. Lowentropy regions admit more coherent semantic fields, whereas highentropy regions correspond to semantic ambiguity or degeneracy. Formally, semantic fields minimize an effective potential determined by the entropy $S$:

$$\delta_m \int_X V(m, S) = 0.$$

**Proposition 5.3.** *If $V(m, S)$ is strictly convex in $m$ for fixed $S$, then the semantic field on $X$ is uniquely determined by boundary data on $\partial X$.*

*Sketch.* Strict convexity implies uniqueness of minimizers, while entropyadmissibility ensures that boundary conditions suffice to select a unique minimizer. □

## 5.4 Semantic Phase Transitions

We now consider semantic fields as order parameters. When lamphrodynamic entropy passes critical thresholds, semantic coherence may undergo a phase transition analogous to spontaneous symmetry breaking:

$$m \mapsto m',$$

where $m'$ lies in a different semantic equivalence class.

**Definition 5.4.** A *semantic phase transition* occurs at critical entropy levels $S_c$ for which the semantic potential develops degenerate minima.

*Remark* 5.5. Semantic phase transitions provide a physical mechanism for qualitative changes in meaning—e.g. emergence of new semantic categories or conceptual reorganizations.

## 5.5 Resonances and Semantic Modes

Lamphrodynamic resonance modes (cf. Chapter 8 below) induce oscillatory semantic patterns. These correspond to eigenmodes of an entropyweighted Laplacian acting on the semantic bundle:

$$\Delta_S m = \lambda m.$$

**Proposition 5.6.** *Eigenmodes of $\Delta_S$ classify lamphrodynamically admissible semantic oscillations, with eigenvalues determined by entropy gradients.*

*Sketch.* Entropy weighting ensures that the Laplacian is selfadjoint with respect to an entropyweighted inner product, giving rise to an orthonormal basis of semantic eigenmodes. □

## 5.6 Semantic Evolution in L-Systems

For entropic Lsystems (cf. Chapter 4), semantic fields evolve with rewriting dynamics. Each production rule induces a transformation

$$m \longmapsto P^*(m),$$

where $P^*$ is the induced pullback along the derived morphism representing the rule.

**Proposition 5.7.** *Lsystem rewriting induces a lamphrodynamically admissible evolution of semantic fields, determined by entropyweighted pullbacks.*

*Idea.* Each rewriting step defines a derived morphism compatible with boundary entropy. Pullback along this morphism yields the stated semantic evolution. □

## 5.7 Towards Cognitive and Linguistic Applications

This geometric theory of semantic fields suggests a lamphrodynamic origin of meaning in natural cognition. Semantic evolution tracks entropy gradients, while meaning emerges from lamphrodynamic coherence rather than symbolic assignment. Applications include:

1. phase transitions in conceptual categories,
2. semantic resonance in neural oscillations,
3. entropic models of linguistic meaning.

*Remark* 5.8. While highly speculative, this approach offers a geometric, physically motivated account of semantic emergence consistent with lamphrodynamic constraints.

## 5.8 Open Problems

1. Classification of semantic phase transitions in lamphrodynamic theories.
2. Determination of critical entropy thresholds $S_c$.
3. Relationship between semantic eigenmodes and neural oscillations.
4. Computational semantics based on lamphrodynamic sigma models.

This completes Part II. We now turn to distributed computation and CRDTs in Part III.

# Part III

# Distributed Computation and CRDTs

# Chapter 6

# The TARTAN–RSVP Framework

## 6.1 Motivation

Traditional distributed computation concerns consistency and synchronization across spatially separated agents. Conflict–free replicated data types (CRDTs) guarantee eventual consistency under concurrent updates without requiring global coordination. In the lamphrodynamic setting, distributed geometric computation is realized as a field theory of interacting configurations. The guiding question of this chapter is:

> Can CRDT semantics be recovered from lamphrodynamic evolution?

We will show that specific CRDTs can be modeled as *field configurations* whose update rules correspond to lamphrodynamically admissible transformations. Eventual consistency emerges from entropy dissipation and geometric collapse.

## 6.2 CRDTs as Field Configurations

Let $\mathsf{CRDT}$ denote a category of CRDTs, with morphisms given by admissible updates. We construct a functor

$$: \mathsf{CRDT} \longrightarrow,$$

assigning to each CRDT a derived configuration object. Updates are mapped to lamphrodynamic transformations respecting entropy admissibility.

**Definition 6.1.** A *lamphrodynamic CRDT* is a CRDT whose state space is realized as a subset of , and whose update rules correspond to merge or collapse operations.

## 6.3 Eventual Consistency

CRDTs guarantee eventual consistency: given sufficiently many updates and message deliveries, all replicas converge to the same state. In lamphrodynamic terms, convergence arises from entropy minimization under repeated merge and collapse operations.

**Proposition 6.2.** *Let $X, Y \in$ represent two CRDT replicas. If updates are entropy–admissible and lamphrodynamically bounded, then repeated merge and collapse operations converge to a common configuration $Z \in$.*

*Sketch.* Each merge reduces discrepancy while preserving entropy admissibility. Collapse eliminates internal inconsistency while preserving boundary conditions. Lamphrodynamic energy estimates guarantee convergence to a fixed point. □

## 6.4 Gray-Code Rewriting

TARTAN incorporates Gray–code rewriting to minimize semantic conflict and maintain lamphrodynamic coherence across distributed computational paths. Let $G$ denote a Gray–code rewrite operator acting on configurations.

**Definition 6.3.** A *lamphrodynamic Gray–code path* is a sequence of configurations connected by Gray–code rewrites which preserve entropy admissibility.

**Proposition 6.4.** *Gray–code lamphrodynamic rewrites minimize semantic distortion under concurrent merge operations.*

*Idea.* Gray–code ordering ensures that successive rewrites differ in only one bit of boundary entropy, minimizing lamphrodynamic discrepancy. □

## 6.5 Formal Statement of Convergence

**Theorem 6.5** (Lamphrodynamic Eventual Consistency)**.** *Let $\{X_i\}$ be a family of lamphrodynamic CRDT replicas. Assume update rules are merge–collapse admissible, and communication is eventual. Then there exists a limiting configuration $X_\infty \in$ such that $X_i \rightsquigarrow X_\infty$ for all $i$.*

*Sketch.* Repeated merge produces a decreasing lamphrodynamic discrepancy. Collapse removes internal semantic conflict. Standard fixed–point techniques for entropy dissipative systems yield a common limit. □

## 6.6 Distributed SpherePop

Distributed SpherePop generalizes merge–collapse computation to multiple agents. Each agent holds a derived configuration; updates propagate via entropy–admissible messages; merge and collapse maintain local coherence.

*Remark* 6.6. Unlike traditional distributed computation, lamphrodynamic propagation imparts geometric structure to communication and ensures physical coherence across replicas.

## 6.7 Open Problems

1. Characterize necessary and sufficient conditions for lamphrodynamic convergence.

2. Develop complexity bounds for distributed SpherePop.

3. Explore applications to decentralized lamphrodynamic inference.

# Chapter 7

# Ethical Constraints in Geometric Computation

## 7.1 Motivation

Geometric computation via merge and collapse lacks an inherent notion of *ethical admissibility*. Yet distributed computation, especially in multi–agent systems, must incorporate ethical constraints that prevent harmful or undesirable outcomes. The central hypothesis of this chapter is that *ethical norms* can be encoded as *geometric constraints* on lamphrodynamic evolution, inducing an admissible subset of configurations.

We investigate the mathematical structure of such constraints and explore philosophical implications for distributed geometric computation.

## 7.2 Ethical Admissibility

Let  denote a collection of ethical conditions expressible as constraints on lamphrodynamic fields. A configuration is *ethically admissible* if it satisfies all conditions in .

**Definition 7.1.** An ethical constraint is a lamphrodynamic inequality of the form

$$F(\Phi, \boldsymbol{v}, S) \geq 0,$$

preserved under merge and collapse.

*Remark* 7.2. Ethical admissibility resembles entropy admissibility, but may involve additional state variables or derived invariants.

## 7.3 Collapse and Salience

Collapse abstracts internal details. Ethical collapse assigns greater weight to boundary conditions corresponding to ethically salient features, effectively reweighing lamphrodynamic fields by ethical importance.

**Proposition 7.3.** *Under ethical reweighting, collapse preserves ethically salient boundary conditions while eliminating ethically neutral internal variations.*

## 7.4 Distributed Ethics

Distributed lamphrodynamic computation raises the question of whether ethical admissibility can be guaranteed across multiple agents. Consistency requires:

1. local ethical admissibility,
2. global lamphrodynamic compatibility,
3. admissibility under merge and collapse.

**Theorem 7.4** (Ethical Consistency)**.** *If each agent satisfies ethical admissibility locally, and updates are lamphrodynamically admissible, then distributed merge–collapse evolution preserves ethical admissibility.*

*Sketch.* Ethical admissibility is preserved by local updates and merge; collapse removes ethically neutral detail; lamphrodynamic invariance ensures closure. □

## 7.5 Philosophical Implications

Interpreting ethical norms as geometric constraints suggests that ethics is not merely external to computation, but internal to lamphrodynamic structure. Ethical admissibility becomes a physical constraint analogous to conservation laws or entropy conditions.

*Remark* 7.5. This perspective hints at a lamphrodynamic foundation for ethical agency, linking physical admissibility to normative structure.

## 7.6 Open Problems

1. Characterize ethical invariants preserved by collapse.
2. Develop categorical semantics for ethical admissibility.
3. Investigate lamphrodynamic foundations of moral salience.

This completes Part III. We now turn to spectral geometry and cymatics.

# Part IV

# Spectral Geometry and Cymatic Resonance

# Chapter 8

# Entropy–Weighted Spectral Theory

## 8.1 Motivation

The spectral properties of lamphrodynamic operators encode geometric and physical information about the entropy distribution $S$. Eigenfunctions of entropy–weighted Laplacians describe resonant modes of lamphrodynamic evolution and provide a bridge between field theory, spectral geometry, and observable resonance phenomena ("cymatics").

This chapter establishes the analytic foundations of entropy–weighted spectral theory, proves self–adjointness of the main operators, and derives spectral asymptotics under lamphrodynamic regularity assumptions. These results will later inform Chapters 9 and 10, where we relate spectral modes to cognitive and aesthetic structure.

## 8.2 Entropy–Weighted Laplacian

Let $(M, g)$ be a Riemannian manifold with entropy density $S$. The entropy–weighted Laplacian is defined by

$$\Delta_S u := e^{-S}\nabla_a(e^S\nabla^a u).$$

*Remark* 8.1. Entropy weighting biases diffusion so that entropy gradients induce preferred propagation directions. This reflects lamphrodynamic flow of semantic or physical information.

## 8.3 Self–Adjointness

**Theorem 8.2** (Self–Adjointness). *Assume $S \in C^1(M)$ and $S$ is bounded from below. Then $\Delta_S$ is essentially self–adjoint on $C_c^\infty(M)$ with respect to the weighted measure $e^S \mathrm{dvol}_g$.*

*Sketch.* Integration by parts with weight $e^S$ shows symmetry. Standard elliptic estimates and Friedrichs extension yield essential self–adjointness. □

## 8.4 Spectral Properties

Consider the spectral problem

$$\Delta_S u = \lambda u.$$

**Proposition 8.3.** *Under the assumptions of ??, the spectrum of $\Delta_S$ is real and bounded below, with discrete spectrum if $M$ is compact.*

*Sketch.* Self–adjointness implies real spectrum. Compactness of $M$ yields discreteness by Rellich compactness. □

## 8.5 Heat Kernel Asymptotics

The heat kernel associated to $\Delta_S$, $K_S(t, x, y)$, satisfies

$$\partial_t K_S = \Delta_S K_S.$$

**Theorem 8.4.** *As $t \to 0^+$,*

$$K_S(t, x, x) \sim (4\pi t)^{-n/2} \sum_{k=0}^{\infty} a_k(x) t^k,$$

*where $a_k$ depend on curvature of $g$ and derivatives of $S$.*

*Idea.* Adapt classical heat kernel asymptotics to weighted Laplacians using perturbative expansions in $S$ and its derivatives. □

## 8.6 Spectral Asymptotics

**Proposition 8.5.** *Let $\lambda_k$ denote eigenvalues of $\Delta_S$ on compact $M$. Then*

$$\lambda_k \sim C k^{2/n}$$

*for $k \to \infty$, with constant $C$ depending on $g$ and $S$.*

*Sketch.* Entropy weighting modifies volume density in Weyls law. Leading order depends on $e^S \mathrm{dvol}_g$. □

## 8.7 Physical Interpretation

Eigenfunctions represent resonant lamphrodynamic patterns. High entropy biases localization, while entropy gradients induce directional propagation. Cymatic resonance arises when lamphrodynamic excitation matches eigenvalues of $\Delta_S$.

*Remark* 8.6. Entropy weighting provides a physical mechanism for emergent resonance patterns in lamphrodynamic systems, with potential applications to cognition, neural oscillations, and aesthetic perception.

## 8.8 Open Problems

1. Spectral gap estimates under entropy gradients.

2. Relation to lamphrodynamic phase transitions.

3. Experimental verification of entropy–weighted resonance.

# Chapter 9

# Cymatics and Cognitive Modes

## 9.1 Motivation

Cymatics refers to the emergence of resonant geometric patterns in oscillating media. In the lamphrodynamic setting, resonant modes of $\Delta_S$ give rise to oscillatory patterns which may correspond to cognitive modes. This suggests a connection between spectral geometry and cognition, mediated by entropy distributions.

We approach this proposal cautiously: while speculative, it offers a physically motivated link between lamphrodynamics and cognitive phenomena.

## 9.2 Entropy–Driven Resonance

Given eigenpairs $(\lambda_k, u_k)$ of $\Delta_S$, resonant excitation produces oscillatory patterns proportional to $u_k$. Entropy gradients modulate resonant frequencies via $\lambda_k$.

**Proposition 9.1.** *Lamphrodynamic excitation of eigenmode $u_k$ produces a resonant pattern whose spatial localization depends on the entropy gradient $\nabla S$.*

*Sketch.* Weighted Laplacian localizes eigenfunctions toward regions of lower effective potential $-S$. Entropy gradients shift localization centers. □

## 9.3 Neural Oscillations

Neural oscillations exhibit frequency bands (theta, alpha, beta, gamma) which may correspond to spectral bands of $\Delta_S$ acting on cognitive manifolds. While speculative, this provides a geometric framework for understanding neural resonance.

*Remark* 9.2. Entropy–regulated spectral geometry may underlie cognitive coherence and coordination across neuronal populations.

## 9.4 Aesthetic and Philosophical Aspects

Cymatic patterns have long been associated with aesthetic and symbolic interpretations. Entropy–weighted spectral geometry provides a physical basis for such interpretations, linking aesthetic resonance to lamphrodynamic structure.

*Remark* 9.3. This perspective suggests that aesthetic phenomena reflect physical resonances rather than arbitrary symbolic associations.

## 9.5 Open Problems

1. Experimental investigation of lamphrodynamic resonance in neural systems.
2. Relationship between semantic phase transitions and cognitive modes.
3. Aesthetic evaluation from entropy–weighted spectral analysis.

This completes Part IV. We now turn to amplitwistors and scattering.

# Part V

# Amplitwistors and Scattering

# Chapter 10

# Twistor Theory and RSVP

## 10.1 Motivation

Twistor theory provides a holomorphic description of four–dimensional spacetime geometry, in which points of spacetime correspond to certain holomorphic curves in twistor space. Ambitwistors generalize this construction to encode scattering amplitudes and null geodesics. Lamphrodynamic fields exhibit structural similarities to twistor variables: entropy–regulated flows and derived geometric structure suggest a natural twistor–like representation of lamphrodynamics.

In this chapter we outline a speculative correspondence between lamphrodynamic configurations and (ambi)twistor geometry, with the goal of connecting RSVP dynamics to scattering theoretic phenomena.

## 10.2 Ambitwistor Space

Ambitwistor space is the cotangent bundle of projective twistor space, encoding null geodesics in spacetime. Points in ambitwistor space represent null momentum directions, and provide a natural setting for scattering amplitudes.

**Definition 10.1.** Let $\mathbb{PT}$ denote projective twistor space. The *ambitwistor space* is the holomorphic cotangent bundle $T^*\mathbb{PT}$, endowed with its canonical holomorphic symplectic structure.

*Remark* 10.2. Ambitwistors provide a gauge–invariant description of null geodesics and play a central role in recent developments in scattering amplitudes.

## 10.3 Lamphrodynamic Fields in Twistor Variables

Lamphrodynamic flows $\boldsymbol{v}$ may be locally decomposed into null and non–null parts. Restricting to null flows suggests a twistor parametrization via incidence relations between spacetime and twistor coordinates. Entropy $S$ then acts as a deformation parameter modifying holomorphic structures.

**Proposition 10.3** (Heuristic)**.** *Locally null lamphrodynamic flows admit a twistor parametrization*

$$\Psi = (\Phi, \boldsymbol{v}, S) \longmapsto (Z, W)$$

*where $(Z, W)$ satisfy deformed twistor incidence relations depending on $S$.*

*Idea.* Null flows determine projective spinors; entropy deforms corresponding holomorphic data. Details require a lamphrodynamic generalization of Penrose transform. □

## 10.4 Penrose Transform for Lamphrodynamics

The Penrose transform relates holomorphic data on twistor space to space–time fields. A lamphrodynamic version would map holomorphic data on (ambi)twistor space to lamphrodynamic fields, possibly incorporating entropy–induced deformations.

There exists a lamphrodynamic Penrose transform mapping holomorphic sections on ambitwistor space to solutions of lamphrodynamic equations.

*Remark* 10.4*.* This conjecture extends classical twistor theory to lamphrodynamic settings, potentially yielding novel scattering descriptions.

## 10.5 Scattering Interpretation

Lamphrodynamic excitations propagate along null characteristics. In ambitwistor space, scattering corresponds to intersection of null geodesics. Entropy modifies scattering geometries by deforming null structures and holomorphic data.

**Proposition 10.5** (Speculative)**.** *Lamphrodynamic scattering amplitudes admit a representation as integrals over deformed ambitwistor space weighted by entropy–dependent measures.*

*Heuristic.* Ambitwistor scattering formulas integrate holomorphic data over null geodesics. Entropy modifies such integrals via lamphrodynamic weighting. □

## 10.6 Open Problems

1. Construct lamphrodynamic twistor and ambitwistor spaces explicitly.

2. Develop entropy–modified Penrose transform.

3. Relate lamphrodynamic scattering to amplituhedron–like geometry.

We now discuss speculative scattering constructions via derived mapping stacks.

# Chapter 11

# Derived Mapping Stacks and Scattering

## 11.1 Motivation

Derived mapping stacks provide a powerful framework for describing field configurations as morphisms between derived geometric objects. In the lamphrodynamic setting, scattering processes may be understood as derived morphisms between boundary configurations, with amplitudes encoded by mapping spaces in the derived category.

This chapter outlines a speculative AKSZ–style construction for lamphrodynamic scattering, emphasizing conceptual connections rather than detailed computations.

## 11.2 Derived Mapping Stacks

Given derived stacks $X$ and $Y$, the mapping stack $\mathrm{Map}(X, Y)$ parameterizes morphisms $X \to Y$ up to homotopy. Scattering processes correspond to derived maps from incoming to outgoing boundary configurations.

**Definition 11.1.** The *lamphrodynamic scattering stack* is

$$:= \mathrm{Map}(B_{\mathrm{in}}, B_{\mathrm{out}})$$

where $B_{\mathrm{in}}, B_{\mathrm{out}}$ are derived lamphrodynamic boundary stacks.

*Remark* 11.2. generalizes traditional scattering amplitudes by incorporating derived geometry and lamphrodynamic structure.

## 11.3 AKSZ–like Construction

Analogous to Volume I, Chapter 7, an AKSZ–type construction yields a BV action

$$S_{\mathrm{sc}}\colon\ \mathrm{Map}(B_{\mathrm{in}}, B_{\mathrm{out}}) \to \mathbb{R}$$

whose critical points describe scattering processes.

**Proposition 11.3** (Heuristic)**.** *Assuming lamphrodynamic boundary stacks carry a* $(-1)$*–shifted symplectic structure, the BV action* $S_{\mathrm{sc}}$ *satisfies the classical master equation* $\{S_{\mathrm{sc}}, S_{\mathrm{sc}}\} = 0$*.*

## 11.4 Speculative Directions

1. Entropy–dependent deformation of scattering geometry.
2. Relation to ambitwistor scattering integrals.
3. Potential amplituhedron–like lamphrodynamic polytopes.

## 11.5 Conclusion

While speculative, these ideas connect lamphrodynamic field theory to modern scattering formalisms, suggesting deep geometric relationships between entropy, derived stacks, and holomorphic scattering structures.

This completes Part V. We now turn to creative and philosophical dimensions.

# Part VI

# Creative and Philosophical Dimensions

# Chapter 12

# The Call from Ankyra: Mathematical Commentary

## 12.1 Introduction

In this chapter we provide a mathematically informed commentary on the creative work *The Call from Ankyra*. The purpose is not literary criticism per se, but rather to identify and articulate lamphrodynamic themes embedded in the narrative structure. Our guiding principle is that ideas expressed in poetic or narrative form may anticipate technical concepts later formalized in field theory and geometry.

## 12.2 Entropy Smoothing as Narrative Device

One of the recurring motifs in *Ankyra* is the notion of a gradual smoothing of hidden structure, corresponding in the mathematical setting to entropy–driven regularization of lamphrodynamic fields. Entropy dissipation introduces smoothing effects, as proved rigorously in Volume I, Chapters 9–10. These mathematical results offer a precise counterpart to the narrative process of gradual revelation in the text.

## 12.3 Ethical Choice and Collapse

The narrative includes moments of choice represented as irreversible collapses. In lamphrodynamics, collapse corresponds to abstraction of internal field complexity (see Volume III, Chapter 2 for a formal discussion). This suggests an interpretation in which ethical decisions correspond to irreversible projections in configuration space.

## 12.4 Literary Structure as Lamphrodynamic Flow

Plot progression in *Ankyra* may be viewed as a lamphrodynamic flow through a configuration landscape, with entropy guiding narrative possibilities. Although speculative, this reading reflects core themes in Volume III: emergence of meaning, collapse, and semantic fields.

# Chapter 13

# Visions of a Spirit Seer: Visual–Mathematical Analysis

## 13.1 Introduction

This chapter discusses connections between Swedenborg's visionary texts and lamphrodynamic structures. Our aim is to highlight formal analogies and structural resonances, not to assert direct influence.

## 13.2 Cymatic Visual Patterns

Swedenborg describes visual patterns that resemble contemporary cymatic images generated by spectral resonances. Volume III, Chapter 9 develops an entropy–weighted spectral theory whose eigenmodes produce distinctive visual patterns. These patterns provide a mathematical interpretation of certain visionary imagery.

## 13.3 Art as Mathematical Intuition

Mathematical structures may emerge through intuitive or visionary imagery before they are formalized. This perspective aligns with the broader theme of Volume III: creative works as speculative precursors to formal structures in lamphrodynamics and semantic computation.

# Chapter 14

# Philosophical Foundations

## 14.1 Fields and Substances

RSVP adopts a field–theoretic ontology: fundamental reality is described by fields (or stacks) rather than point–like substances. This stands in contrast to traditional substance ontology in philosophy, but aligns with modern physics and structural realism (cf. Volume III, Appendix L).

## 14.2 Process Philosophy

Process philosophy emphasizes becoming over static being. Lamphrodynamics embodies this view: entropy and vector flow generate dynamic structure. Connections with Whitehead and other process thinkers suggest a philosophical framework for lamphrodynamic emergence.

## 14.3 Measurement and Observers

Lamphrodynamics introduces observer–dependent collapse (Volume III, Chapter 2). Philosophically, this aligns partially with relational interpretations, though the formalism is distinct from quantum mechanics.

## 14.4 Consciousness (Speculative)

Speculatively, lamphrodynamic fields may provide a substrate for conscious processes. Volume III explores these ideas while maintaining a clear separation between rigorous mathematics and speculative extensions.

# Part VII

# Open Problems and Future Directions

# Chapter 15

# Critical Open Problems

## 15.1 Overview

We list central open problems arising from the lamphrodynamic program. Some are mathematical, others physical, and some philosophical.

## 15.2 Open Problems

1. Prove global well–posedness for nonlinear lamphrodynamic equations.
2. Classify entropic singularities in derived geometric terms.
3. Extend AKSZ construction to lamphrodynamic scattering.
4. Develop entropy–dependent Penrose transform.
5. Establish rigorous correspondence between RSVP and emergent gravity.
6. Formulate Turing–complete SpherePop rigorously.

# Chapter 16

# Speculative Extensions

## 16.1 Quantum Gravity via RSVP

RSVP suggests a geometric origin for gravity through entropy and vector fields. Extending this to quantum gravity requires quantizing lamphrodynamic fields via derived geometry and BV quantization.

## 16.2 String Theory Connections

Potential links to string theory may arise through shifted symplectic structures and AKSZ sigma models (Volume I, Chapter 7). The precise correspondence remains speculative.

## 16.3 Beyond Physics

Finally, lamphrodynamic principles may apply to complex systems outside physics, including computation, cognition, and social systems. These speculative directions offer promising opportunities for future work.

# Appendix A

# Category Theory Reference

This appendix collects a compact set of categorical notions used throughout Volume III, with an emphasis on those aspects most relevant to homotopy theoretic formulations of geometric computation, SpherePop calculus, and lamphrodynamic semantic evolution. It is not intended as a complete introduction to category theory, but as a technical reference summarizing definitions, constructions, and theorems that are explicitly invoked in the main text.

We assume basic familiarity with ordinary categories and functors. Throughout, $\mathsf{Cat}$ denotes the 2–category of (small) categories, and  denotes the category of sets. When needed, we indicate size issues by assuming the existence of universes, although these subtleties do not affect the results presented here.

## A.1 Categories and Functors

**Definition A.1.** A *category* $\mathbb{C}$ consists of a class of objects $(\mathbb{C})$, a class of morphisms $_{\mathbb{C}}(X,Y)$ for each pair of objects $(X,Y)$, identity morphisms $_X \in_{\mathbb{C}} (X,X)$, and a composition law $_{\mathbb{C}}(X,Y) \times_{\mathbb{C}} (Y,Z) \to_{\mathbb{C}} (X,Z)$ that is associative and unital.

**Definition A.2.** A *functor* $F : \mathbb{C} \to$ assigns to each object $X \in \mathbb{C}$ an object $F(X) \in$ and to each morphism $f : X \to Y$ a morphism $F(f) : F(X) \to F(Y)$, preserving identities and composition.

**Definition A.3.** A *natural transformation* $\eta : F \Rightarrow G$ between functors $F, G : \mathbb{C} \to$ assigns to each $X \in \mathbb{C}$ a morphism $\eta_X : F(X) \to G(X)$ such that for every $f : X \to Y$ in $\mathbb{C}$, the naturality square $G(f) \circ \eta_X = \eta_Y \circ F(f)$ commutes.

## A.2 Limits and Colimits

Limits and colimits abstract universal constructions. These are essential in SpherePop, where merge operations correspond to homotopy colimits of geometric data, and collapse operations correspond to truncation functors.

**Definition A.4.** Given a diagram $D : J \to \mathbb{C}$, a *limit* $\varprojlim D$ is a universal cone over $D$. A *colimit* $\varinjlim D$ is a universal cocone.

**Example A.5.** Products, equalizers, pullbacks are special cases of limits; coproducts, coequalizers, and pushouts are colimits.

*Remark* A.6. In homotopy theoretic settings (model categories or $\infty$–categories) one replaces limits and colimits with *homotopy* limits and colimits, which rectify the failure of strict universal properties under weak equivalences. This distinction is central to the formal proof that SpherePop merge operations realize homotopy colimits rather than strict colimits.

## A.3 Adjunctions

**Definition A.7.** An *adjunction* between categories $\mathbb{C}$ and  is a pair of functors $F : \mathbb{C} \to$ and $G :\to \mathbb{C}$ together with natural bijections

$${}_{(}FX, Y) \cong_{\mathbb{C}} (X, GY),$$

natural in $X \in \mathbb{C}$ and $Y \in$. We write $F \dashv G$.

Adjunctions generate many fundamental constructions in homotopy theory, including derived functors and Quillen adjunctions.

## A.4 Model Categories

To work with explicit homotopy theoretic constructions (beyond derived $\infty$–categorical formalism), the notion of a model category is often useful.

**Definition A.8.** A *model category* $(, , , )$ consists of a category  together with three classes of morphisms: weak equivalences , fibrations , and cofibrations , satisfying the Quillen axioms.

Every model category admits derived functors and homotopy limits/colimits via cofibrant and fibrant replacements. In Volume III we use model structures primarily as a conceptual bridge to $\infty$–categorical homotopy theory, rather than as a computational tool.

## A.5 $\infty$–Categories

**Definition A.9.** An $\infty$–category is a higher category whose $n$–morphisms are all invertible for $n > 1$. Many equivalent models exist: quasi–categories, complete Segal spaces, simplicial categories, etc.

*Remark* A.10. For most of this volume, the intended model is that of quasi–categories, as in Lurie. Homotopy limits and colimits are encoded by mapping spaces rather than strict diagrams.

**Definition A.11.** A functor between $\infty$–categories preserves equivalences and coherent higher morphisms. Natural transformations generalize to higher homotopies between such functors.

## A.6 Derived Functors

Let $F : \mathbb{C} \to$ be a functor between model categories. Its *left derived functor* $\mathbf{L}F$ is obtained by applying $F$ to a cofibrant replacement; the *right derived functor* $\mathbf{R}F$ uses fibrant replacement.

*Remark* A.12. In $\infty$–categorical settings, derived functors are simply the homotopy correct functors defined on $\infty$–categories themselves; no replacements are needed if one works natively in quasi–categories.

## A.7 Homotopy Colimits

**Definition A.13.** Given a diagram $D : J \to$ in a model category, the *homotopy colimit* $\operatorname{hocolim} D$ is the left derived colimit, typically computed via cofibrant replacement or via simplicial resolutions.

*Remark* A.14. In Volume III, Theorem 24.2 identifies SpherePop merge operations with homotopy colimits in a suitable $\infty$–categorical framework. This justifies the intuition that merge is a higher–categorical "gluing" operation preserving homotopical information.

## A.8 Quillen Adjunctions and Equivalences

**Definition A.15.** A *Quillen adjunction* between model categories  and $\mathbb{N}$ is an adjunction $F \dashv G$ such that $F$ preserves cofibrations and acyclic cofibrations (equivalently, $G$ preserves fibrations and acyclic fibrations).

**Definition A.16.** A Quillen adjunction $F \dashv G$ is a *Quillen equivalence* if the derived unit and counit maps are weak equivalences. In this case,  and $\mathbb{N}$ have equivalent homotopy categories.

*Remark* A.17. Quillen equivalence is the key mechanism by which SpherePop configurations are shown to be *computationally invariant* under change of model, thereby allowing derived truncation and homotopy colimit constructions to be interpreted in multiple equivalent frameworks.

## A.9 Truncation and Postnikov Towers

**Definition A.18.** For an $\infty$–category $\mathbb{C}$, the $n$–truncation $\tau_{\leq n}\mathbb{C}$ is obtained by forcing all homotopy groups above degree $n$ to vanish. At the level of objects, this corresponds to retaining data up to homotopies of order $n$.

*Remark* A.19. Collapse operations in SpherePop are conjecturally equivalent to derived truncation functors, eliminating higher–order homotopic detail while preserving essential information. Conjecture 20.3 provides a precise formulation and evidence for this interpretation.

## A.10 Monoidal Structures

Many constructions in RSVP and SpherePop rely on monoidal structures, such as tensor products of complexes and monoidal $\infty$–categories.

**Definition A.20.** A *monoidal category* $(\mathbb{C}, \otimes, \mathbf{1})$ is a category equipped with a bifunctor $\otimes : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$, associativity constraints, and a unit object $\mathbf{1}$ satisfying coherence conditions.

*Remark* A.21. The monoidal structure on derived categories of sheaves underlies the AKSZ construction; similarly, monoidal structures on $\infty$–categories support the compositional semantics of SpherePop computations.

## References

For further reading:


- Lurie: *Higher Topos Theory*, *Higher Algebra*.
- Hovey: *Model Categories*.
- Dwyer–Hirschhorn–Kan–Smith: *Homotopy Theories and Model Categories*.
- Riehl: *Category Theory in Context*.

# Appendix B

# Computational Complexity Background

This appendix summarizes the classical theory of computational complexity required for the formal development of SpherePop, semantic computation, and distributed lamphrodynamic processes in Volume III. The treatment emphasizes structural complexity (reductions, completeness, hierarchies) and those aspects which reappear in homotopy–theoretic or geometric encodings. We work predominantly in the classical Turing model and remark on geometric representations when appropriate.

## B.1 Deterministic and Nondeterministic Computation

**Definition B.1** (Turing machine)**.** A (deterministic) Turing machine $M$ consists of a finite set of states, a finite tape alphabet with a distinguished blank symbol, a transition function, and a distinguished halting configuration. Given an input $x \in \{0,1\}^*$, $M$ either halts with an output or continues indefinitely.

**Definition B.2** (Time complexity)**.** The (worst–case) time complexity of $M$ is the function

$$T_M(n) := \max\{\text{number of steps } M \text{ uses on input } x \ : \ |x| = n\}.$$

We say that $M$ runs in time $T(n)$ if $T_M(n) \leq T(n)$ for all $n$.

**Definition B.3** (Complexity classes)**.**

1. $\mathsf{P}$ consists of all problems decidable in polynomial time by a deterministic Turing machine.
2. $\mathsf{NP}$ consists of all problems decidable in polynomial time by a nondeterministic Turing machine.
3. $\mathsf{PSPACE}$ consists of problems decidable using polynomial space.

*Remark* B.4. The containment $\mathsf{P} \subseteq \mathsf{NP} \subseteq \mathsf{PSPACE}$ is known, but no strict separation of these classes has been proved. The open question $\mathsf{P} \stackrel{?}{=} \mathsf{NP}$ is central to classical complexity theory.

## B.2 Reductions and Completeness

**Definition B.5** (Polynomial–time reduction)**.** A language $A$ reduces to $B$ (written $A \leq_p B$) if there exists a polynomial–time computable function $f$ such that $x \in A$ iff $f(x) \in B$. A problem is $\mathsf{NP}$–$\mathsf{hard}$ if every problem in $\mathsf{NP}$ reduces to it, and $\mathsf{NP}$–$\mathsf{complete}$ if it is both $\mathsf{NP}$–$\mathsf{hard}$ and in $\mathsf{NP}$.

*Remark* B.6. For geometric computation models, one typically constructs an encoding $\mathrm{Enc}(x)$ of classical input $x$ as a geometric configuration such that logical operations correspond to merge or collapse operations. Polynomial simulation requires that the size and depth of the encoding scale polynomially with $|x|$.

## B.3 Alternation and Space Complexity

**Definition B.7** (Alternating Turing machine)**.** An alternating Turing machine is a nondeterministic Turing machine whose states are partitioned into *existential* and *universal* states, with acceptance defined inductively on the computation tree.

**Theorem B.8** (Chandra–Kozen–Stockmeyer)**.** *For $k \geq 1$,*

$$\mathsf{AP} = \mathsf{PSPACE}.$$

*More precisely, polynomial–time alternating computation equals polynomial space deterministic computation.*

*Remark* B.9. This equivalence plays a role in geometric encodings where "branching" is represented topologically (e.g. via choice of merge sequencing), and universal quantification appears as a geometric constraint.

## B.4 Space–Time Hierarchies

**Theorem B.10** (Time hierarchy)**.** *For reasonable $t(n)$,*

$$\mathrm{DTIME}(t(n)) \subsetneq \mathrm{DTIME}(t(n) \log t(n)).$$

**Theorem B.11** (Space hierarchy)**.** *For space–constructible $s(n)$,*

$$\mathrm{DSPACE}(s(n)) \subsetneq \mathrm{DSPACE}(s(n) \log s(n)).$$

*Remark* B.12. Such strict inclusions demonstrate that increasing resources truly increases computational power in the Turing model; analogous results for geometric computations require careful resource measures (size of configuration, dimensionality, number of merge operations).

## B.5 Circuit Complexity and Uniformity

**Definition B.13** (Boolean circuit)**.** A Boolean circuit is a directed acyclic graph of logical gates computing a Boolean function $f : \{0,1\}^n \to \{0,1\}$. Its size is the number of gates and its depth is the length of the longest directed path.

**Definition B.14** (Circuit classes)**.** 1. $\mathsf{AC}^0$ are constant–depth, polynomial–size circuits with unbounded fan–in AND/OR gates.

2. $\mathsf{NC}^k$ are poly–size circuits of depth $O(\log^k n)$.

*Remark* B.15. Circuit classes give an alternative viewpoint for merge–collapse systems: merge corresponds to parallel composition, while collapse relates to depth reduction or abstraction. Establishing lower bounds in this setting is an open problem (see **??**).

## B.6 Encoding Lambda Calculus and Turing Machines

**Definition B.16** (Lambda calculus)**.** The untyped lambda calculus consists of variables, abstractions $\lambda x.M$, and applications $M\,N$. Reduction rules include $\beta$–reduction $(\lambda x.M)\,N \to M[x := N]$.

**Theorem B.17** (Church–Turing thesis (informal))**.** *The lambda calculus, Turing machines, and other standard models compute the same class of (partial) functions on natural numbers.*

*Remark* B.18. In SpherePop, lambda terms will be represented as geometric configurations, where application is modeled by merge, and abstraction corresponds to a collapse or quotient identifying a boundary component. Complexity bounds require that the encoding and its evolution remain polynomially controlled.

## B.7 Complexity in Geometric and Homotopy–Theoretic Models

**Definition B.19** (Size of a geometric configuration)**.** Fix a class of finite geometric objects (e.g. triangulated manifolds, or finite unions of balls). The *size* of a configuration is the number of primitive pieces needed to describe it (cells, simplices, generators).

*Remark* B.20. A merge–collapse system is computationally *efficient* if each elementary operation transforms a configuration of size $n$ into one of size $O(n^c)$ for some fixed $c$, and the number of operations needed to simulate a Turing step is polynomial in $n$. This is a key requirement in the proof of Turing completeness in **??**.

## B.8 Distributed Complexity and CRDTs

**Definition B.21** (Distributed complexity (informal))**.** Given a network of processes updating a shared state, the distributed complexity measures the number of rounds or messages required to achieve consistency or agreement under specified failure models.

*Remark* B.22. CRDTs (Conflict–Free Replicated Data Types) provide eventual consistency using commutative operations. Interpreting these operations as lamphrodynamic flows introduces a geometric semantics in which convergence properties become statements about dissipative field evolution.

## B.9 Equivalence Problems and Higher Categories

*Remark* B.23. Determining equivalence of objects in an $\infty$–category is in general far more complex than equality of classical data. Complexity grows rapidly with dimension and coherence conditions. At present, only partial complexity classifications are known; these motivate the conjectured hardness results in Volume III for geometric computation of equivalence.

## B.10 Summary

This appendix has outlined the classical foundations of computational complexity relevant to geometric and homotopy–theoretic computation. The key concepts used in later chapters are reductions, completeness, complexity classes, hierarchy theorems, and the geometric interpretation of merge–collapse operations. Further references are provided in the unified bibliography.

# Appendix C

# Philosophy of Physics Primer

This appendix provides a concise orientation to selected themes in the philosophy of physics that are most relevant to the interpretation of RSVP. It is not intended as a comprehensive survey, but rather as a focused guide to the metaphysical and epistemological commitments implicit in a lamphrodynamic, field–theoretic ontology. Our aim is to clarify (i) what counts as a physical theory in this context, (ii) how mathematical structure relates to physical ontology, and (iii) to indicate the points of entry for structural realism, process metaphysics, and the interpretation of emergent spacetime.

## C.1 What Counts as a Physical Theory?

From a minimal perspective, a physical theory consists of a domain of application ("what it talks about"), a family of mathematical objects (in which the relevant quantities reside), a dynamics formulated in terms of those objects, and an interpretation connecting the formalism to empirical content. Modern practice tends to regard mathematical structures as indispensable: the *content* of a theory is exhausted by its formal and interpretive apparatus. Nevertheless, a persistent tension remains between purely *mathematical* structure and putative *physical* existence.

Classical accounts often treat spacetime as a primitive arena and physical fields as structures defined over this arena. In contrast, derived geometry suggests that the "arena" might itself be a higher–categorical construct, calibrated to the local and global behavior of fields. In RSVP, the lamphrodynamic fields carry the primary ontological load, and spacetime geometry emerges as a secondary, sometimes approximate, construct.

## C.2 Mathematical Structure and Physical Ontology

### C.2.1 The dilemma

A recurring dilemma concerns whether a mathematical structure *is* the physical world (structural realism), or whether mathematical structure merely *represents* the physical world (more traditional realism). The former position avoids dualism between mathematics and physics, but must account for empirical and modal features of scientific practice. The latter retains an ontological gap between mathematics and reality, but risks proliferating unobservable substrates.

In RSVP, the choice is sharpened by the use of derived stacks and shifted symplectic structures. If these objects are taken merely as formal tools, the resulting theory offers a sophisticated representation of physics without asserting that these higher–categorical structures exist in the world. If, however, one takes the lamphrodynamic plenum as the ontic source of spacetime geometry and physical interaction, then the derived structures are themselves objects of physical ontology.

### C.2.2 Interpretive stance

The position taken in this monograph is cautious: we adopt a *moderate structural realism* in which the lamphrodynamic fields constitute the ontological substrate, while the higher–categorical apparatus is regarded as the most adequate mathematical language for describing this substrate. Nothing in the formalism requires an identification of mathematical structure with physical reality, but the consilience of field ontology and higher geometry strongly suggests that the formalism is not merely instrumental.

## C.3 Structural Realism

Structural realism maintains that the *structure* described by a successful physical theory is what exists, rather than the specific objects or substrates posited by earlier ontologies. There are two principal varieties:

1. *Epistemic structural realism*, which holds that we can know only the structure of the world but not its intrinsic nature.

2. *Ontic structural realism*, which maintains that structure is all there is.

For RSVP, epistemic structural realism is too modest: lamphrodynamic dynamics purport to describe the generation of gravitational and cosmological effects from entropy and vector flows. Ontic structural realism is therefore more natural: the plenum and its derived moduli are treated as the structural substrate from which spatiotemporal geometry and emergent gravitational phenomena arise.

## C.4 Process Philosophy and Field Ontology

### C.4.1 Process over substance

A process–theoretic interpretation holds that physical reality consists not of fundamental substances endowed with properties, but rather of processes, interactions, and relations. Classical field theories already move in this direction. What RSVP adds is that the lamphrodynamic fields encode not only energetic and geometric properties, but also *entropy*, construed as a physically ontological quantity.

### C.4.2 Field ontology in RSVP

Field ontology here means that $(\Phi, \boldsymbol{v}, S)$ are not ancillary descriptors of material distribution, but primary elements of physical reality. Spacetime curvature, redshift, and cosmological dynamics are emergent aspects of the lamphrodynamic plenum. The lamphrodynamic field is thus a process ontology, represented in a derived geometric language.

## C.5 Emergence, Explanation, and Reduction

Emergence in the present context refers to the appearance of effective degrees of freedom (metric geometry, gravitational dynamics, cosmology) from the lamphrodynamic fields. Traditional reductionist accounts presuppose a fixed spacetime. Here, spacetime itself is emergent.

An adequate explanation must show:

1. how the lamphrodynamic fields generate effective geometrical structures;
2. how the emergent geometry satisfies approximate Einstein equations or modified field equations;
3. when and why classical spacetime breaks down or reconfigures.

Reduction is therefore asymmetric: RSVP reduces geometry to lamphrodynamics, but lamphrodynamics cannot be reduced to classical geometry without loss of essential structure.

## C.6 Measurement and Observers

The status of measurement in RSVP differs from quantum mechanics, although similar philosophical questions arise. The lamphrodynamic fields encode thermodynamic and informational structure, but measurement is a classical act extracting effective geometric and energetic quantities. Observers are not fundamental objects, but emergent dynamical systems embedded in the lamphrodynamic plenum. Nothing in the theory requires an observer–centric ontology.

## C.7 Relevance to RSVP

The foregoing themes clarify several interpretive claims:

- spacetime is not fundamental;
- entropy and vector flow are ontic, not merely epistemic;
- gravitational and cosmological phenomena are emergent from lamphrodynamics;
- higher–categorical and derived structures encode ontic relations among lamphrodynamic configurations.

This philosophical stance does not prejudge the correctness of RSVP, but provides a coherent interpretive background against which the theory is articulated.

## C.8 Further Reading

Suitable introductions include Ladyman and Ross (*Every Thing Must Go*), French (*The Structure of the World*), and selected works of Whitehead and Rescher for process metaphysics. Readers interested in structural realism in physics may consult the extensive literature on ontic structural realism and the philosophy of quantum field theory.

# Appendix D

# Full Proofs of Technical Lemmas

This appendix records several technical proofs and auxiliary results that were deferred in the body of Volume III for reasons of readability. They concern three main themes: homotopy–theoretic foundations of the SpherePop calculus; computational expressiveness and Turing completeness; and convergence results for distributed lamphrodynamic semantics (TARTAN–RSVP). We adopt the notation and indexing conventions of the main text; all cross–references are to Volume III unless stated otherwise.

**Structure of this appendix.** The material is organized as follows.

1. Detailed proofs for **Theorem 24.2** (merge as homotopy colimit), including auxiliary lemmas on cofibrancy, latching objects, and pushout constructions in model categories of topological configurations.

2. Supplementary results supporting the informal evidence for **Conjecture 20.3** (collapse as derived truncation), including worked examples in low dimensions and a characterization of boundary invariants preserved by collapse.

3. Full simulation lemmas used in the proof of **Turing completeness** in **??**, including the reduction of lambda calculus to geometric rewrite systems and complexity–theoretic embeddings.

4. Technical convergence results for distributed computation in **CRDT–RSVP** semantics (Chapter 6), including homotopy fixed–point proofs and Gray–code acyclicity.

## D.1 Proof of Merge as Homotopy Colimit (Theorem 24.2)

Recall the statement of **??**:

**Theorem D.1** (Merge as Homotopy Colimit)**.** *Let $\{{}_{\alpha}\}$ be a diagram of SpherePop configuration spaces indexed by a small category $I$ and let $({}_{\alpha})$ denote the merge configuration obtained by taking geometric unions of spherical regions with compatible boundary data. Then under the model category structure of* **??**, *the natural comparison map*

$$\operatorname{hocolim}_{I\,\alpha} \longrightarrow ({}_{\alpha})$$

*is a weak equivalence whenever each $_\alpha$ is cofibrant and all transition maps are cofibrations.*

*Proof.* We sketch the conceptual structure and refer to standard model category arguments for routine verifications. Let  denote the model category of SpherePop configurations as in **??**. By construction, objects of  represent finite unions of labeled spherical regions with admissible boundary data; cofibrations are generated by cellular inclusions together with gluing operations along sub-spheres.

Given a diagram $\{_\alpha\}$, construct the Reedy model structure associated to $I$; this produces matching and latching objects $M_\alpha, L_\alpha$. Because transitions are cofibrations by assumption, latching maps $L_\alpha \to_\alpha$ are cofibrations. Therefore, the canonical Reedy cofibrant replacement coincides with the original diagram.

Now form the homotopy colimit via the Bousfield–Kan formula

$$\mathrm{hocolim}_{I\ \alpha} \simeq |N_\bullet(I) \times_{I\ \alpha}|$$

and observe that, at the level of geometric realization, this aligns with the sequential composition of gluing operations implemented in the merge construction. In the presence of cofibrancy, the comparison map is induced by the universal property of pushouts along cofibrations. Weak equivalence follows from the fact that geometric union does not introduce new homotopy data beyond that captured by the simplicial enhancement of the diagram. □

*Remark* D.2. Standard references include Hirschhorn (*Model Categories*), Hovey (*Model Categories*), and Dwyer–Hirschhorn–Kan–Smith (*Homotopy Limit Functors*). Our argument follows classical patterns but applied to configuration–type objects.

## D.2 Collapse as Derived Truncation (Conjecture 20.3)

We supply evidence and partial results toward **??**. Let $\kappa()$ denote the collapse of a configuration and let $t_0()$ denote its 0–truncation in the associated $\infty$–topos.

**Proposition D.3.** *If  admits a filtration by spherical cells whose attaching maps are $n$–connected for all $n > 0$, then*

$$t_0() \simeq \kappa().$$

*Proof.* Filtrations of this form imply that higher homotopy data is carried only by attachments of strictly positive connectivity. Collapsing sphere boundaries removes these attachments but leaves the 0–level data invariant. Therefore, collapse agrees with truncation. □

*Remark* D.4. The conjecture holds in all examples computed in low dimensions (**??**). Full generality remains open.

## D.3 Turing Completeness: Simulation Lemmas

We record two key simulation lemmas used in the proof of Turing completeness.

**Lemma D.5** (Simulation of Lambda Calculus)**.** *For every lambda expression $e$, there exists a finite SpherePop configuration $_e$ and a corresponding rewrite rule such that $_e$ reduces to $_{e'}$ if and only if $e$ reduces to $e'$ in the standard lambda calculus.*

*Proof.* Encode lambda abstraction and application as labeled spherical constructors with directed attachment data. Beta–reduction corresponds to a local merge–collapse operation along the interface encoding the application. Correctness follows from structural induction on $e$. □

**Lemma D.6** (Space and Time Overhead)**.** *The geometric reduction of $_e$ simulates lambda reduction with at most polynomial overhead in the size of $e$.*

*Proof.* Each geometric rewrite touches a bounded number of spherical generators; descendants grow linearly in redex size, and collapse reduces spatial footprint. Induction on reduction length gives the desired bound. □

## D.4 CRDT–*RSVP* Convergence

**Proposition D.7** (Eventual Consistency)**.** *Under lamphrodynamic update semantics, any diagram of CRDT–encoded configurations converges to a homotopy fixed point whenever transition maps satisfy Gray–code acyclicity.*

*Proof.* Construct an $\infty$–categorical diagram of lamphrodynamic update rules and apply homotopy limit comparison; Gray–code constraints prevent the introduction of directed cycles, ensuring convergence to a terminal object in the homotopy category. Details follow the line of reasoning in **??**. □

*Remark* D.8*.* Analogous results hold for PN–Counters and LWW–Registers provided their update operations satisfy lamphrodynamic ordering; see **??** for a catalogue of cases.

## D.5 Further Reading

Relevant texts include Quillen (*Homotopical Algebra*), Hirschhorn (*Model Categories and Their Localizations*), Hovey (*Model Categories*), and standard references on rewriting systems and CRDTs. Interpretational details for RSVP appear throughout the corresponding chapters of Volume III.